Conhecendo os principais protocolos de comunicação da internet

HTTP | Web socket

Juliana Mascarenhas

Tech Education Specialist / Sócia - criadora de conteúdo no @Simplificandoredes e @SimplificandoProgramação

Mestre em modelagem computacional

@in/juliana-mascarenhas-ds/



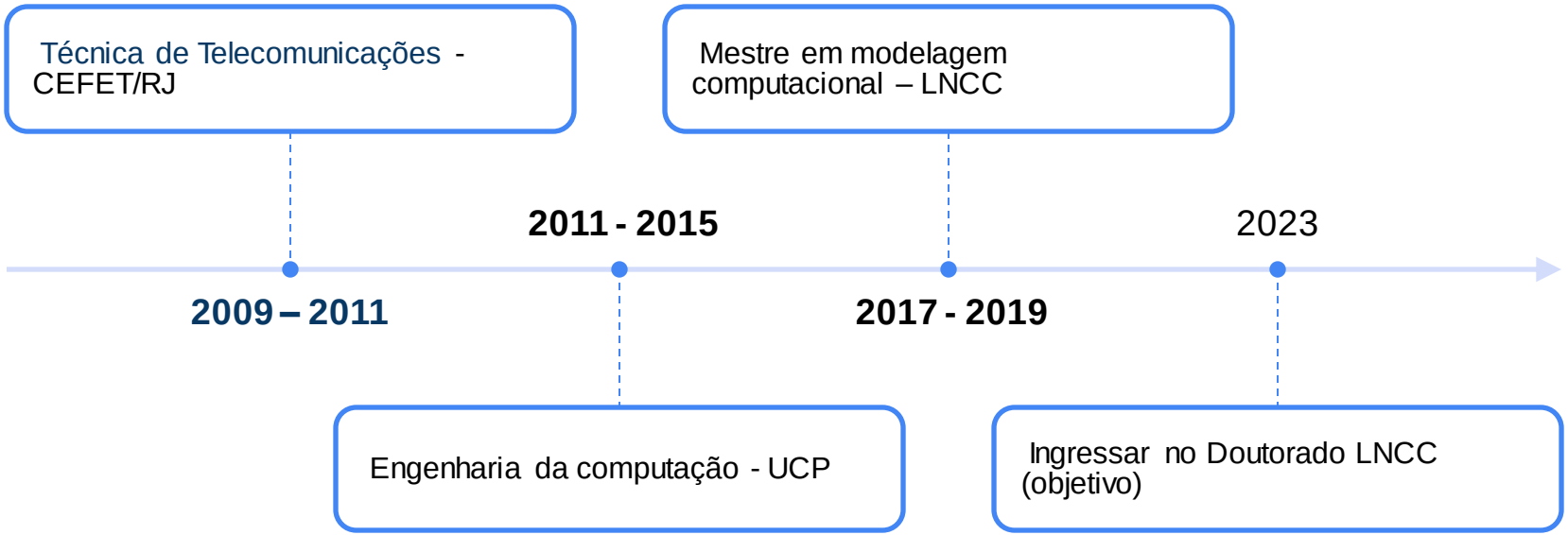
https://github.com
/julianazanelatto

Juliana Mascarenhas

Tech Education Specialist
Sócia no @SimplificandoRedes

Cientista de dados
Desenvolvedora Java/Python
Me Modelagem Computacional - LNCC

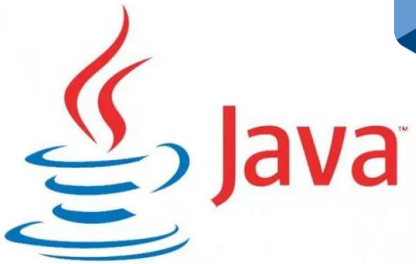
Sobre Mim


Técnica de Telecomunicações - CEFET/RJ
Mestre em modelagem computacional – LNCC
2011 - 2015
2023
2009 – 2011
2017 - 2019
Engenharia da computação - UCP
Ingressar no Doutorado LNCC (objetivo)

Sobre Mim















APACHE

HTTP SERVER PROJECT



ZABBIX



Sobre Mim


Redes de computadores
2009
SNORT
ZABBIX
Gerenciamento de Projetos
2014
PMI
Project Management Institute
Java
2012
Programação
PostgreSQL
2017
NETWORK SCIENCE
Ciencia de dados
python

Cursos


DIGITAL INNOVATION ONE
Criando um Microsserviço de Upload de Imagens com o Amazon S3
Juliana Mascarenhas
Me. Modelagem Computacional


Introdução à Programação e Pensamento Computacional
Curso base para iniciante
Juliana Mascarenhas
Tech Education Specialist / Sócia (Content Creator) @SimplificandoRedes
Me Modelagem Computacional / Cientista de dados
@in/juliana-mascarenhas-ds/
[1]


DIGITAL INNOVATION ONE
Criando uma API REST conectada a Amazon RDS com Java
Juliana Mascarenhas
Me. Modelagem Computacional


DIGITAL INNOVATION ONE
Java Spring com RabbitMQ
Juliana Mascarenhas
Mestre em modelagem computacional

Objetivo Geral

APIs, servidores, clientes se comunicam na internet através de um protocolo de comunicação. Este protocolo rege a comunicação e determina o comportamento das partes. Esse cenário pode influenciar no desenvolvimento da aplicação do dev. Sendo assim, é preciso conhecer minimamente os cenários existentes a fim de mitigar possíveis erros.

Pré-requisitos

1. Noções de funcionamento do modelo em camadas TCP/IP ou OSI;

2. Conceitos básicos de segurança da informação: autenticação, chave, certificado ...

Percurso

Aula 1	**Protocolo de comunicação Web HTT**
Aula 2	HTTPS, o que muda no protocolo?
Aula 3	Protocolo de comunicação Web Socket

Aula 1

Protocolo de comunicação Web HTTP

//Conhecendo os principais protocolos de comunicação da internet

Percurso

Etapa 1	**Como funciona o protocolo HTTP?**
Etapa 2	Mensagens HTTP – Request & Response
Etapa 3	Para que servem Cookies e Cache?

Percurso

Etapa 4	HTTP 2.0 - Atualizações do protocolo
Etapa 5	Servidores / Sistemas de aplicação

Etapa 1

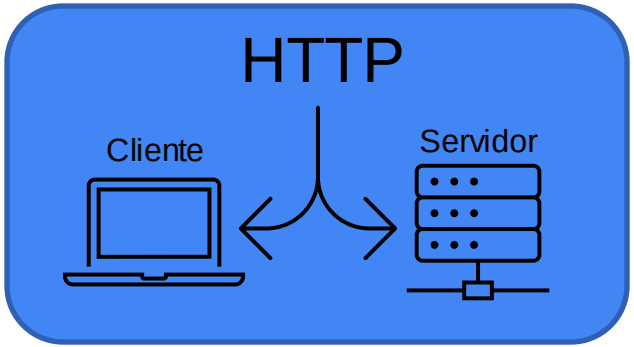
Como funciona o protocolo HTTP?

//Conhecendo os principais protocolos de comunicação da internet / Protocolo HTTP

Protocolo HTTP

HyperText Transfer Protocol (HTTP)

- Definido pelos RFCs 1945 e 2116
- Protocolo de comunicação
- Browser -> implementa o cliente HTTP
- Servidor -> host objetos web



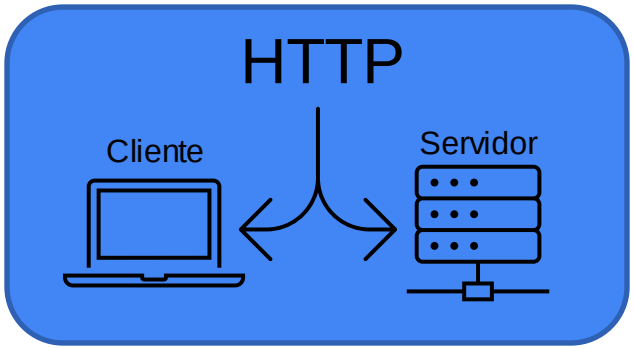
Protocolo HTTP

HyperText Transfer Protocol (HTTP)

Objetos web

- Cliente
  Mensagens - Request HTTP

- Servidor
  Mensagens – Response HTTP



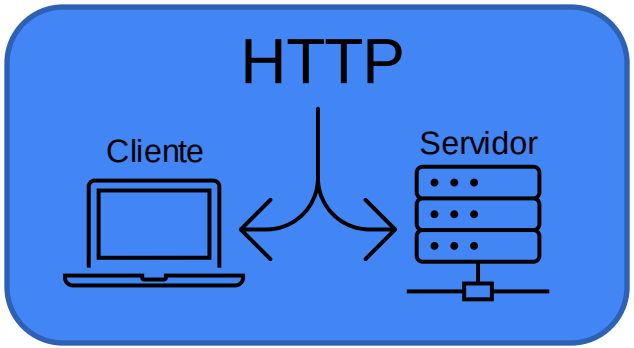
Protocolo HTTP

HyperText Transfer Protocol (HTTP)

- Cliente
  Mensagens - Request HTTP
- Servidor
  Mensagens – Response HTTP

Objetos web

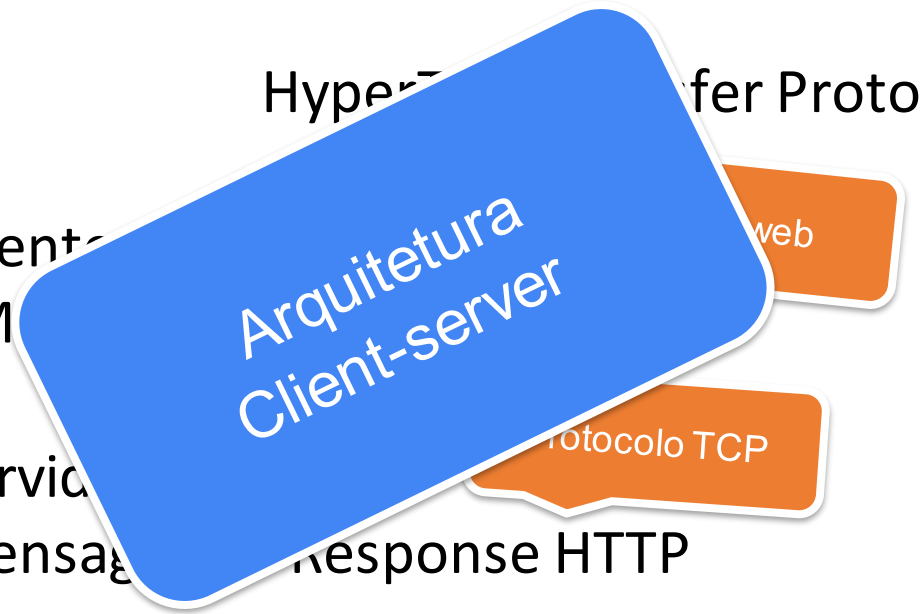
Protocolo TCP

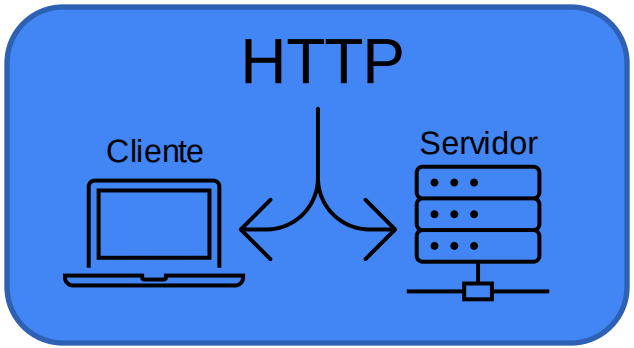


Protocolo HTTP

Hyper… …fer Protocol (HTTP)

- Client… …
  M…
- Servi… …
  Mensag… …Response HTTP

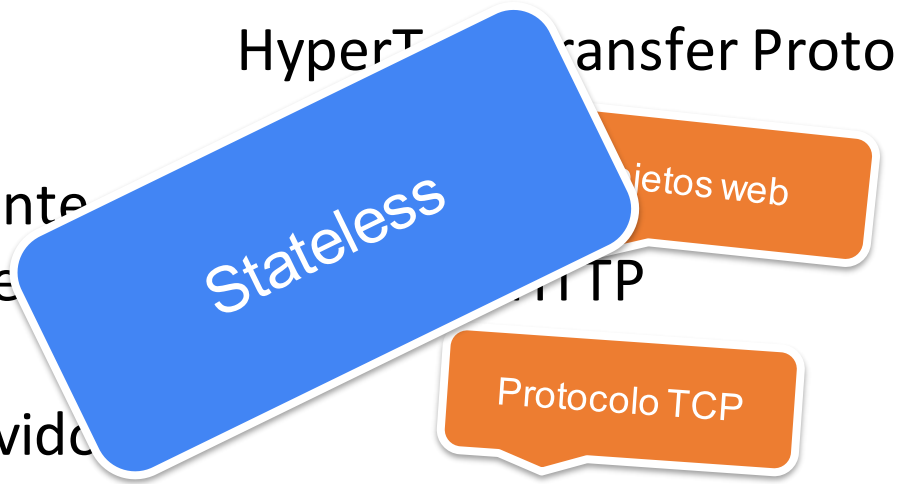


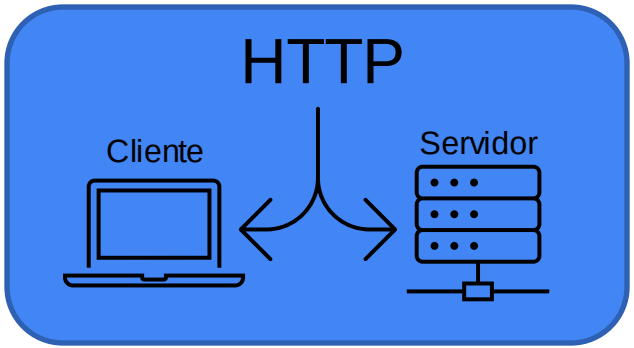


Protocolo HTTP

HyperT ransfer Protocol (HTTP)

- Cliente
  Me HTTP
- Servido
  Mensagens – Response HTTP





Prof

1° webpage

1991

HTTP 0.9

1996

HTTP 1.0

1997

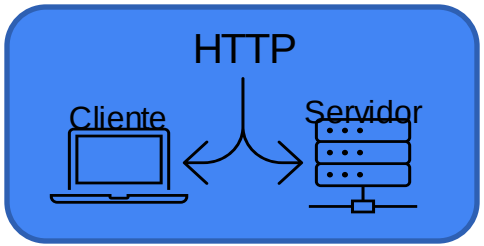
HTTP 1.1

2012

SPDY

2015

HTTP 2.0



Protocolo HTTP

Mudanças na web

1991

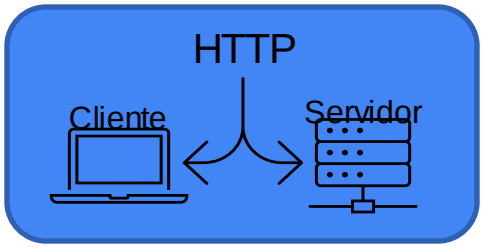
HTTP 0.9

1997

HTTP 1.1

2015

HTTP 2.0



HTTP 1.0

1996

SPDY

2012

CORS

Keep-Alive

Protocolo HTTP

Mensagens HTTP

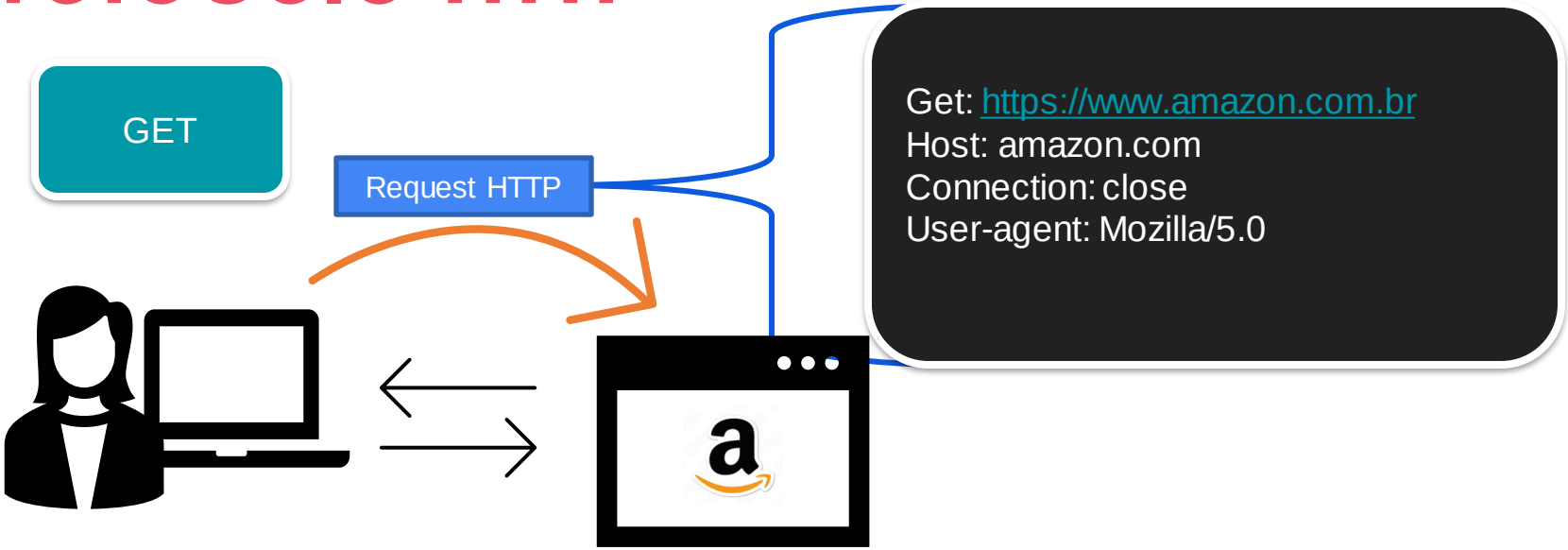
- Tipos: request & response

```
Get: /somedir/page.html HTTP/1.1
Host: www.someschool.edu
Connection: close
User-agent: Mozilla/5.0
Accpet-language: fr
```

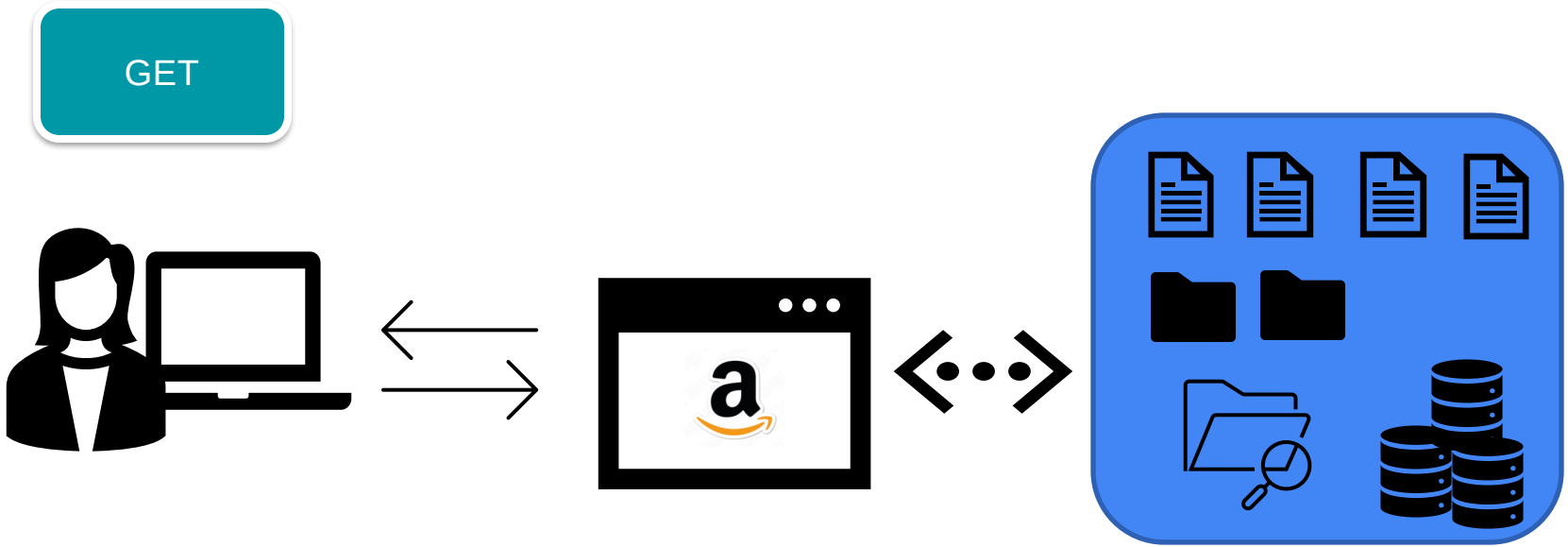
```
HTTP/1.1 200 OK
Connection: close
Date: Tue, 09 Aug 2011 15:44:04 GMT
Server: Apache/2.2.3 (CentOS)
Last-Modified: Tue, 09 Aug 2011 15:11:03 GMT
Content-Length: 6821
Content-Type: text/html

(data , data, ....)
```

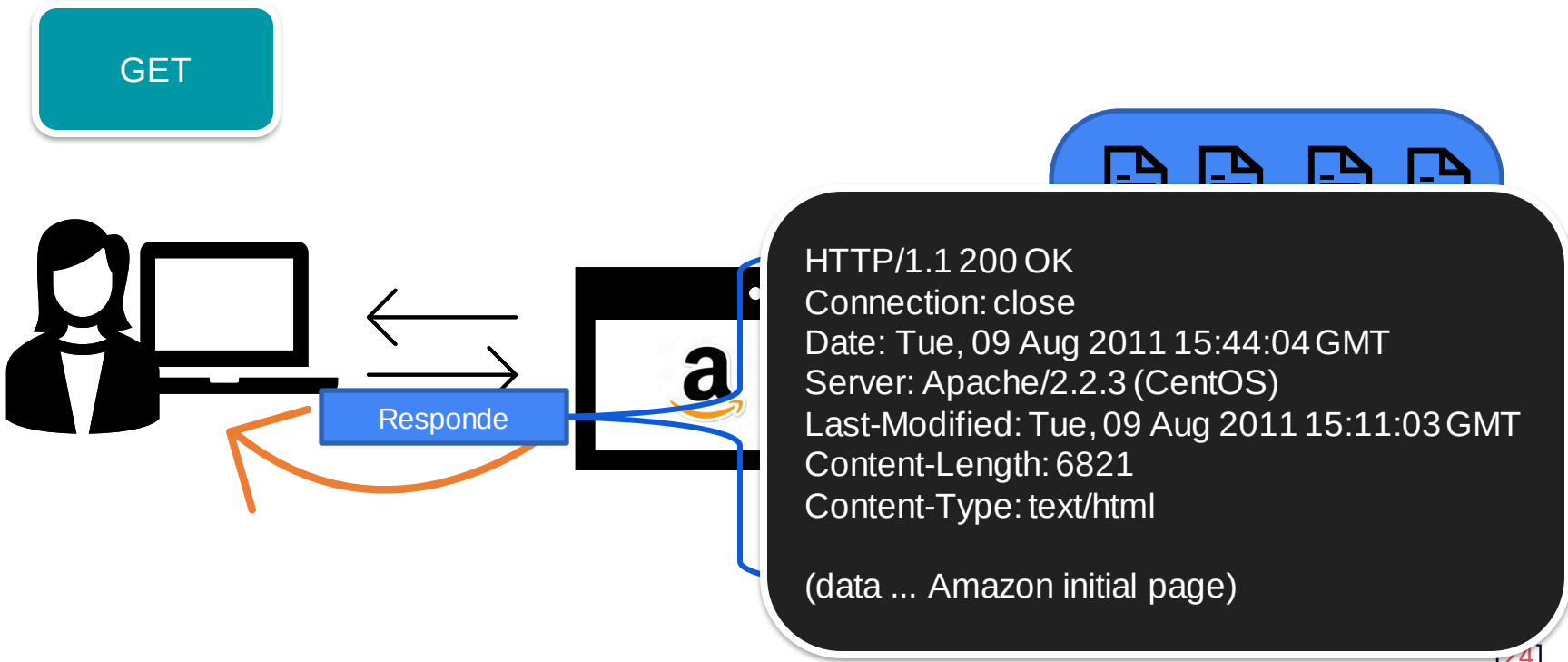
Protocolo HTTP



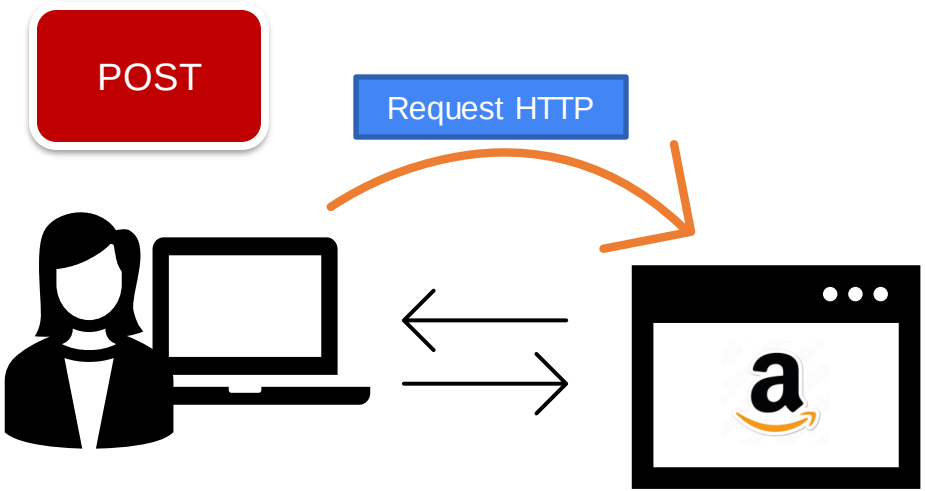
Protocolo HTTP



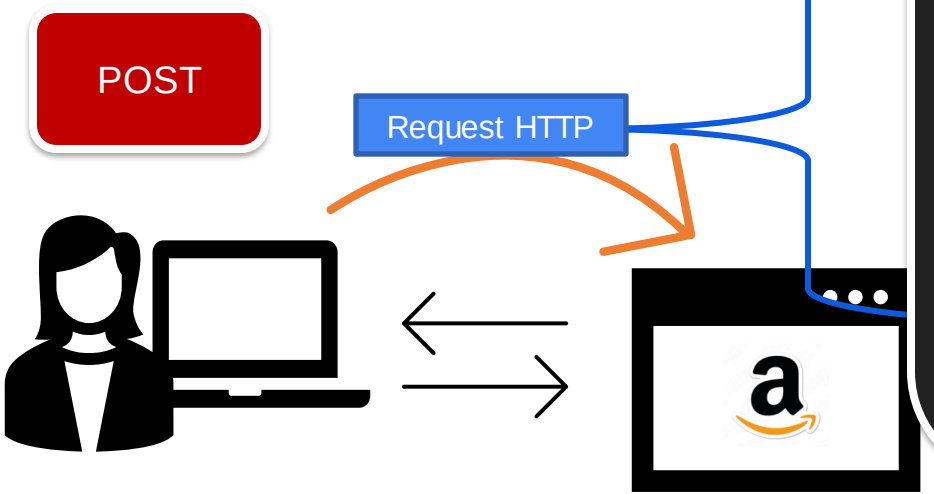
Protocolo HTTP



Protocolo HTTP



Protocolo HTTP

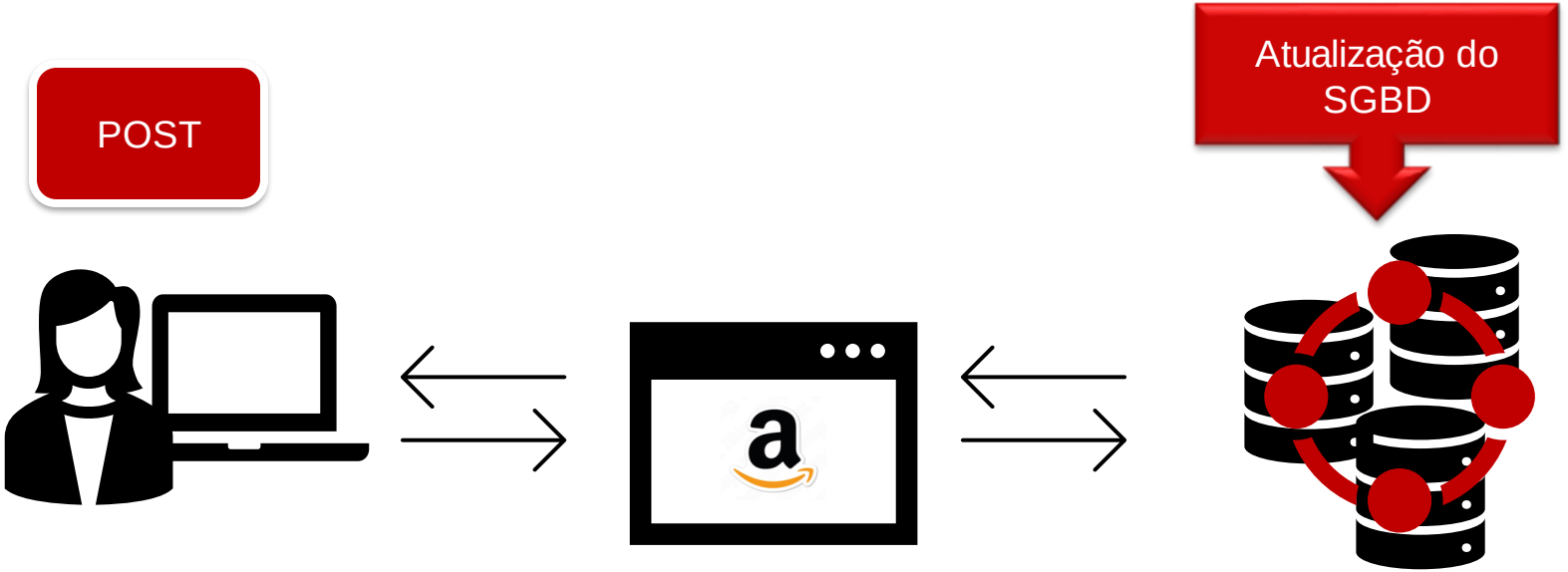


Post:
https://www.amazon.com.br/cadastro
Host: amazon.com
Connection: close
User-agent: Mozilla/5.0

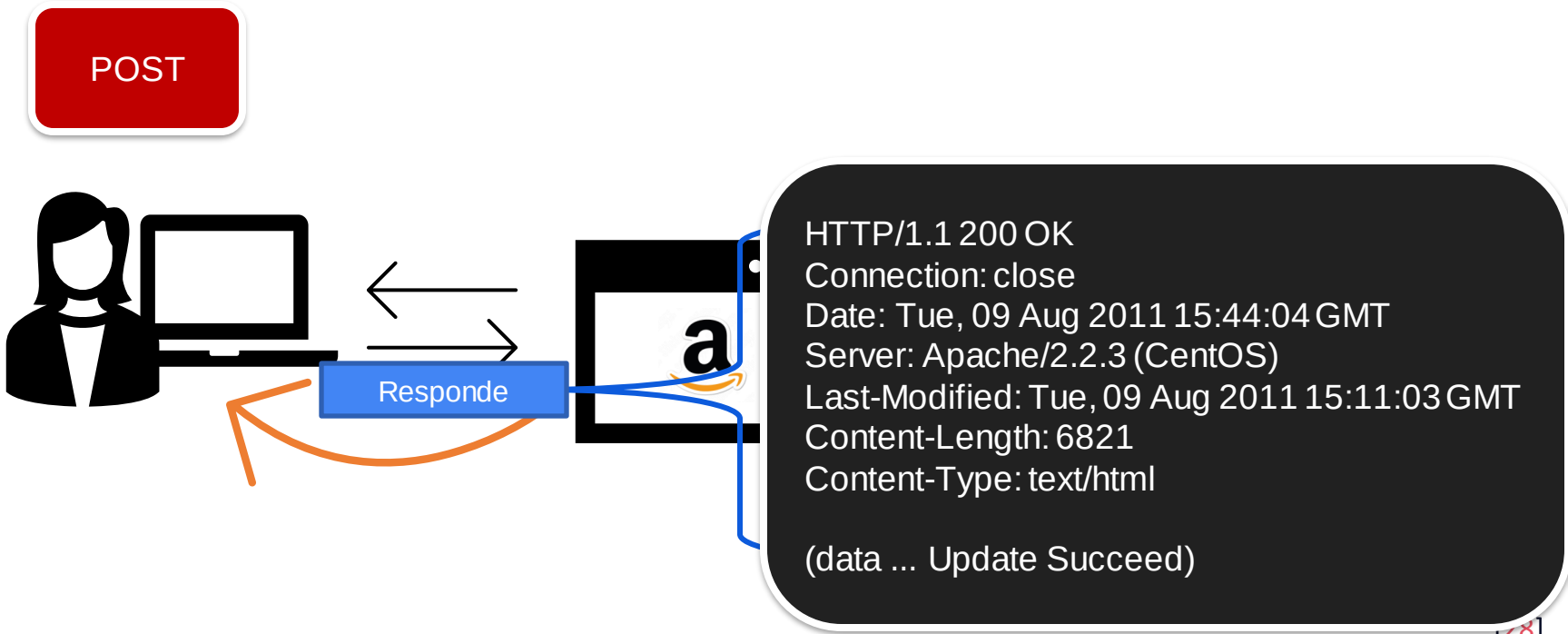
...

Data: nome, data de nascimento, contato
...

Protocolo HTTP



Protocolo HTTP



Protocolo HTTP



HTTP Entity Body

XML

```xml
<?xml version="1.0" encoding="UTF-8"?>
<authentication-context>
  <username>my_username</username>
  <password>my_password</password>
  <validation-factors>
    <validation-factor>
      <name>remote_address</name>
      <value>127.0.0.1</value>
    </validation-factor>
  </validation-factors>
</authentication-context>
```

JSON

```json
{
   "username" : "my_username",
   "password" : "my_password",
   "validation-factors" : {
      "validationFactors" : [
         {
            "name" : "remote_address",
            "value" : "127.0.0.1"
         }
      ]
   }
}
```

Protocolo HTTP

HTTP Entity Body

Body?



ARMAZENAMENTO

TRANSMISSÃO

RECONSTRUÇÃO

XML

```
<?xml version="1.0" encoding="UTF-8"?>
                                    ext>
                                    ame</username>
                                    ord</password>
                                    rs>
  </validation-factors>
</authentication-context>
```

JSON

```
{
   "username" : "my_username",
   "password" : "my_password",
   "validation-factors" : {
      "validationFactors" : [
         {
            "name" : "remote_address",
            "value" : "127.0.0.1"
         }
```

Protocolo HTTP

HTTP Entity Body

Body?



XML

JSON

```xml
<?xml version="1.0" encoding="UTF-8"?>
<authentication-context>
  <username>my
  <password>my
  <validation-
  </validation-factors>
</authentication-context>
```

```json
{
   "username" : "my_username",
   "password" : "my_password",
   "validation-factors" : {
      "validationFactors" : [
         {
            "name" : "remote_address",
            "value" : "127.0.0.1"
         }
      ]
   }
}
```

REQUEST HEADER

Content-Type: application/json
Accept: application/json

Extensible Markup Language



- Serialization
- Comunicação
- Metadados

XML

```xml
<?xml version="1.0" encoding="UTF-8"?>
<authentication-context>
  <username>my_username</username>
  <password>my_password</password>
  <validation-factors>
    <validation-factor>
      <name>remote_address</name>
      <value>127.0.0.1</value>
    </validation-factor>
  </validation-factors>
</authentication-context>
```

JavaScript Object Notation



Características

- Lightweight
- Independete de ling.
- simples

JS

JSON

```
{
   "username" : "my_username",
   "password" : "my_password",
   "validation-factors" : {
      "validationFactors" : [
         {
            "name" : "remote_address",
            "value" : "127.0.0.1"
         }
      ]
   }
}
```

XML e JSON

Comum

- Auto-descritivos
- Hierárquicos
- Independentes de linguagem de prog.
- Vasta utilização

```xml
<?xml version="1.0" encoding="UTF-8"?>
<authentication-context>
  <username>my_username</username>
  <password>my_password</password>
  <validation-factors>
    <validation-factor>
      <name>remote_address</name>
      <value>127.0.0.1</value>
    </validation-factor>
  </validation-factors>
</authentication-context>
```

```json
{
   "username" : "my_username",
   "password" : "my_password",
   "validation-factors" : {
      "validationFactors" : [
         {
            "name" : "remote_address",
            "value" : "127.0.0.1"
         }
      ]
   }
}
```

XML e JSON

Diferenças

- Tags
- Sucinto
- Legibilidade
- Utilização de arrays

```xml
<?xml version="1.0" encoding="UTF-8"?>
<authentication-context>
  <username>my_username</username>
  <password>my_password</password>
  <validation-factors>
    <validation-factor>
      <name>remote_address</name>
      <value>127.0.0.1</value>
    </validation-factor>
  </validation-factors>
</authentication-context>
```

```json
{
   "username" : "my_username",
   "password" : "my_password",
   "validation-factors" : {
      "validationFactors" : [
         {
            "name" : "remote_address",
            "value" : "127.0.0.1"
         }
      ]
   }
}
```

XML e JSON

Por qual optar?

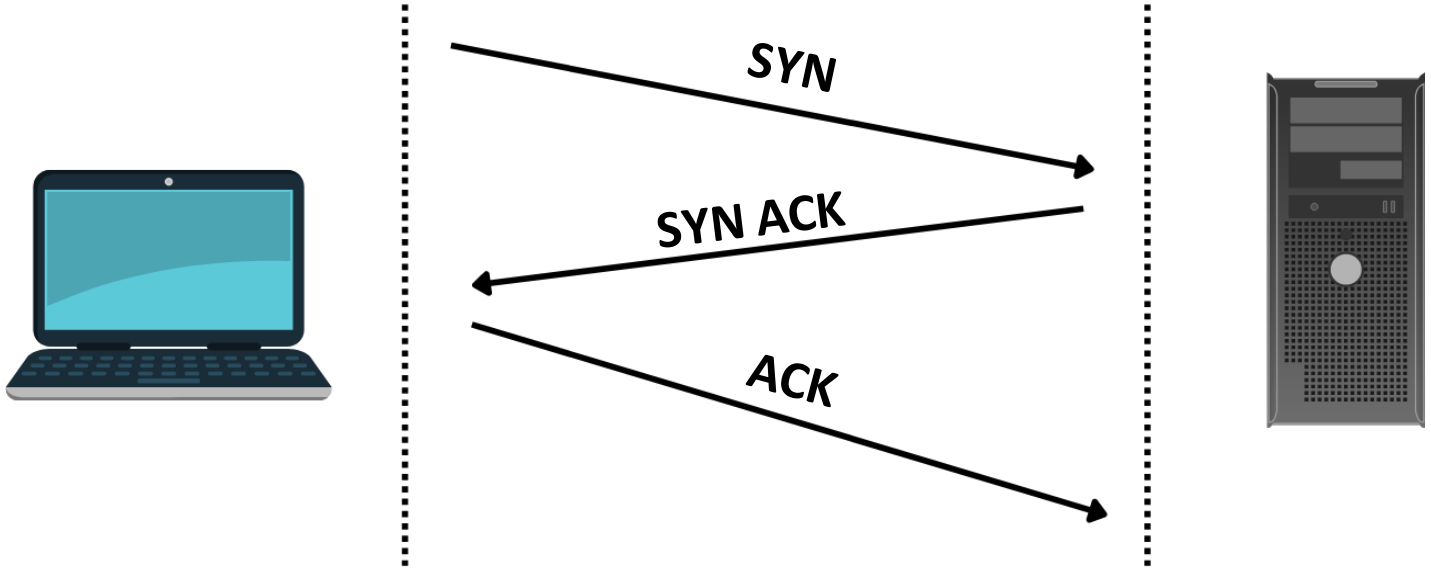
- Tecnologia
- Aplicações leves
- Complexidade
- Metadados

```xml
<?xml version="1.0" encoding="UTF-8"?>
<authentication-context>
  <username>my_username</username>
  <password>my_password</password>
  <validation-factors>
    <validation-factor>
      <name>remote_address</name>
      <value>127.0.0.1</value>
    </validation-factor>
  </validation-factors>
</authentication-context>
```

```json
{
   "username" : "my_username",
   "password" : "my_password",
   "validation-factors" : {
      "validationFactors" : [
         {
            "name" : "remote_address",
            "value" : "127.0.0.1"
         }
      ]
   }
}
```

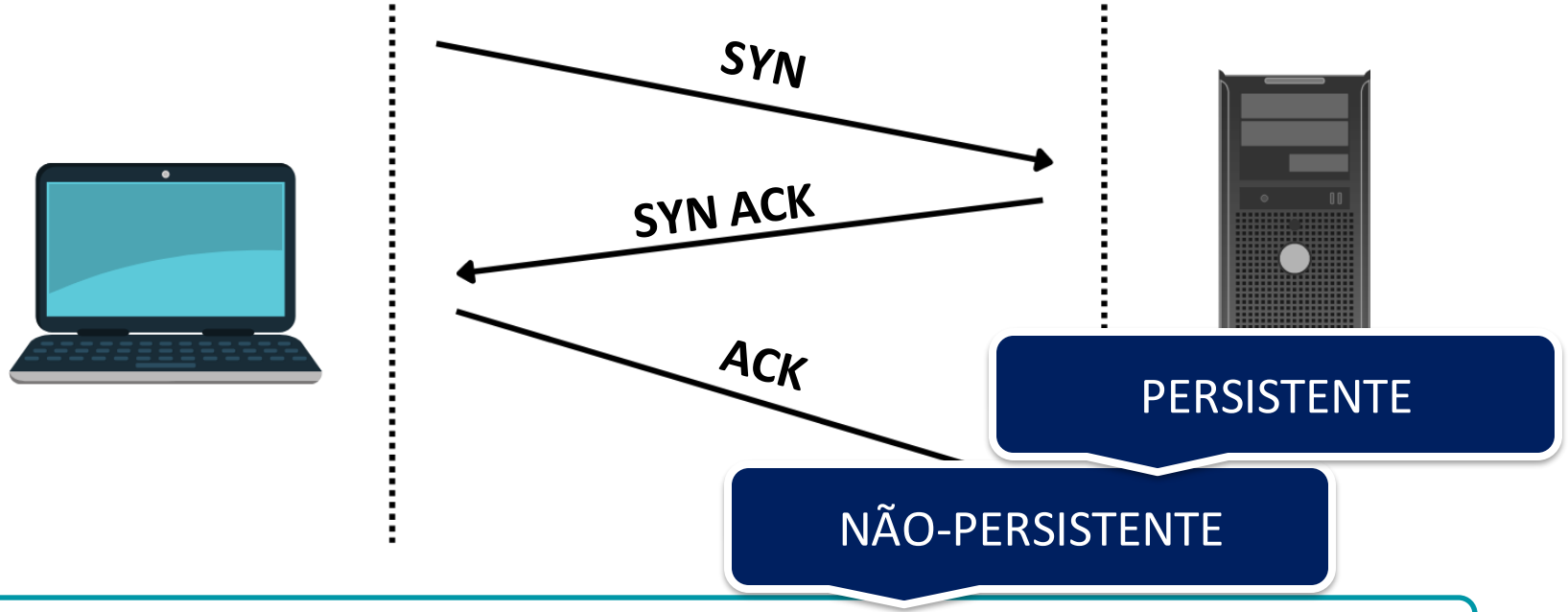
Particularidades do HTTP versão 1.1

Protocolo HTTP



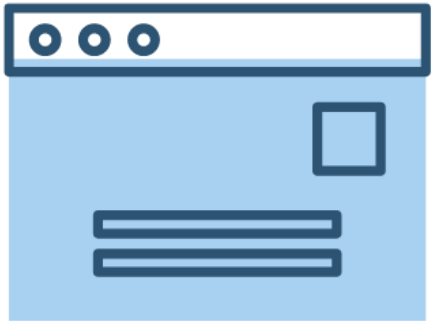
Transmission Control Protocol - TCP

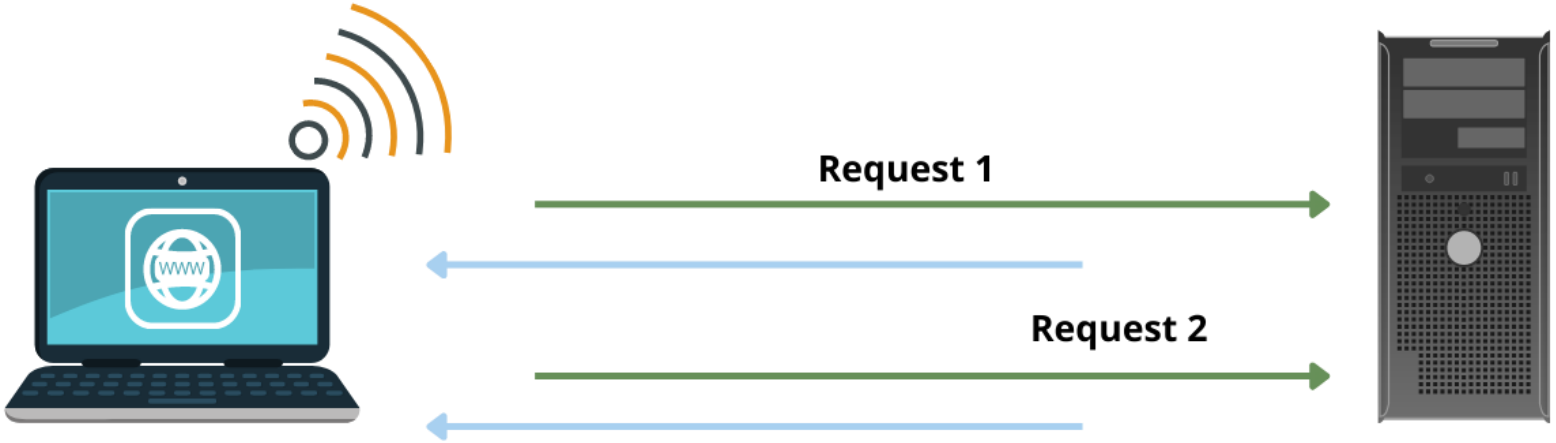
Protocolo HTTP



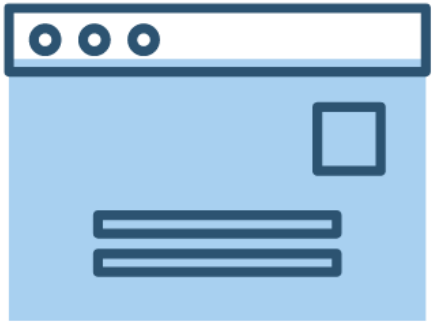
Transmission Control Protocol - TCP

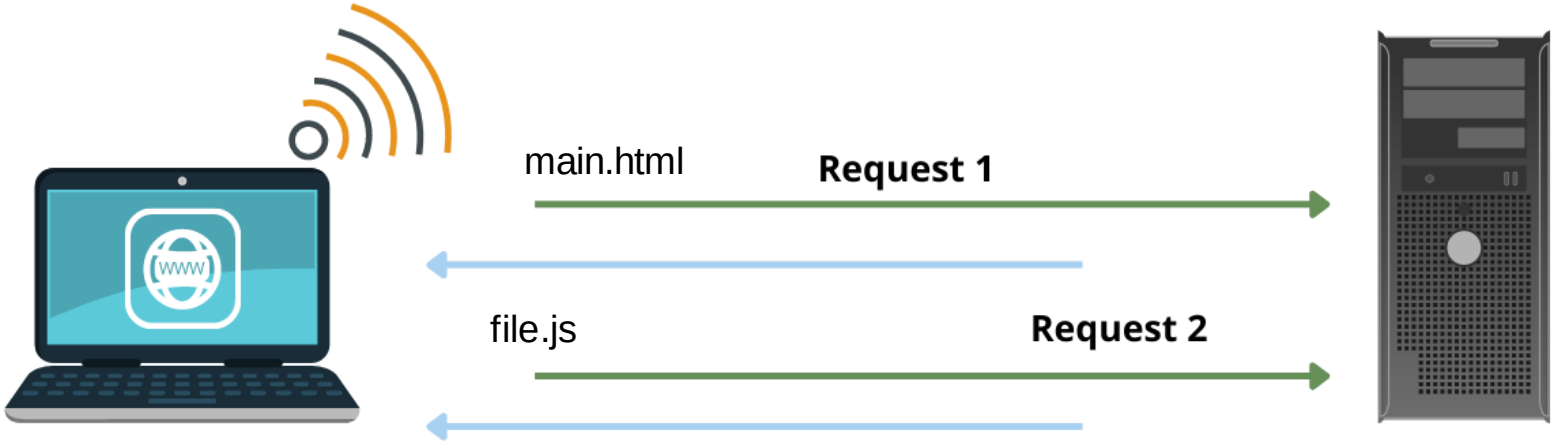
HTTP 1.1



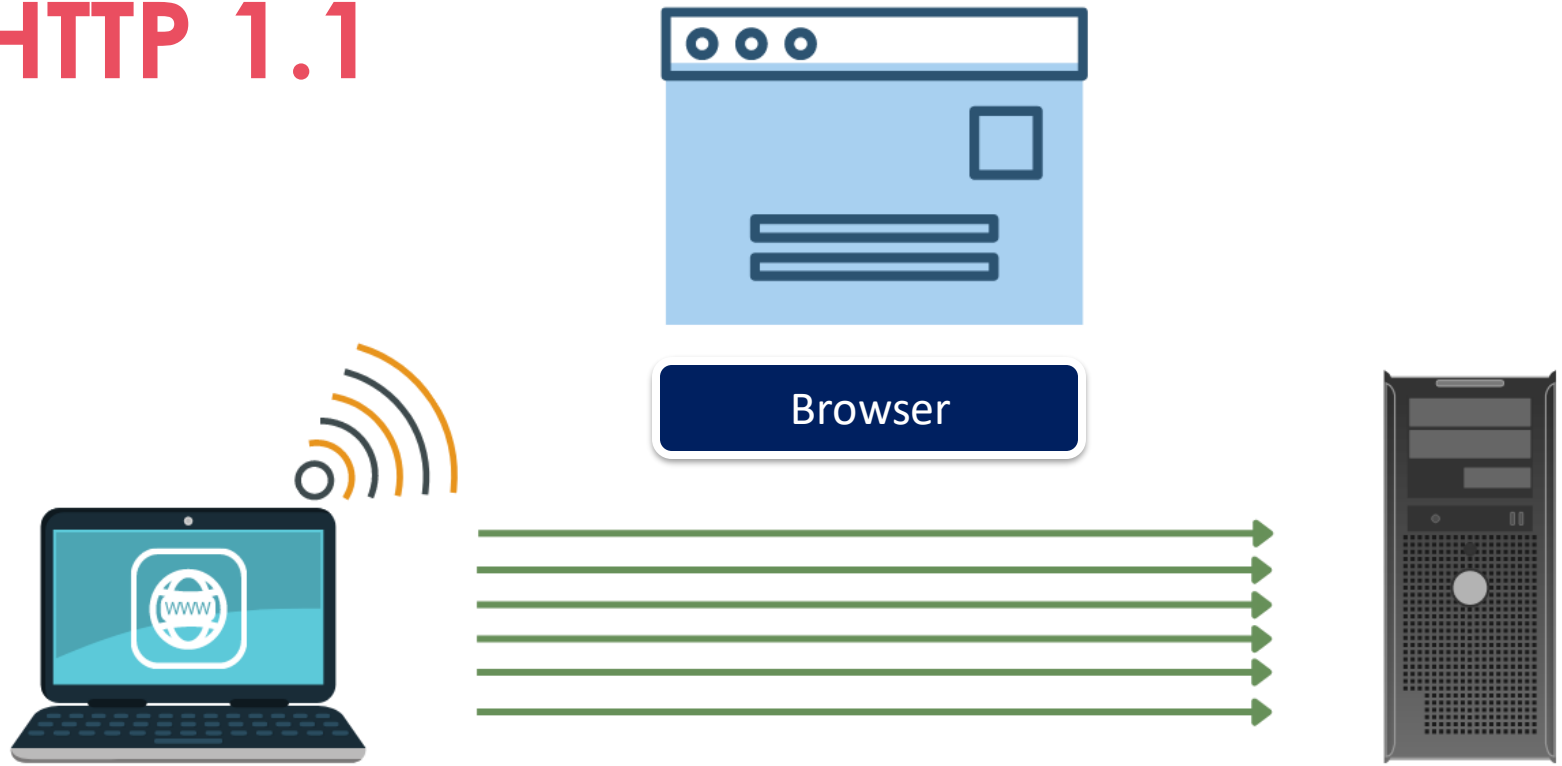


HTTP 1.1

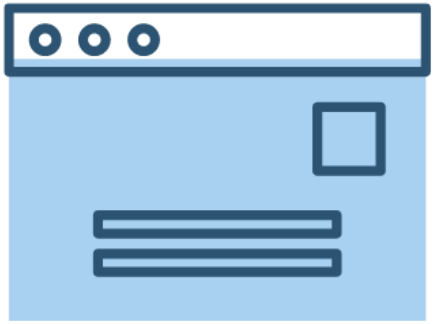




HTTP 1.1

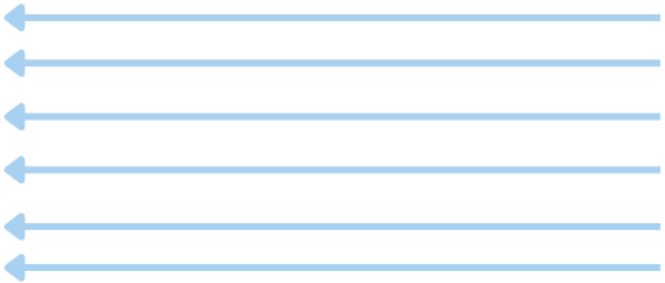


HTTP 1.1



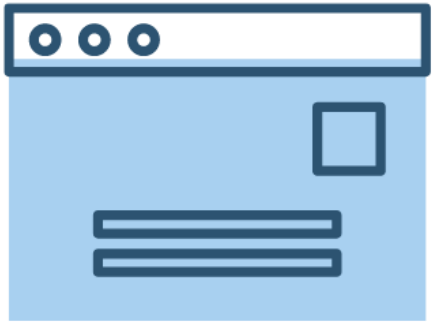
Browser







HTTP 1.1



Permanecer conectado?

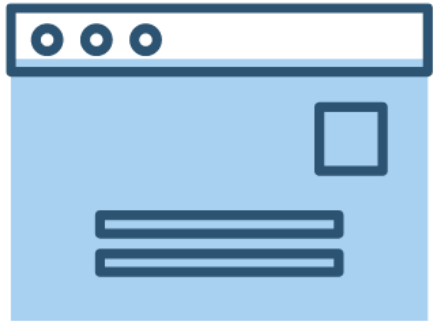
Browser



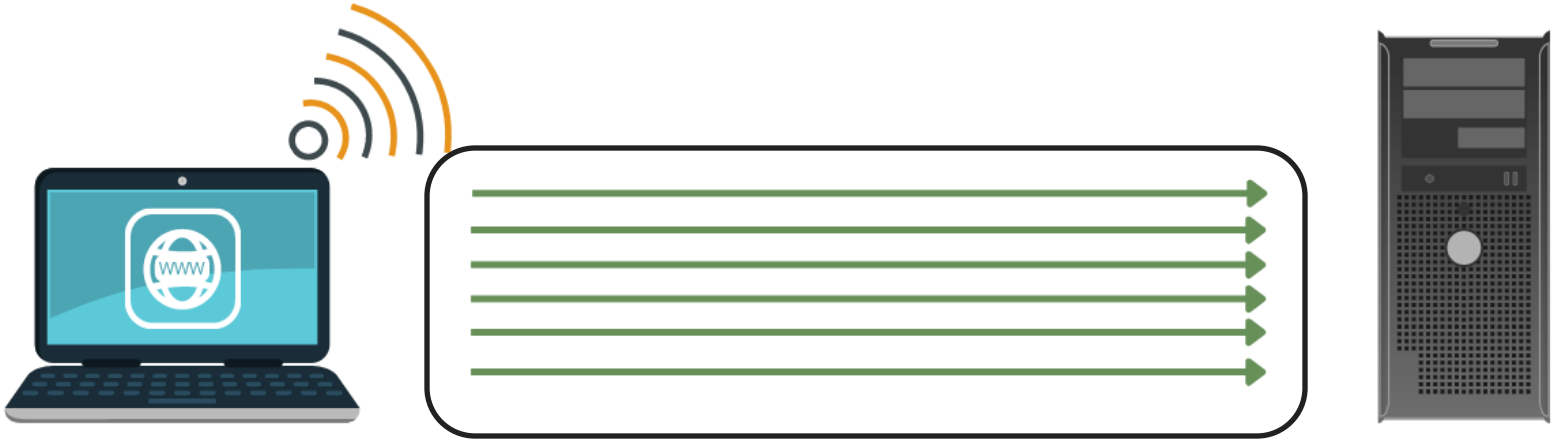




HTTP 1.1



Permanecer conectado?



TCP Persistente

HTTP 1.1



TCP Persistente

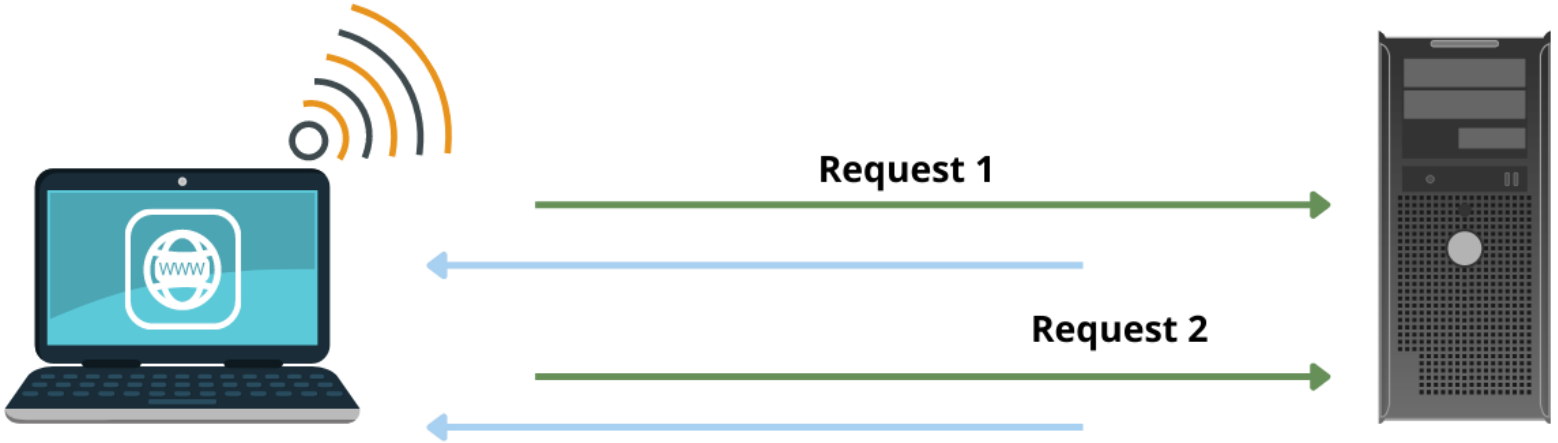
HTTP 1.1

caching

Foco: css/js e gzip

Repetição de dados

HOL – Head of Line Blocking



Etapa 2

Mensagens HTTP – Request & Response

//Conhecendo os principais protocolos de comunicação da internet / Protocolo HTTP

Percurso

~~Etapa 1~~ ~~Como funciona o protocolo HTTP?~~

Etapa 2 **Mensagens HTTP – Request & Response**

Etapa 3 Para que servem Cookies e Cache?

Percurso

Etapa 4 HTTP 2.0 - Atualizações do protocolo

Etapa 5 Servidores / Sistemas de aplicação

Mensagens HTTP

Mensagens HTTP

- Tipos: request & response

```
Get: /somedir/page.html HTTP/1.1
Host: www.someschool.edu
Connection: close
User-agent: Mozilla/5.0
Accpet-language: fr
```

```
HTTP/1.1 200 OK
Connection: close
Date: Tue, 09 Aug 2011 15:44:04 GMT
Server: Apache/2.2.3 (CentOS)
Last-Modified: Tue, 09 Aug 2011 15:11:03 GMT
Content-Length: 6821
Content-Type: text/html

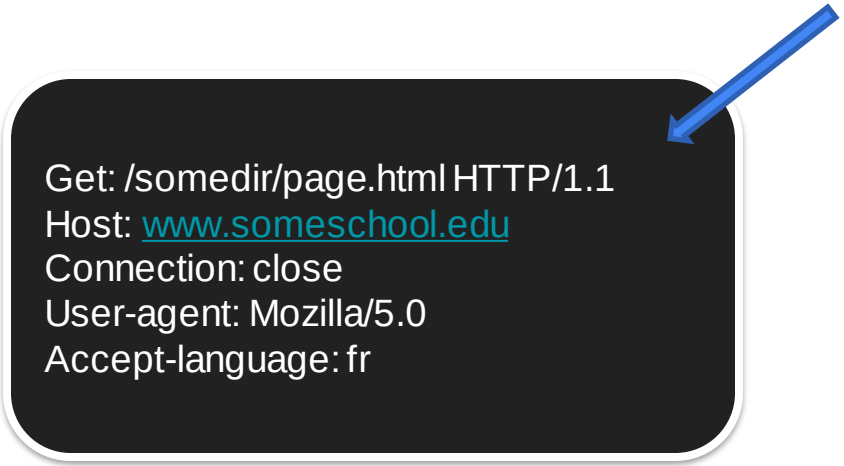
(data , data, ....)
```

HTTP Request

Mensagens HTTP - Request

Texto em ASCII

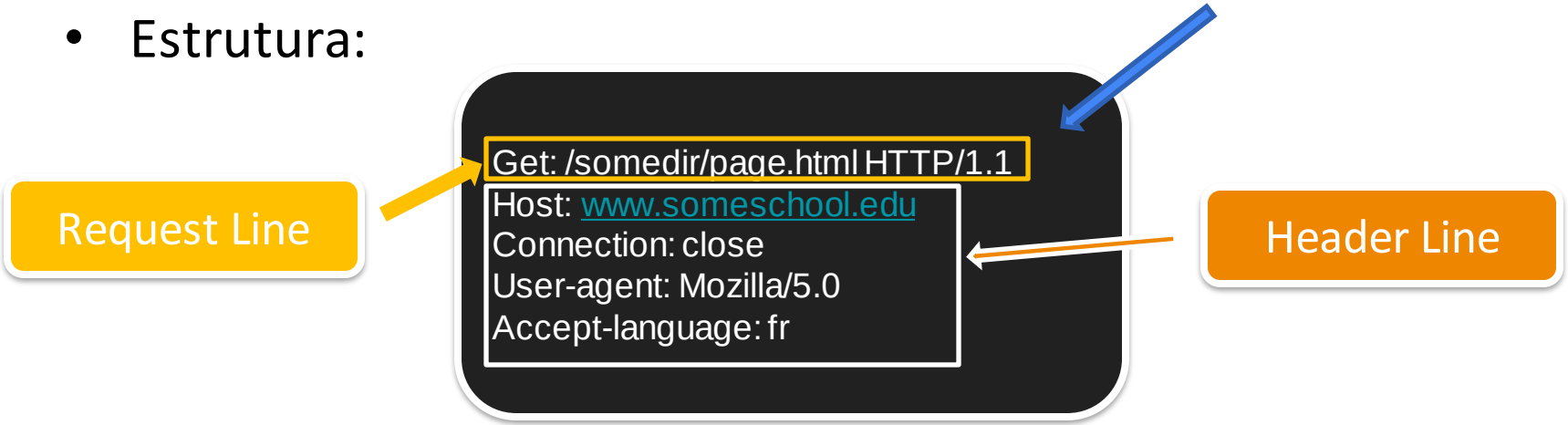
- Estrutura:



HTTP Request

Mensagens HTTP - Request

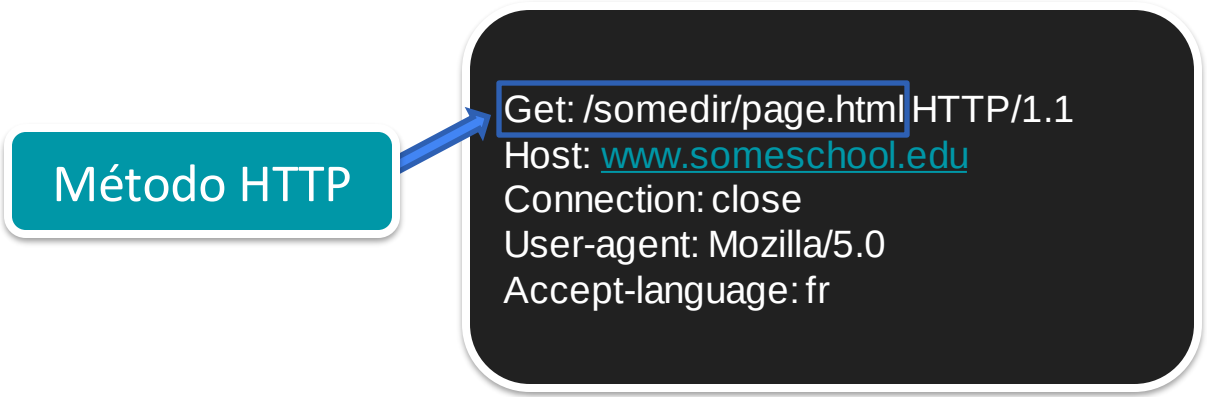
- Estrutura:



HTTP Request

Mensagens HTTP - Request

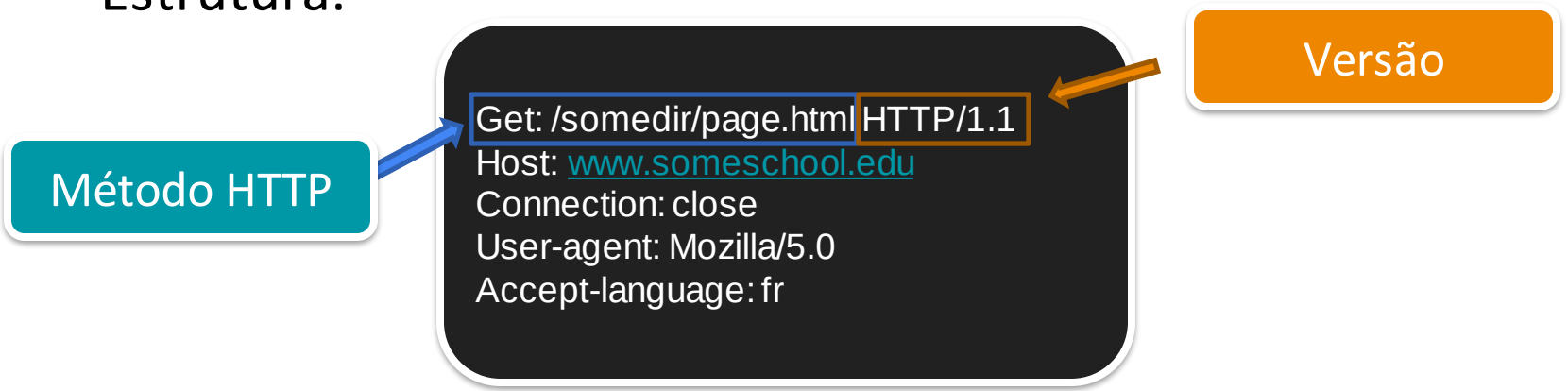
- Estrutura:



HTTP Request

Mensagens HTTP - Request

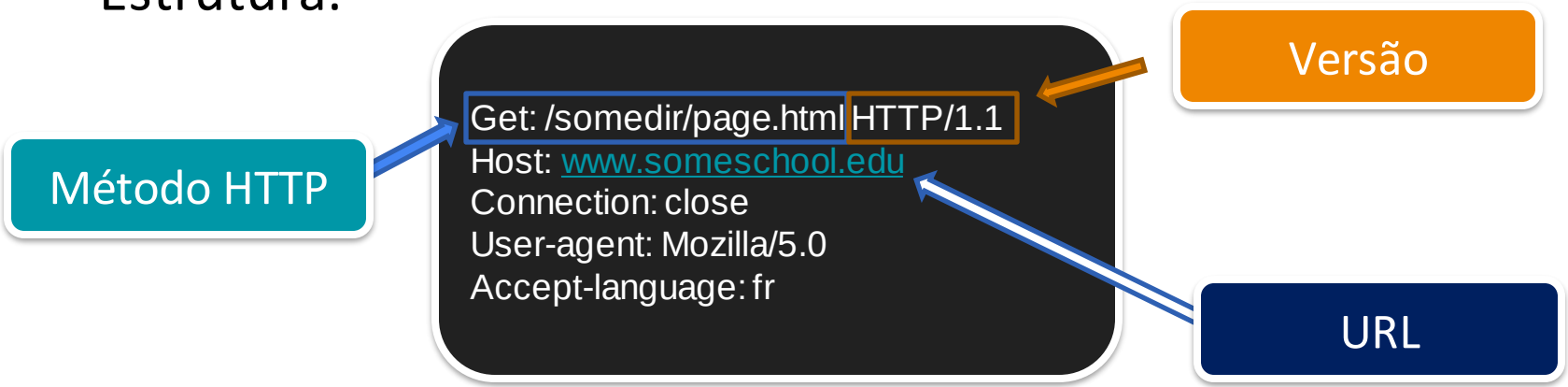
- Estrutura:



HTTP Request

Mensagens HTTP - Request

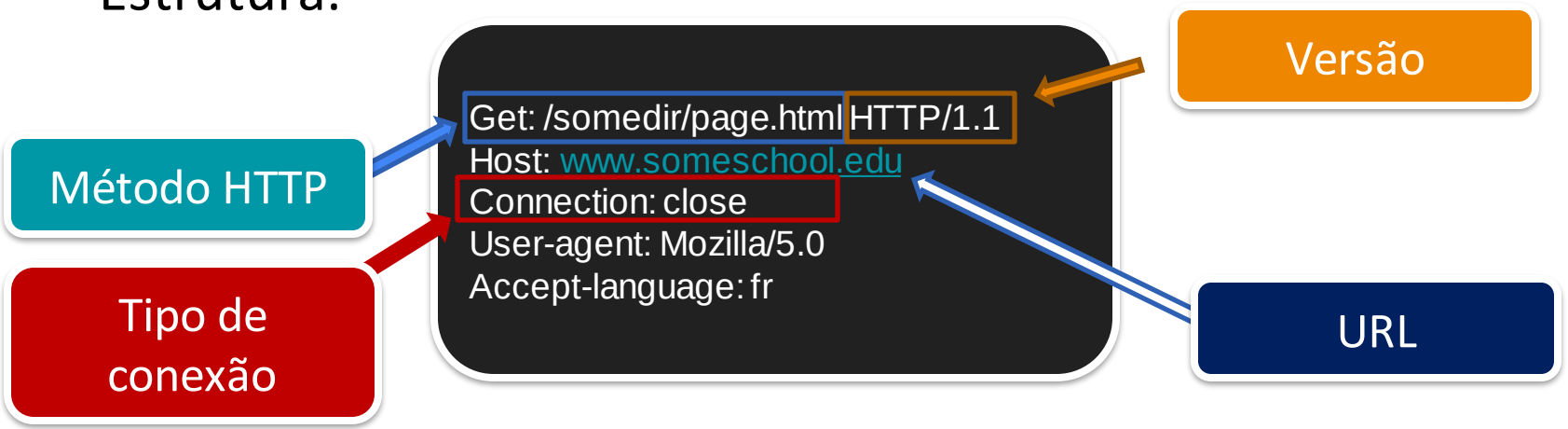
- Estrutura:



HTTP Request

Mensagens HTTP - Request

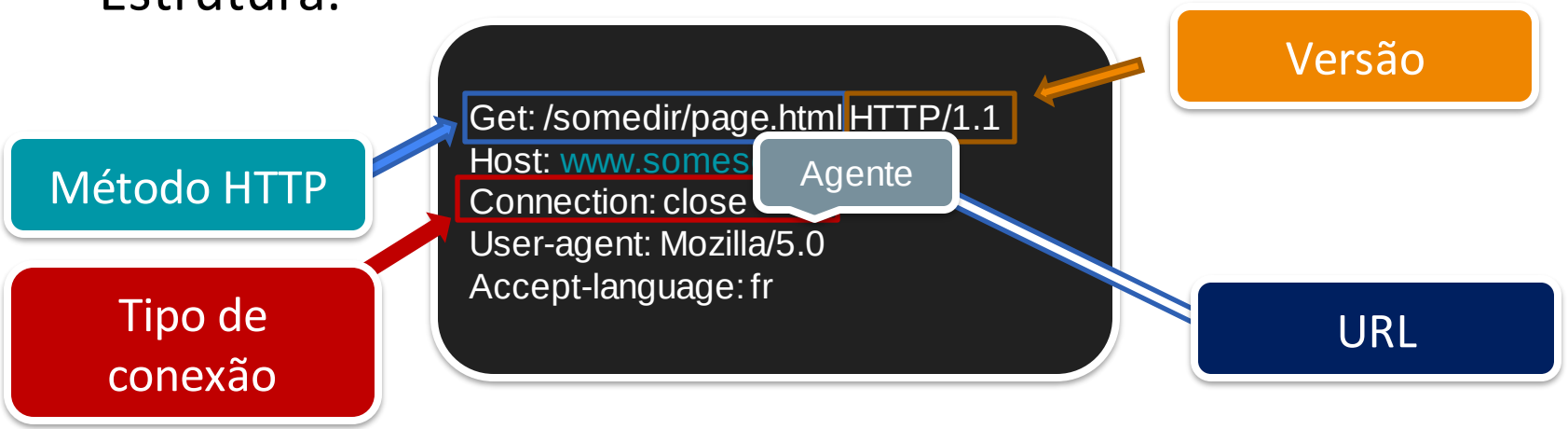
- Estrutura:



HTTP Request

Mensagens HTTP - Request

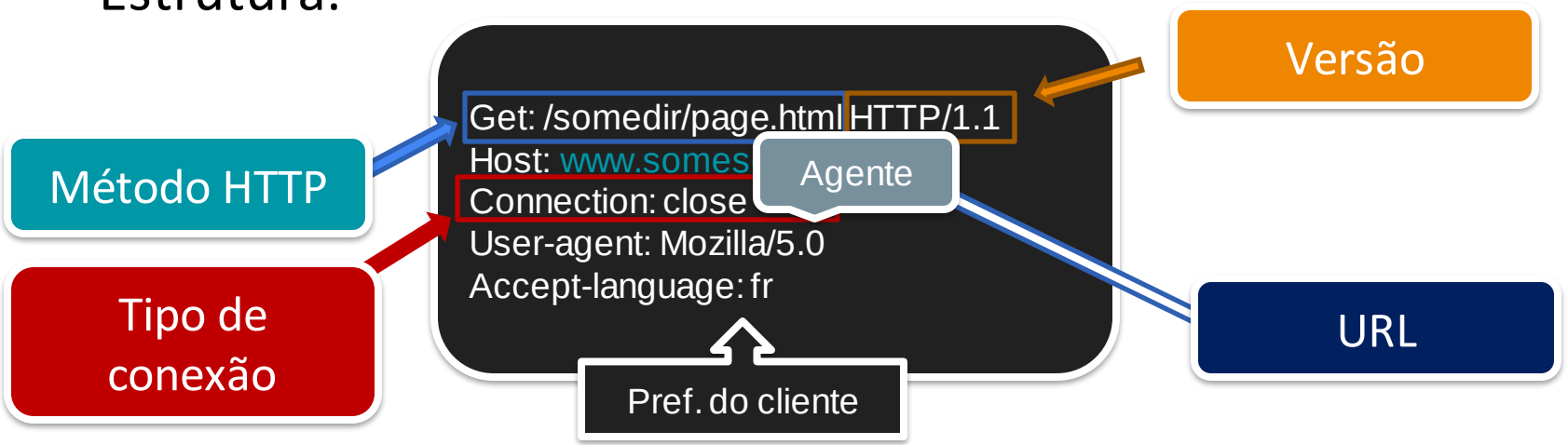
- Estrutura:



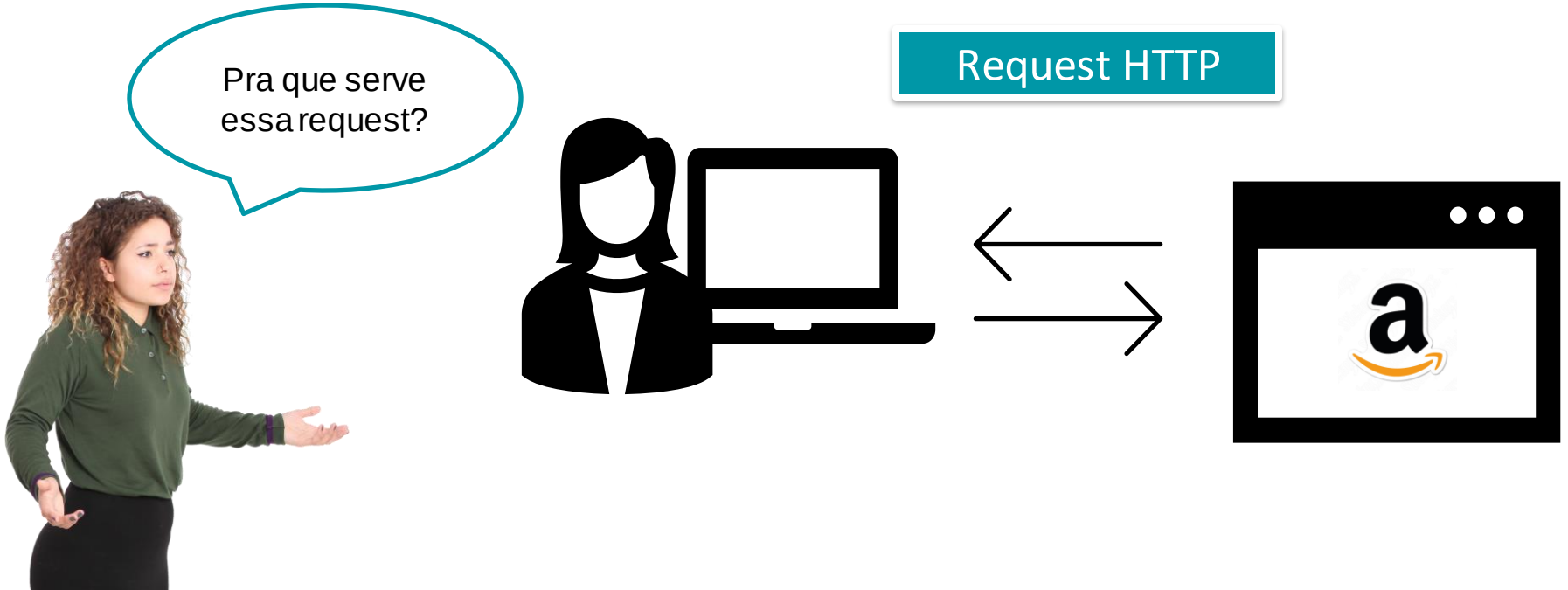
HTTP Request

Mensagens HTTP - Request

- Estrutura:



HTTP Request

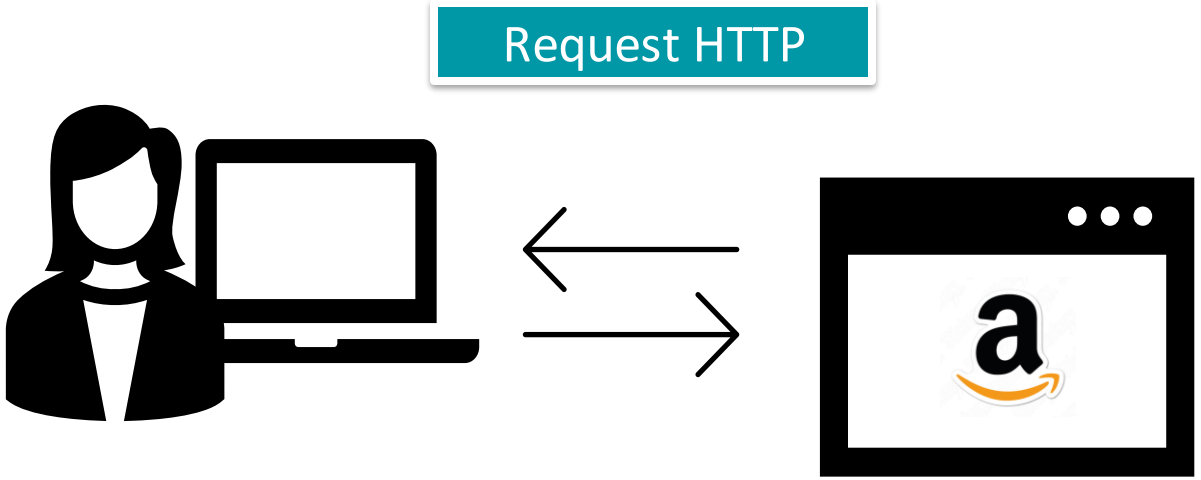


HTTP Request

Métodos mais utilizados

GET

POST



HTTP Request

```
GET: / HTTP/1.1
Host: localhost:8080
Connection: close
User-agent: Mozilla/5.0 (linux;...) Firefox 51.0
Accept: text/html, application/xhtml+xml, ..., */*; q=0.8
Accept-language: em-US, em=0.5
Accept-Enconding: gzip, deflate
```

HTTP Request

```
POST: / HTTP/1.1
Host:       localhost:8080
Connection:     close
User-agent:      Mozilla/5.0 (linux;...)  Firefox 51.0
Accept:          text/html, application/xhtml+xml, ..., */*; q=0.8
Accept-language:     em-US, em=0.5
Accept-Enconding:   gzip, deflate
Contente-Type:       multipart/form-data; boundary=12656974
Content-Length:     345

(more data)
```

GET	solicita um recurso do servidor
HEAD	GET sem corpo de resposta
POST	submete uma entidade a um recurso
PUT	substituição de recursos pelos dados da requisição
DELETE	remoção de um recurso
TRACE	chamada de loop-back a um determinado recurso
OPTION	opções de comunicação com recurso
CONNECT	tunelamento identificado pelo recurso
PATCH	modificação parcial

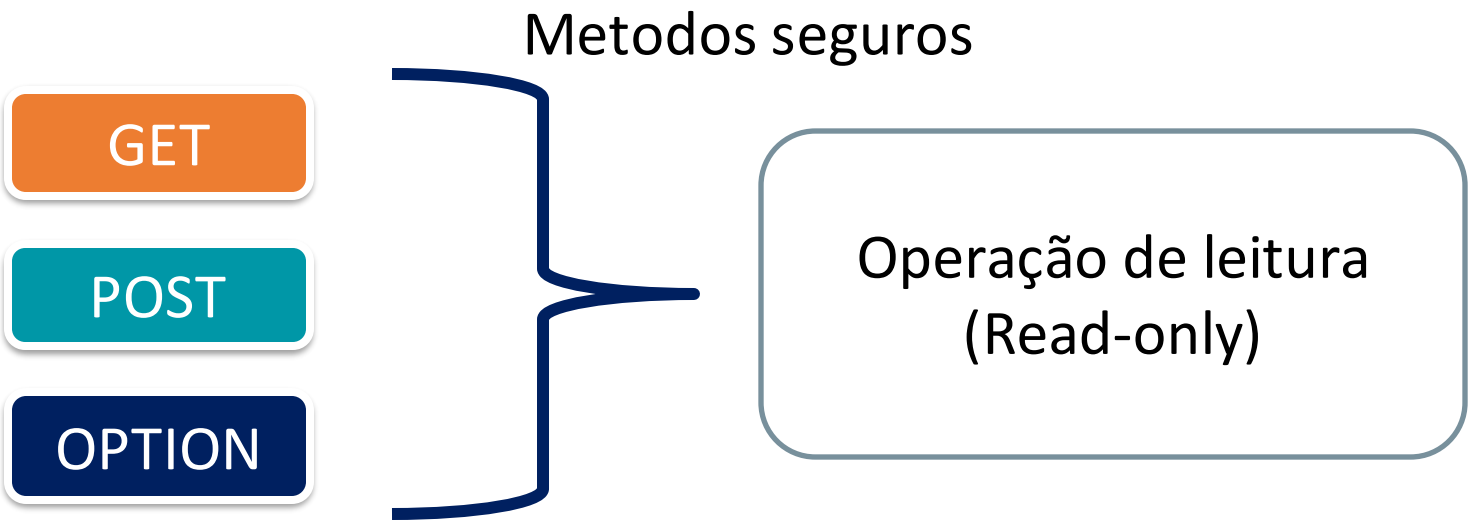
GET	solicita um recurso do servidor
HEAD	GET sem corpo de resposta
POST	submete uma entidade a um recurso
PUT	substituição de recursos pelos dados da requisição
DELETE	remoção de um recurso
TR	da de loop-back a um determinado recurso
OP	de comunicação com recurso
CONN	tunelamento identificado pelo recurso
PATCH	modificação parcial

RFC 5789

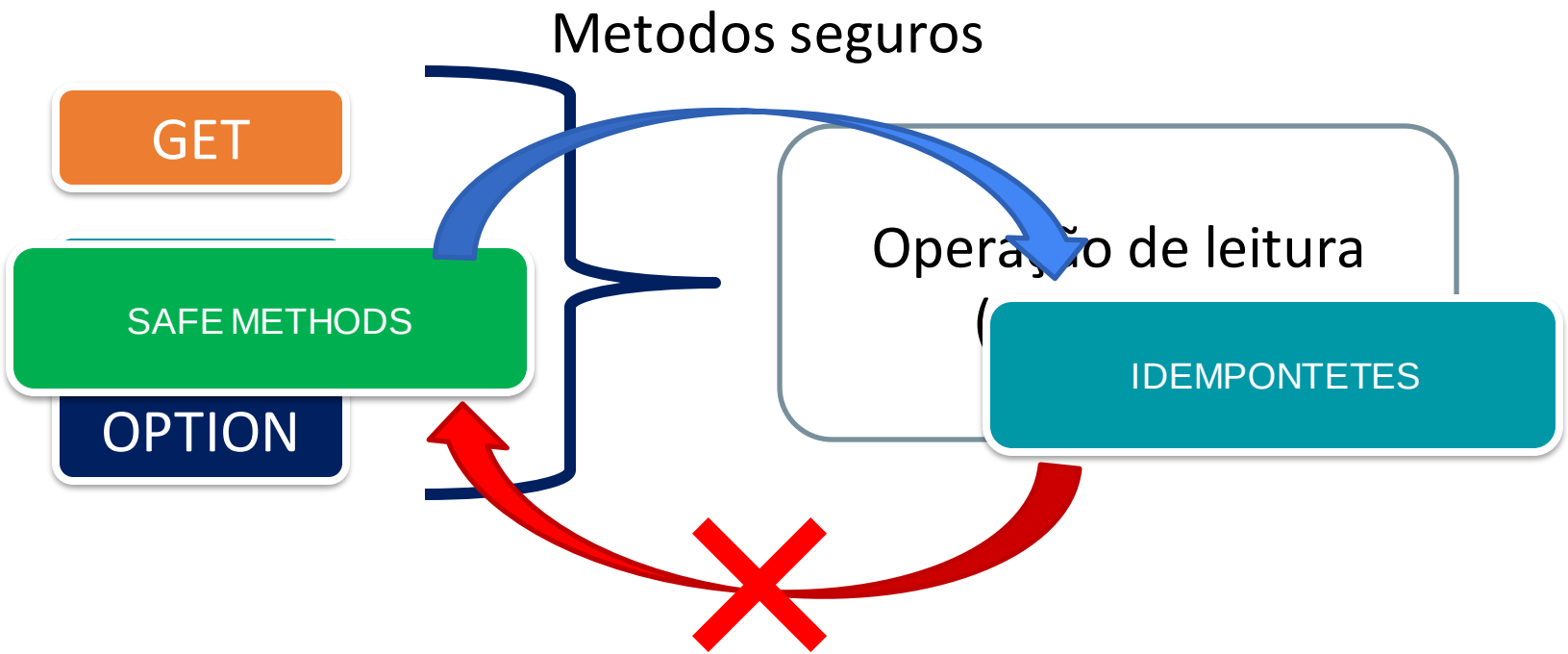
Métodos seguros

GET	solicita um recurso do servidor
HEAD	GET sem corpo de resposta
POST	submete uma entidade a um recurso
PUT	substituição de recursos pelos dados da requisição
DELETE	remoção de um recurso
TRACE	chamada de loop-back a um determinado recurso
OPTION	opções de comunicação com recurso
CONNECT	tunelamento identificado pelo recurso
PATCH	modificação parcial

HTTP Request



HTTP Request



HTTP Request

S/ MUNDAÇAS - CLIENTE

Metodos seguros

GET

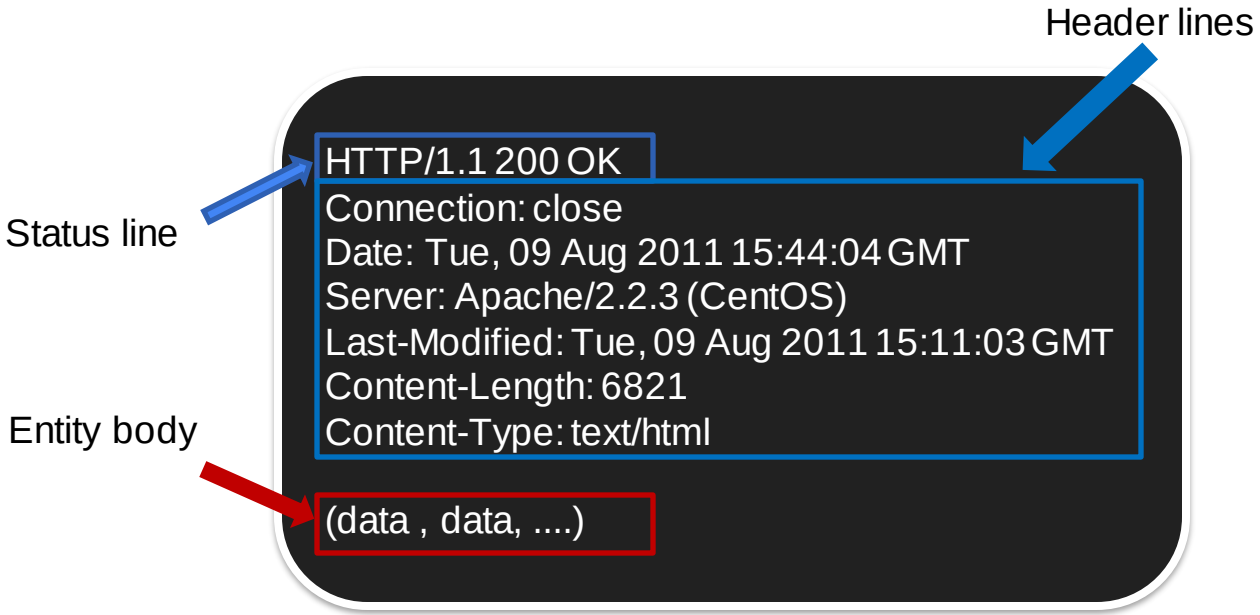
POST

OPTION

Operação de leitura
(Read-only)

Mensagens HTTP: Response

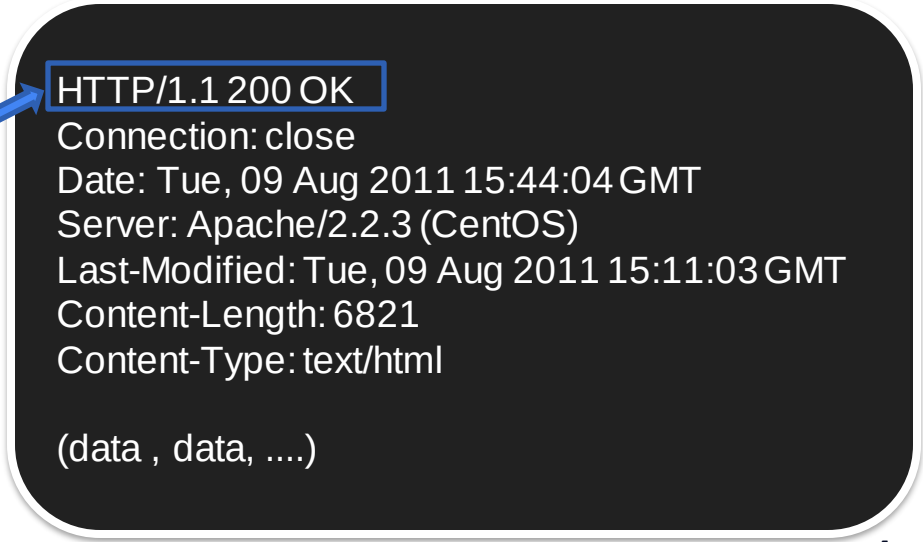
HTTP Response



HTTP Response

Mensagens HTTP - Response

- Versão do protocolo
- Status code
- Status da mensagem



HTTP Response

200 OK:	request bem sucedida e objeto enviado
301 Moved Permanently:	objeto realocado nova URL no campo Location
400 Bad Request:	resposta genérica - servidor não entendeu a mensagem
404 Not Found:	o documento solicitado é inexistente
505 HTTP Version Not Supported:	versão do protocolo não suportada pelo servidor

HTTP Response

Classificação dos Status Code

- Information response (100 – 199)
- Successful response (200 – 299)
- Redirection response (300 – 399)
- Client error response (400 – 499)
- Server error response (500 – 599)

```
HTTP/2 200
```

```
HTTP/1.1 301 Moved Permanently
```

HTTP Response

7231

RFC 2616 – Sec 10

Classificação dos Status Code

- Information response (100 – 199)
- Successful response (200 – 299)
- Redirection response (300 – 399)
- Client error response (400 – 499)
- Server error response (500 – 599)

```
HTTP/2 200
```

```
HTTP/1.1 301 Moved Permanently
```

HTTP Response

Web Distributed Authoring and Versioning

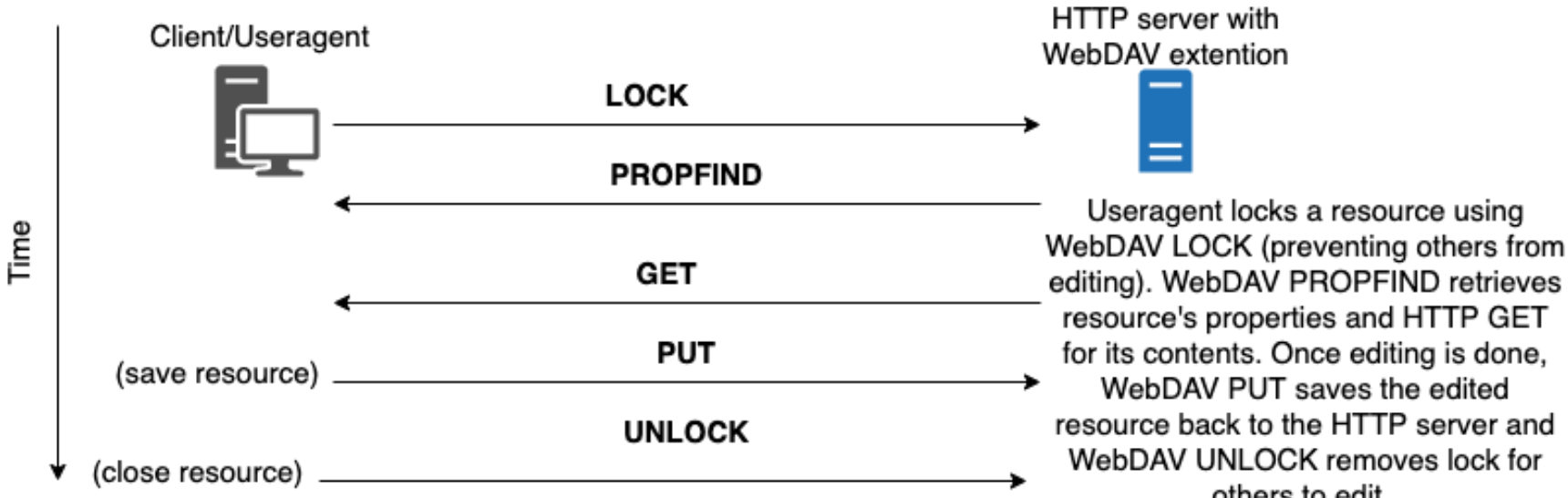
WebDav

- 102 – Processing
- 207 – Multi-status
- 208 – Already Reported
- 422 – Unprocessable Entity
- 423 – Locked
- 424 – Failed Dependency

CalDav

CardDav

HTTP Response



Fonte: Wikipedia

HTTP Response

Web Distributed Authoring and Versioning

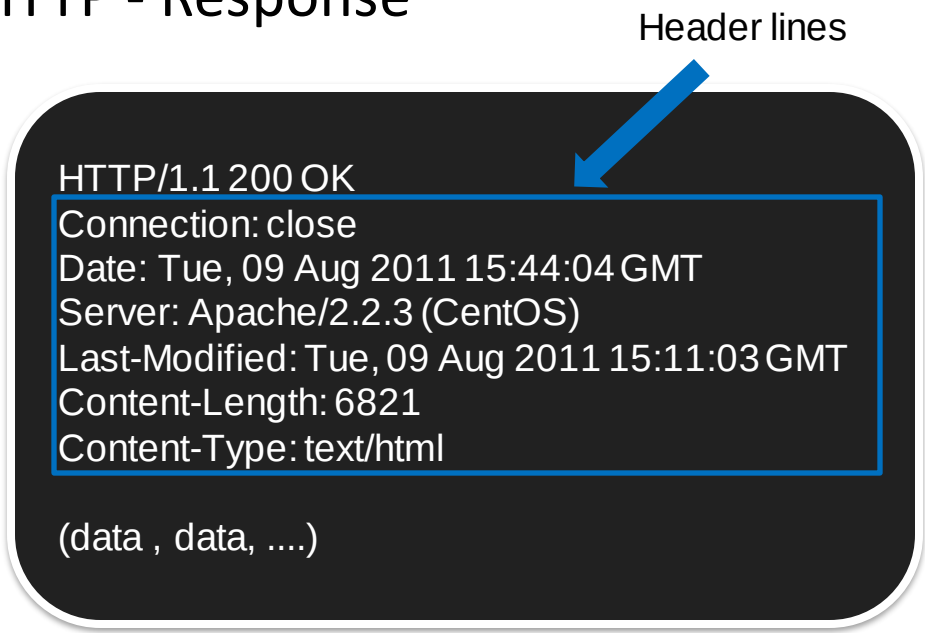
WebDav

- Webpage Meta: Add, deletar, retrieve
- Link pages
- Criação de conj. de documentos
- Copy e move
- Lock: documento editado - 1

HTTP Response

Mensagens HTTP - Response

- Conexão encerrada
- Dados da mensagem:
  Data, servidor, …
- Content-type:
  Tipo de dado

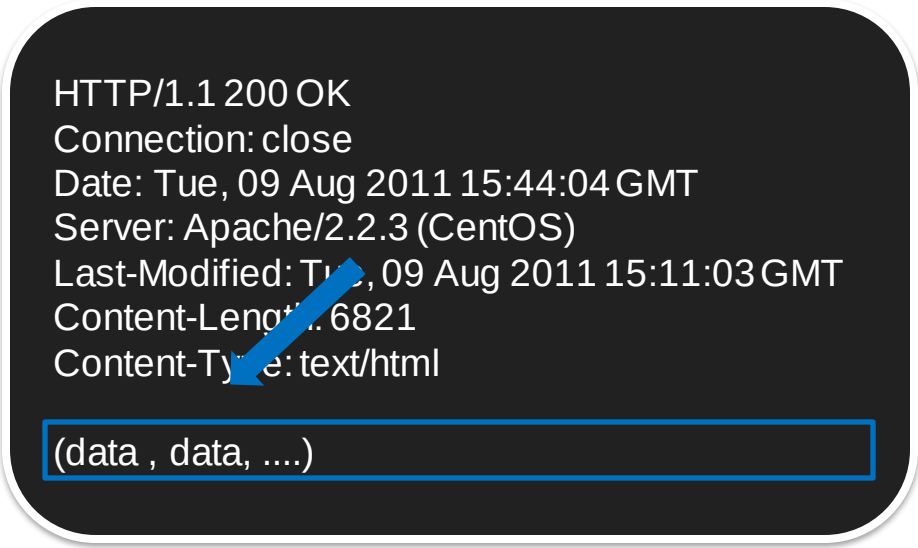


HTTP Response

- Request Method & Status Code

Campos
- Entity Header
- Entity Body

ENTITY



HTTP Response

Entity

- Request Method & Status Code

Campos

- Entity Header
- Entity Body



Definições

```
entity-header  = Allow
               | Content-Encoding
               | Content-Language
               | Content-Length
               | Content-Location
               | Content-MD5
               | Content-Range
               | Content-Type
               | Expires
               | Last-Modified
               | extension-header

extension-header = message-header
```

HTTP Response

Entity

- Request Method & Status Code

Campos

- Entity Header
- Entity Body



Campo: Accept

```
entity-header  = Allow
               | Content-Encoding
               | Content-Language
               | Content-Length
               | Content-Location
               | Content-MD5
               | Content-Range
               | Content-Type
               | Expires
               | Last-Modified
               | extension-header

extension-header = message-header
```

HTTP Response

Entity
- Request Method & Status Code

Campos
- Entity Header
- Entity Body

Campo: Accept

```
Accept          = "Accept" ":"
                  #( media-range [ accept-params ] )

media-range     = ( "*/*"
                  | ( type "/" "*" )
                  | ( type "/" subtype )
                  ) *( ";" parameter )
accept-params   = ";" "q" "=" qvalue *( accept-extension )
accept-extension = ";" token [ "=" ( token | quoted-string ) ]
```

Definições

```
Accept-Language: da, en-gb;q=0.8, en;q=0.7
```

HTTP Response

```
Accept: audio/*; q=0.2, audio/basic
```

```
Accept-Encoding: compress, gzip
Accept-Encoding:
Accept-Encoding: *
Accept-Encoding: compress;q=0.5, gzip;q=1.0
Accept-Encoding: gzip;q=1.0, identity; q=0.5, *;q=0
```

```
Accept-Charset: iso-8859-5, unicode-1-1;q=0.8
```

```
Accept: text/*;q=0.3, text/html;q=0.7, text/html;level=1,
        text/html;level=2;q=0.4, */*;q=0.5
```

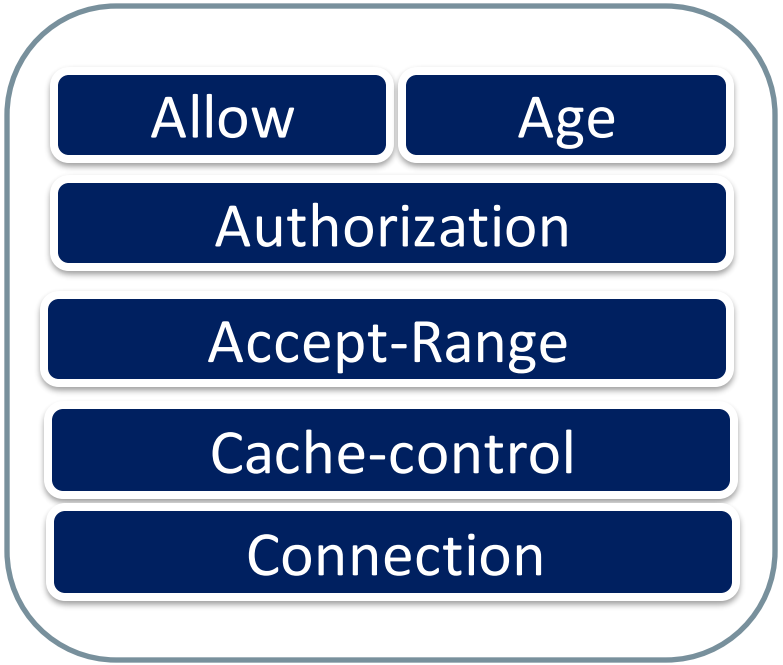
```
Accept: text/plain; q=0.5, text/html,
        text/x-dvi; q=0.8, text/x-c
```

HTTP Response

Entity
- Request Method & Status Code

Campos
- Entity Header
- Entity Body





HTTP Response

Entity

- Request Method & Status Code

Campos

- Entity Header
- Entity Body

Definições

Campo: Content

```
entity-header
                | Content-Encoding
                | Content-Language
                | Content-Length
                | Content-Location
                | Content-MD5
                | Content-Range
                | Content-Type
                | Expires
                | Last-Modified
                | extension-header

extension-header = message-header
```

Etapa 3

Para que serve Cookie e Cache?

// Conhecendo os principais protocolos de comunicação da internet /Protocolo HTTP

Percurso

~~Etapa 1~~ ~~Como funciona o protocolo HTTP?~~

~~Etapa 2~~ ~~Mensagens HTTP – Request & Response~~

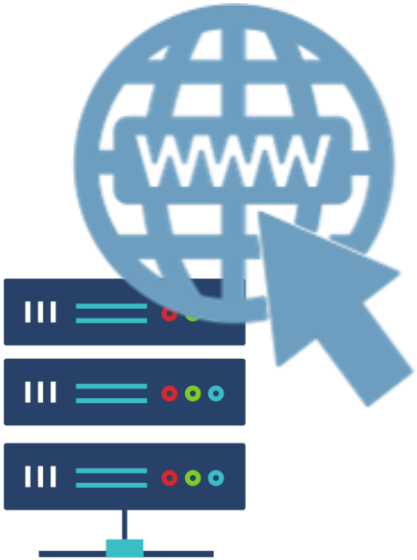
Etapa 3 **Para que servem Cookies e Cache?**

Percurso

Etapa 4 HTTP 2.0 - Atualizações do protocolo

Etapa 5 Servidores / Sistemas de aplicação

Protocolo HTTP

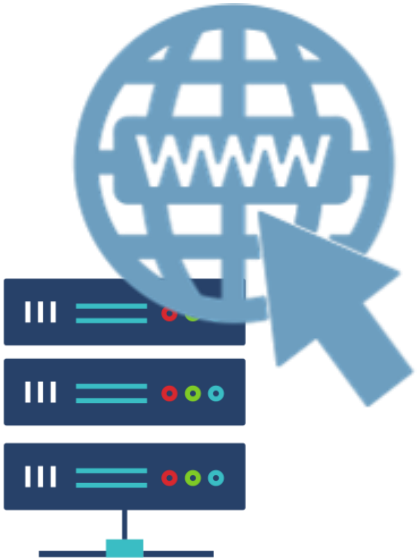


HTTP PROTOCOL

Stateless

Client/Server

Protocolo HTTP



HTTP PROTOCOL

SE PRECISARMOS IDENTIFICAR O CLIENTE?

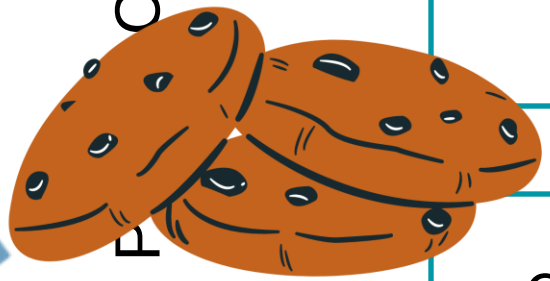
Stateless

Client/Server

EX: Publicidade, formulários, melhor experiência do usuário ...

Cookies





HTTP PROTOCOL

SE PRECISARMOS IDENTIFICAR O CLIENTE?

Stateless

Client/Server

EX: Publicidade, formulários, melhor experiência do usuário ...

O que são Cookies?



TRACK

Pequenos pedaços ou blocos de dados criados e utilizados pelo servidor para persistir dados no dispositivo do cliente.



Especificação: RFC 6265

O que são Cookies?



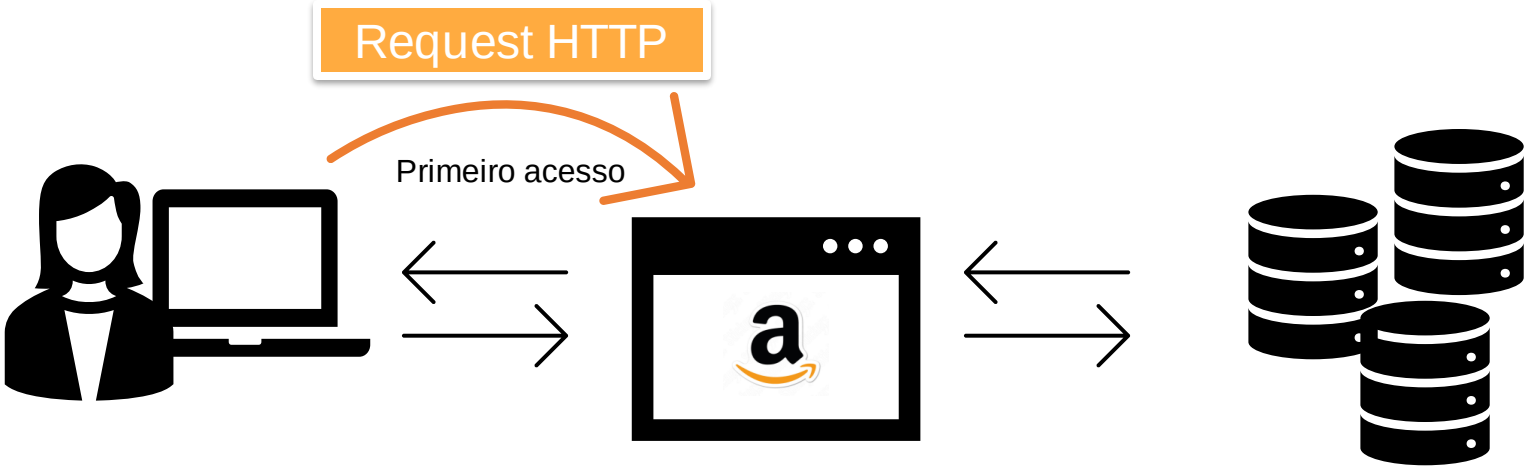




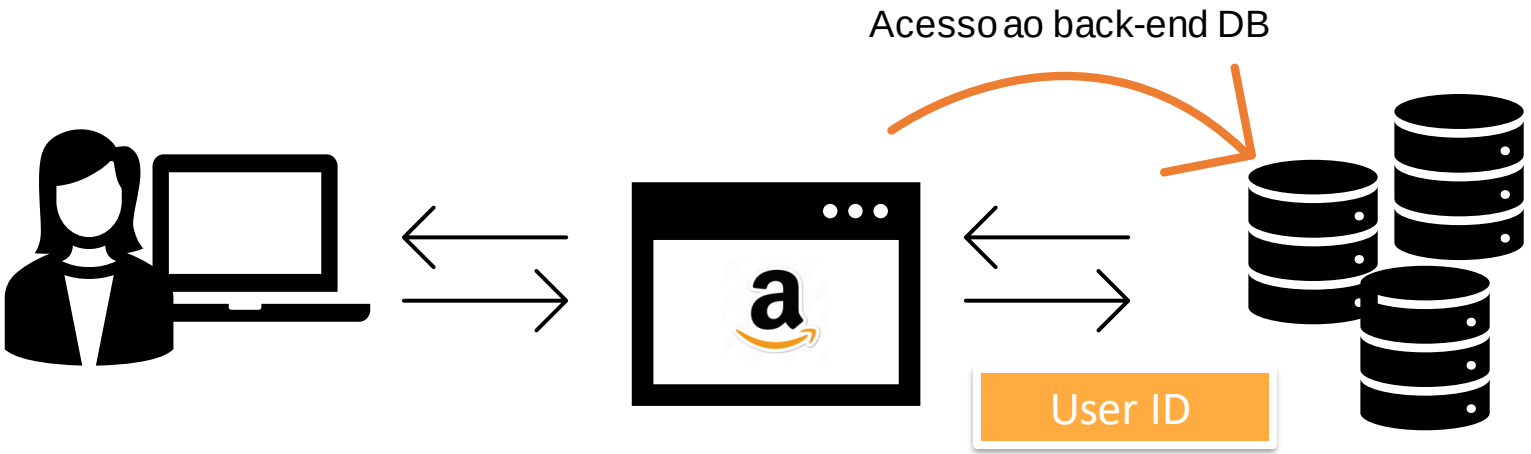
O que são Cookies?



O que são Cookies?



O que são Cookies?



O que são Cookies?



O que são Cookies?

File cookie

Add ...
ID user
hostname

Histórico



Set-Cookie: NAME=VALUE; expires=DATE; path=PATH; domain=DOMAIN_NAME.

O que são Cookies?

File cookie

Add ...
ID user
hostname

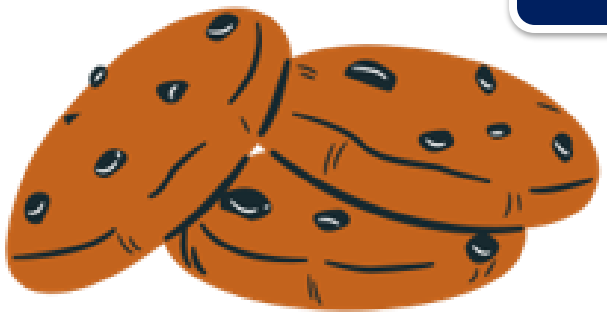
Histórico



Internet Explorer: "C:\Users\username\AppData\Roaming\Microsoft\Windows\Cookies"

O que são Cookies?

Cookies de sessão



Cookies de persistentes

Site: Rediff.com Cookie name: RMID

Name: RMID (Name of the cookie)

Content: 1d11c8ec44bf49e0... (Encrypted content)

Domain: .rediff.com

Path: / (Any path after the domain name)

Send For: Any type of connection

Expires: Thursday, December 31, 2020, 11:59:59 PM

O que são Cookies?

Manter logins ativos

Informação de website



Site: Rediff.com Cookie name: RMID

Name: RMID (Name of the cookie)

Content: 1d11c8ec44bf49e0... (Encrypted content)

Domain: .rediff.com

Path: / (Any path after the domain name)

Send For: Any type of connection

Expires: Thursday, December 31, 2020, 11:59:59 PM

Carrinho de ecommerce

O que são Cookies?

Hacking



LGPD & Cookies

Site: Rediff.com Cookie name: RMID

Name: RMID (Name of the cookie)

Content: 1d11c8ec44bf49e0... (Encrypted content)

Domain: .rediff.com

Path: / (Any path after the domain name)

Send For: Any type of connection

Expires: Thursday, December 31, 2020, 11:59:59 PM

Cookies e Privacidade

HTTP - Caching

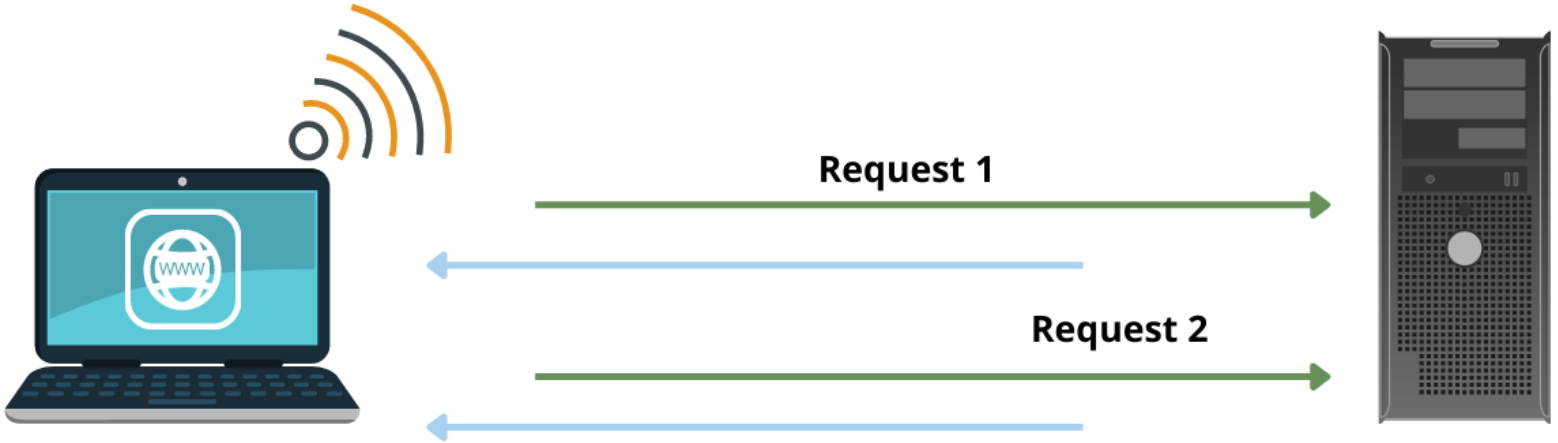
HTTP 1.1

caching

Foco: css/js e gzip

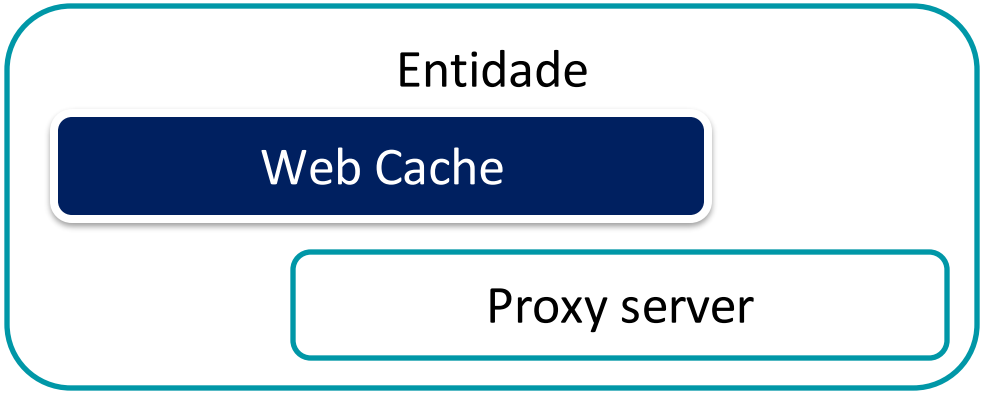
Repetição de dados

HOL – Head of Line Blocking

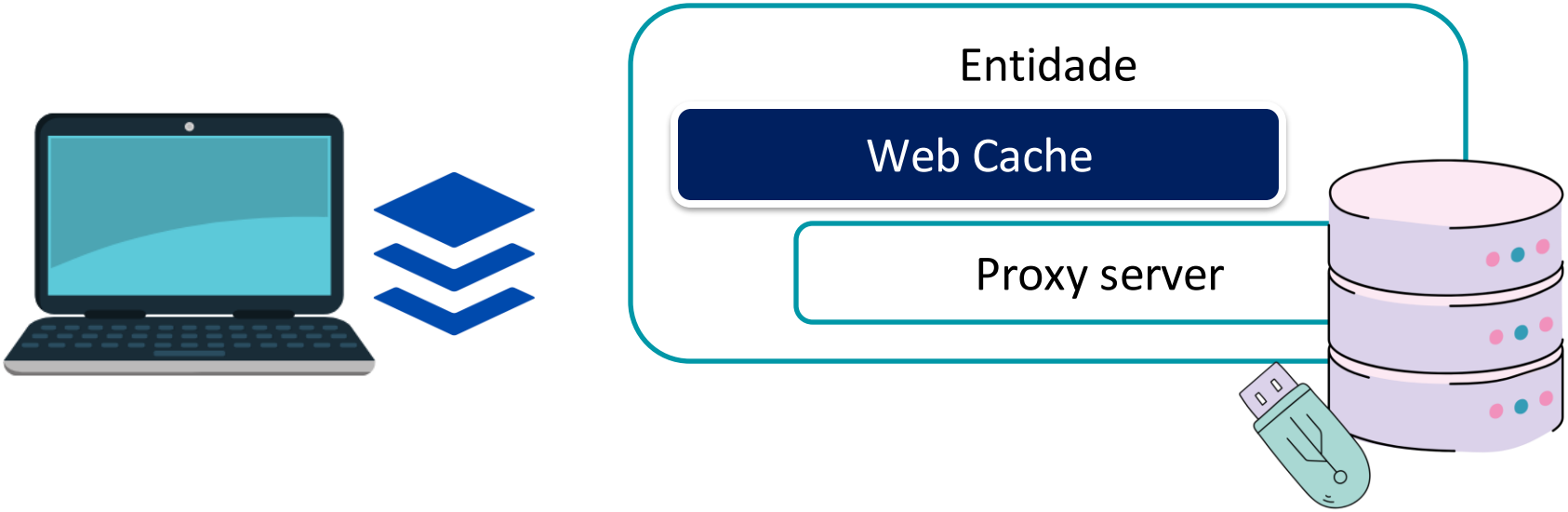


Web Cache?



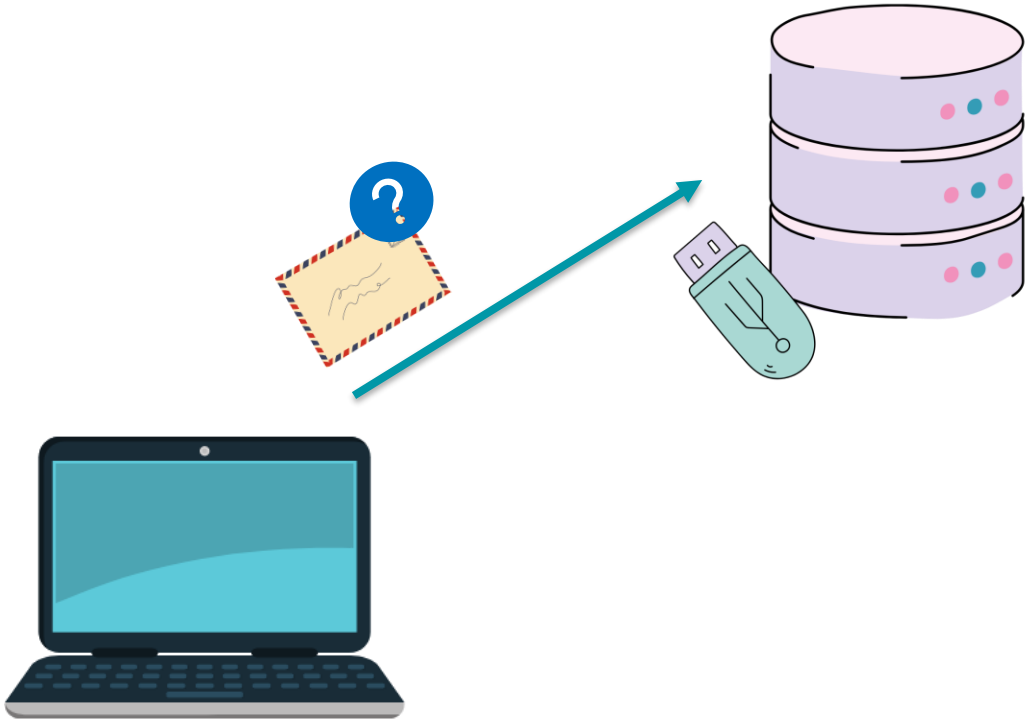


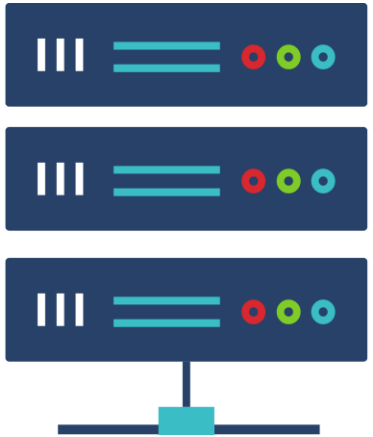
Web Cache?



Web Cache

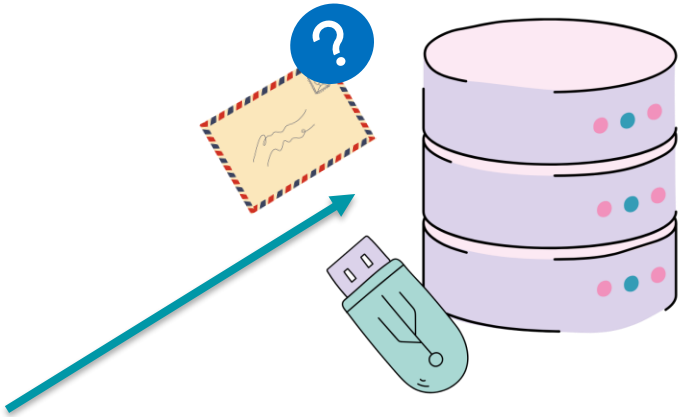
Proxy server



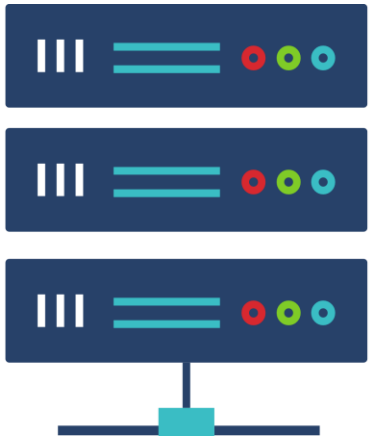


Web Cache:

Proxy server

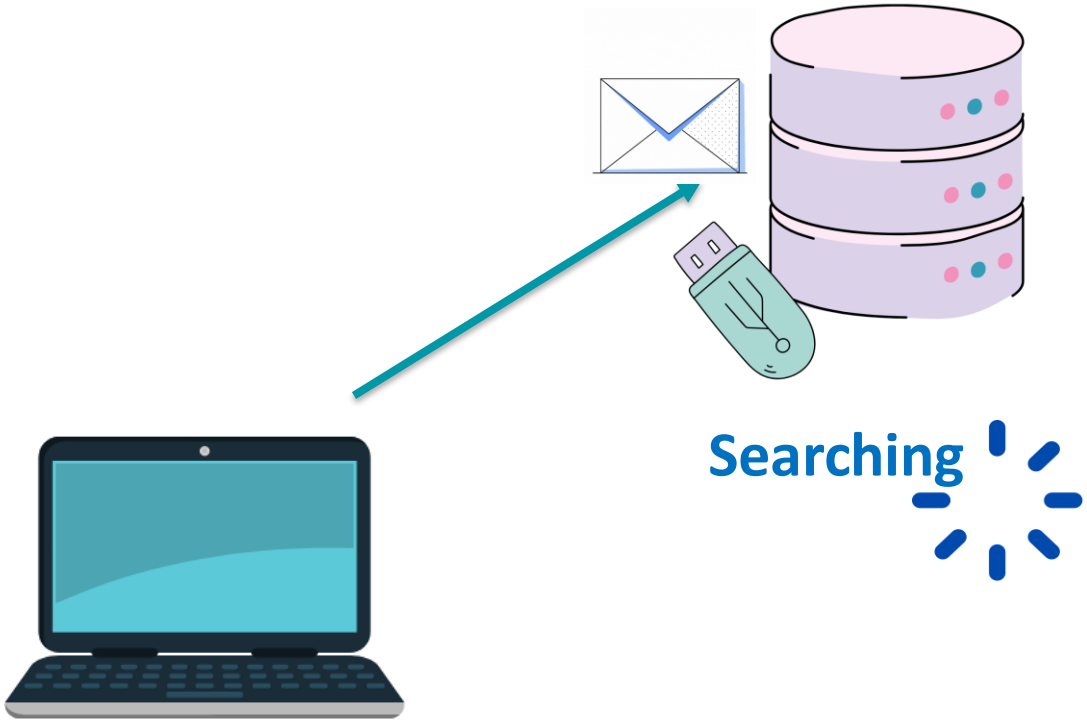






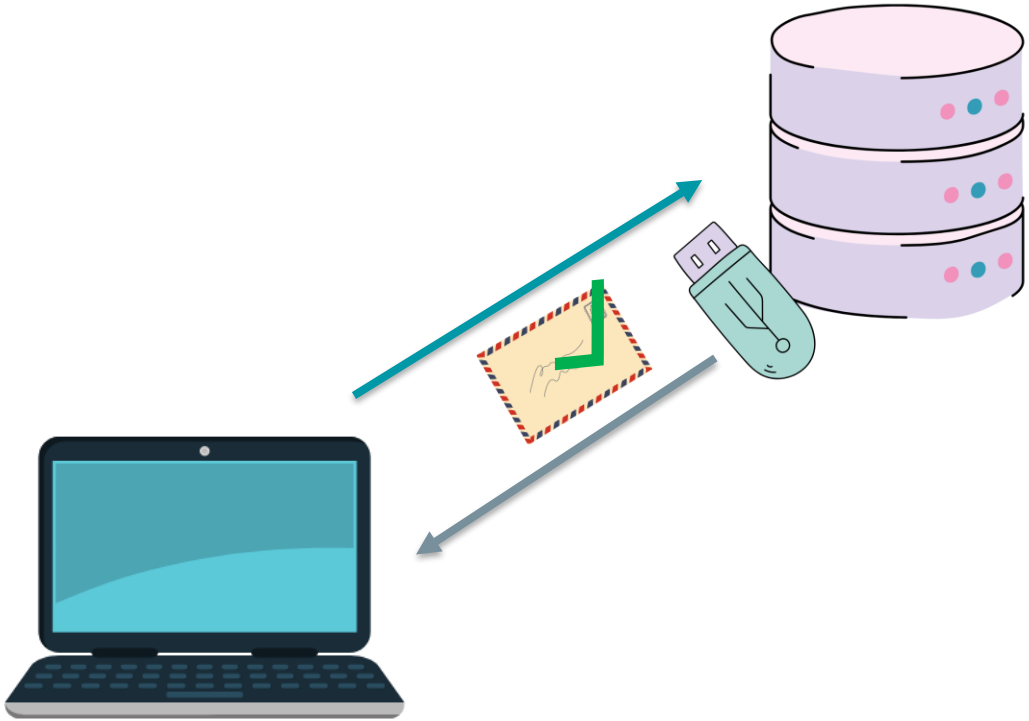
Web Cache

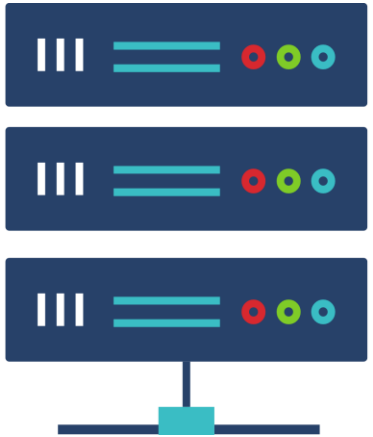
Proxy server



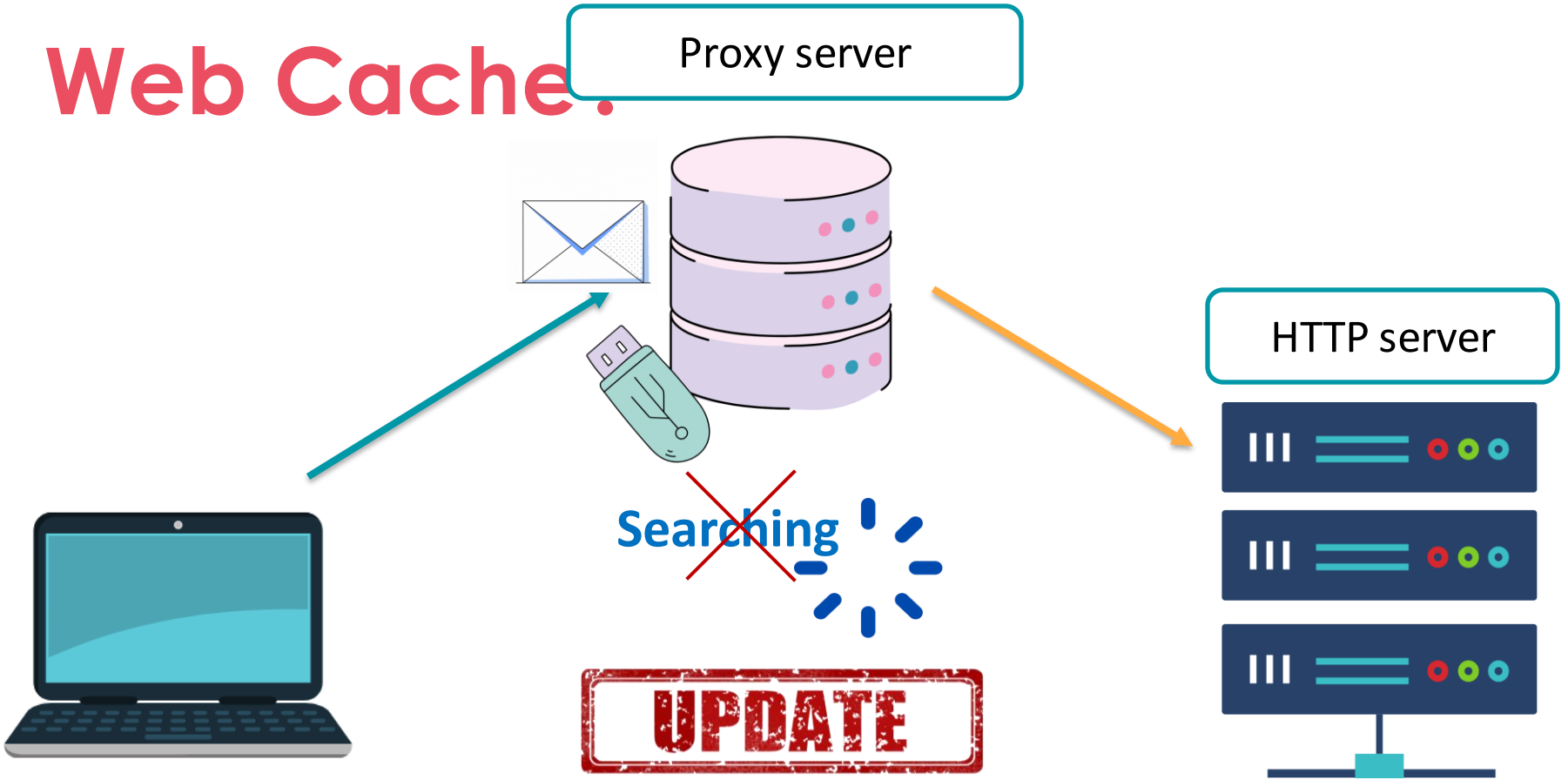
Web Cache

Proxy server

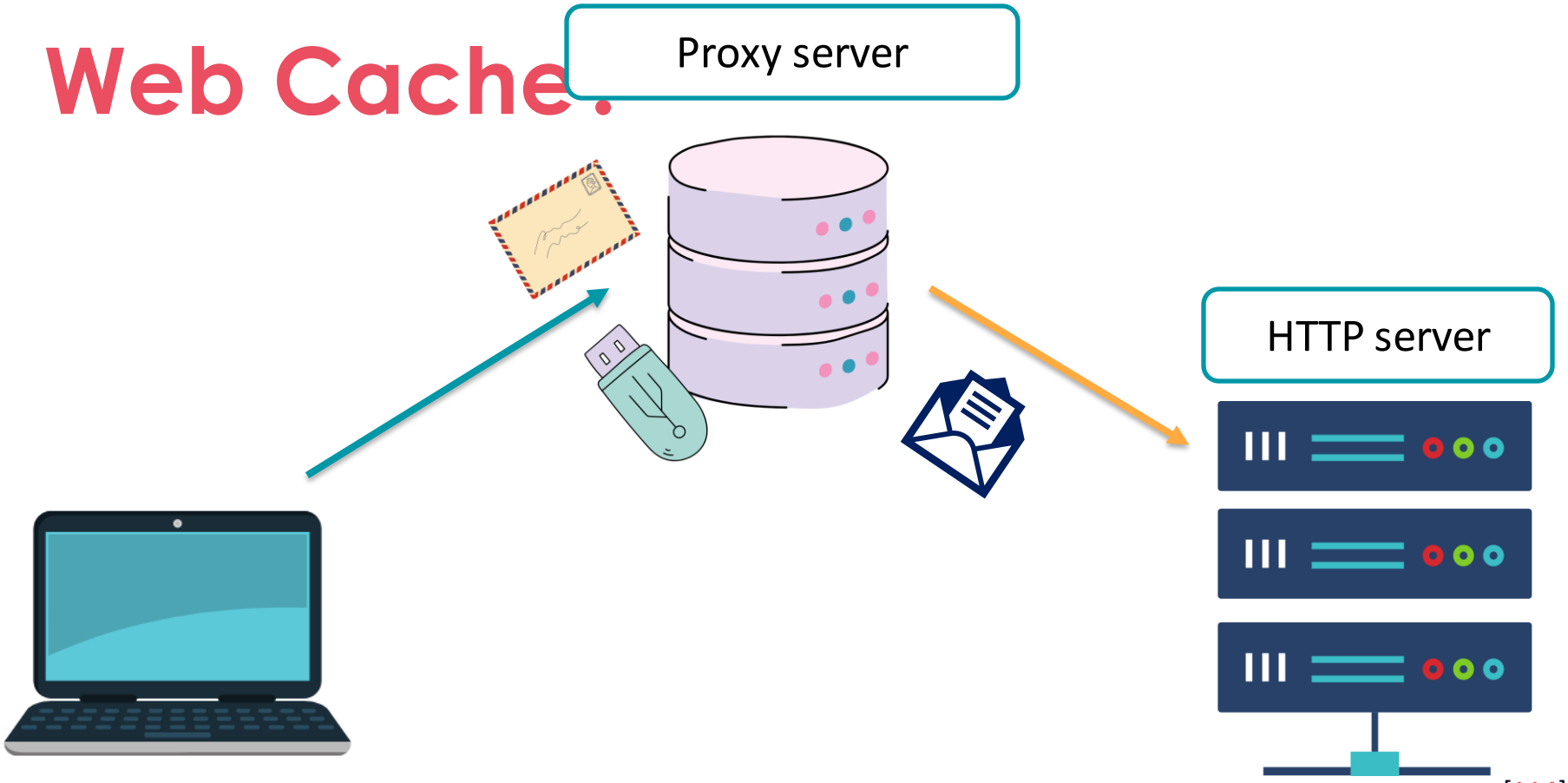




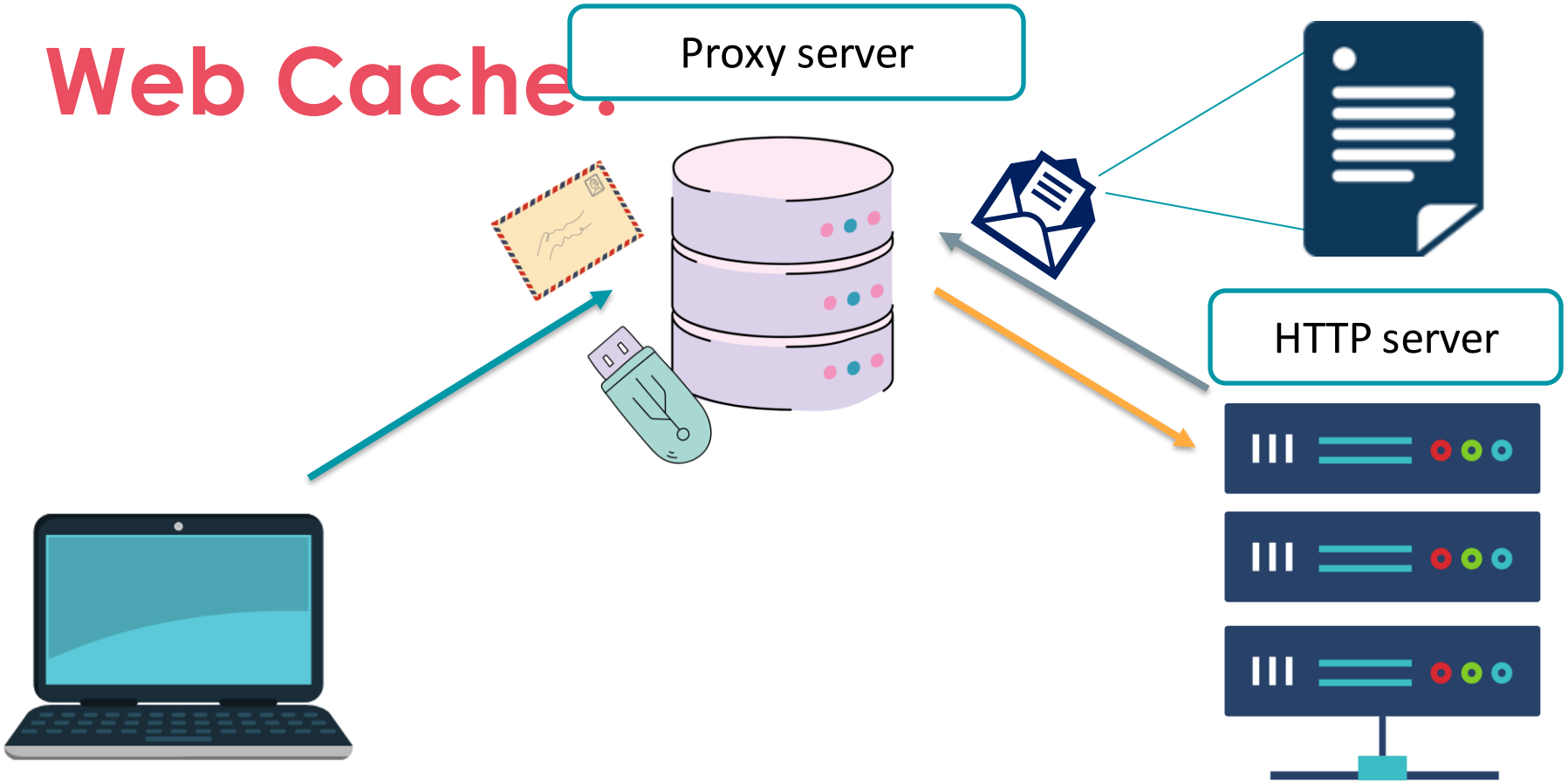
Web Cache



Web Cache

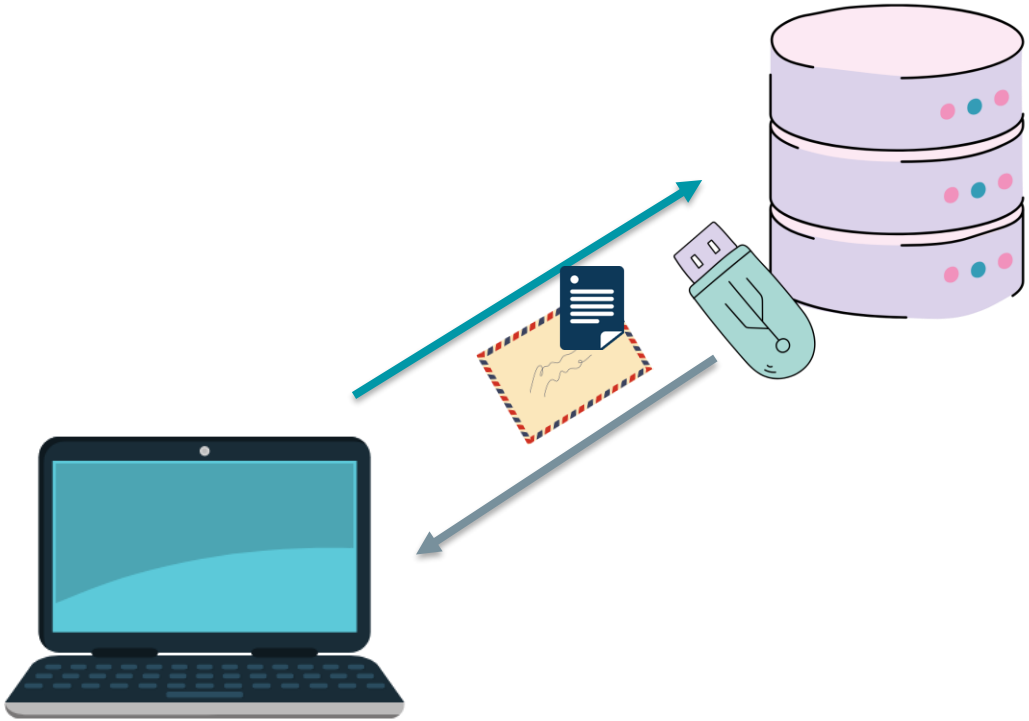


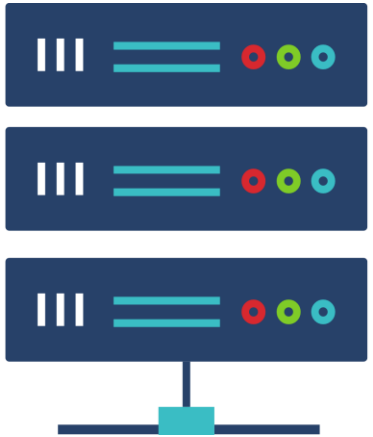
Web Cache



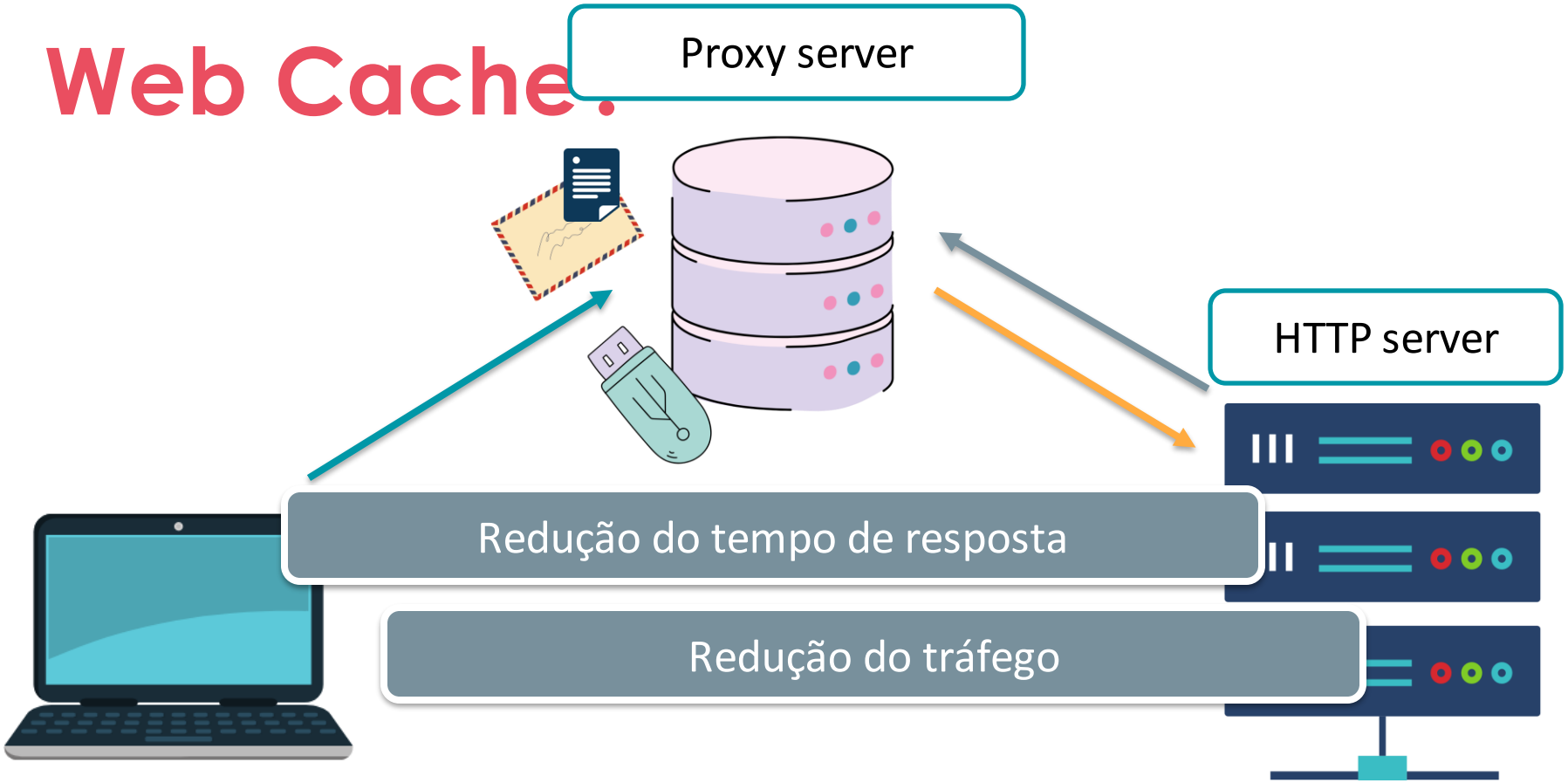
Web Cache:

Proxy server

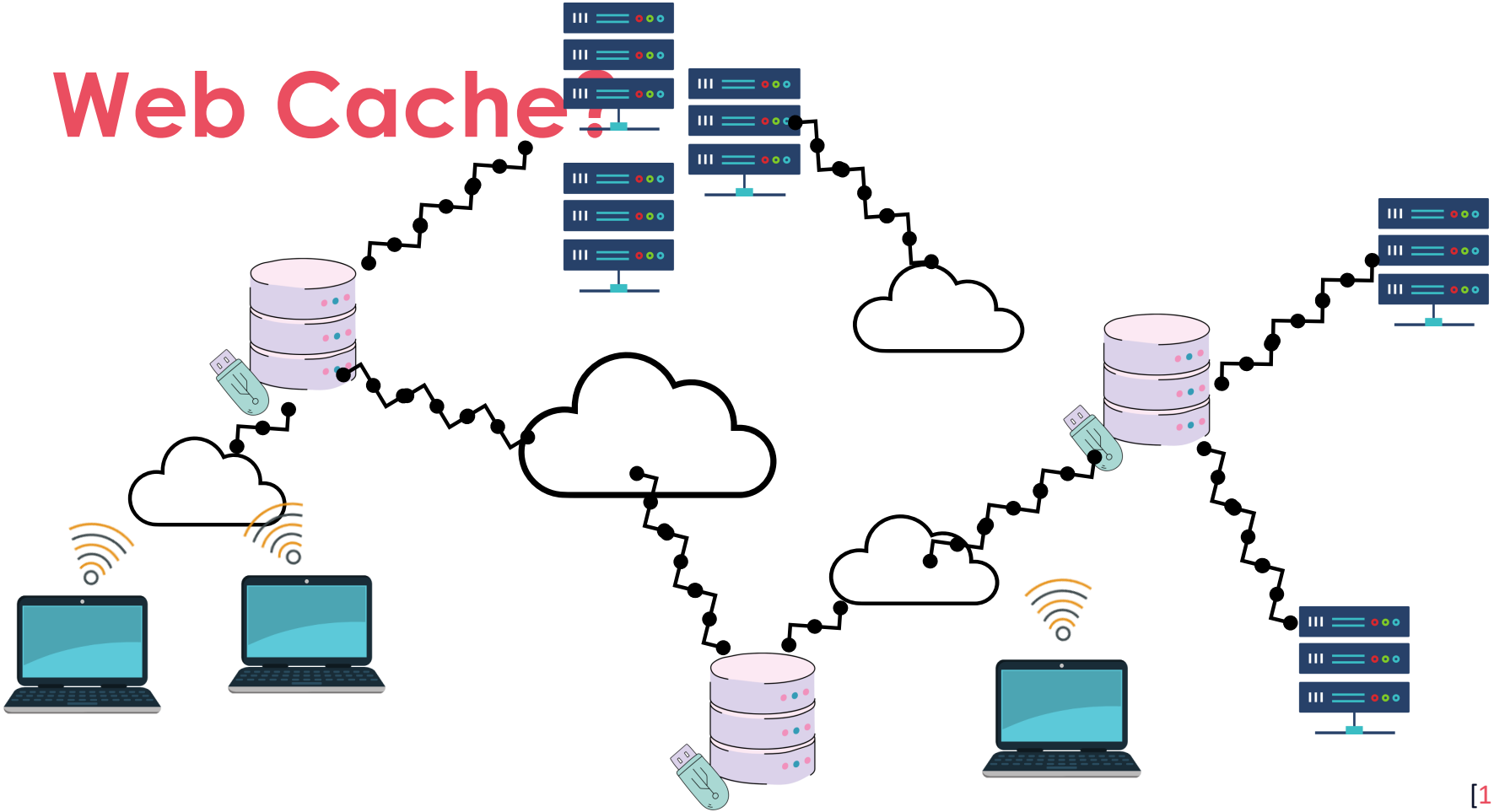




Web Cache


Proxy server
HTTP server
Redução do tempo de resposta
Redução do tráfego

Web Cache?



Etapa 4

HTTP 2.0 - Atualizações do protocolo

//Conhecendo os principais protocolos de comunicação da internet / Protocolo HTTP

Percurso

~~Etapa 1~~ ~~Como funciona o protocolo HTTP?~~

~~Etapa 2~~ ~~Mensagens HTTP – Request & Response~~

~~Etapa 3~~ ~~Para que servem Cookies e Cache?~~

Percurso

Etapa 4 **HTTP 2.0 - Atualizações do protocolo**

Etapa 5 Servidores / Sistemas de aplicação

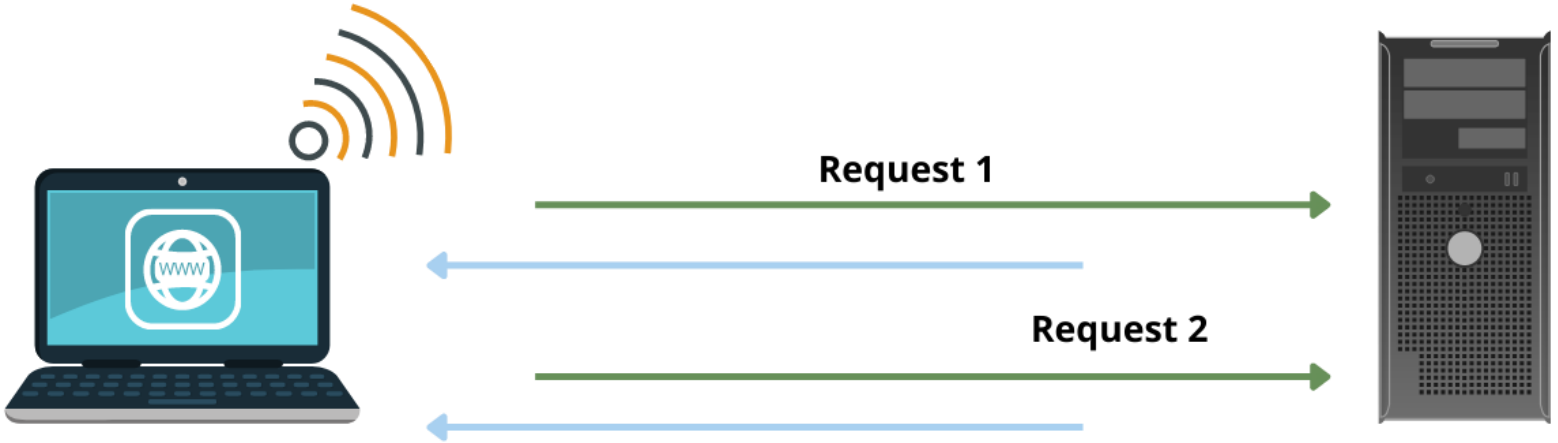
HTTP 1.1

caching

Foco: css/js e gzip

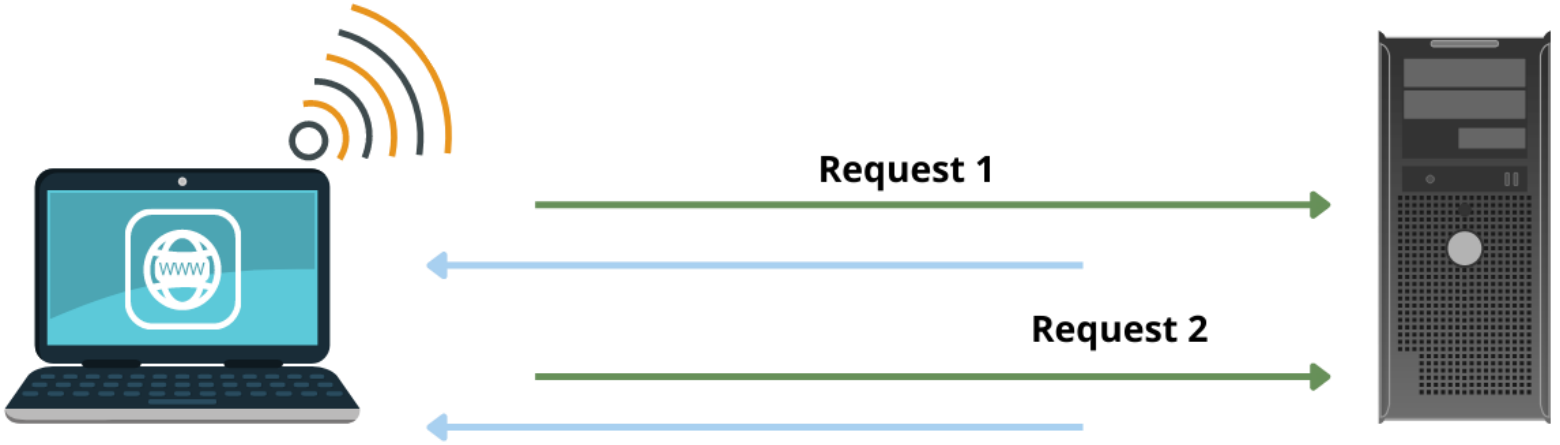
Repetição de dados

HOL – Head of Line Blocking



HTTP 1.1

HOL – Head of Line Blocking

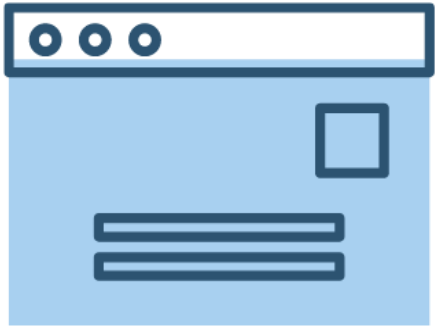


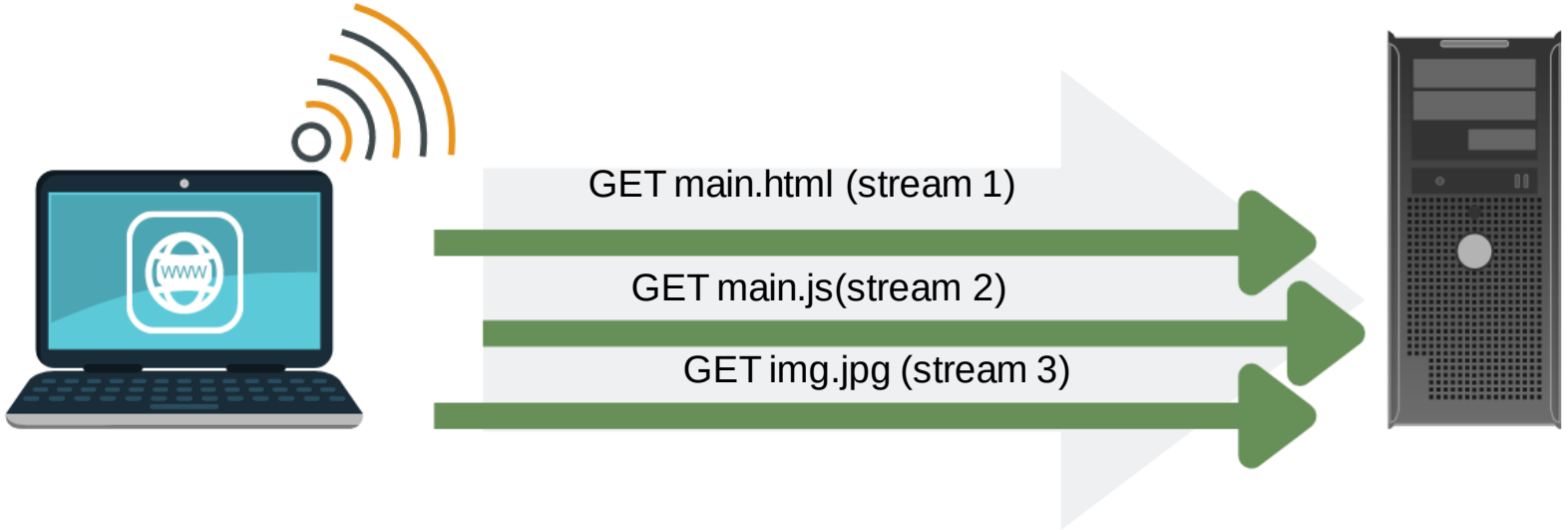
HTTP 2.0



HTTP 2.0

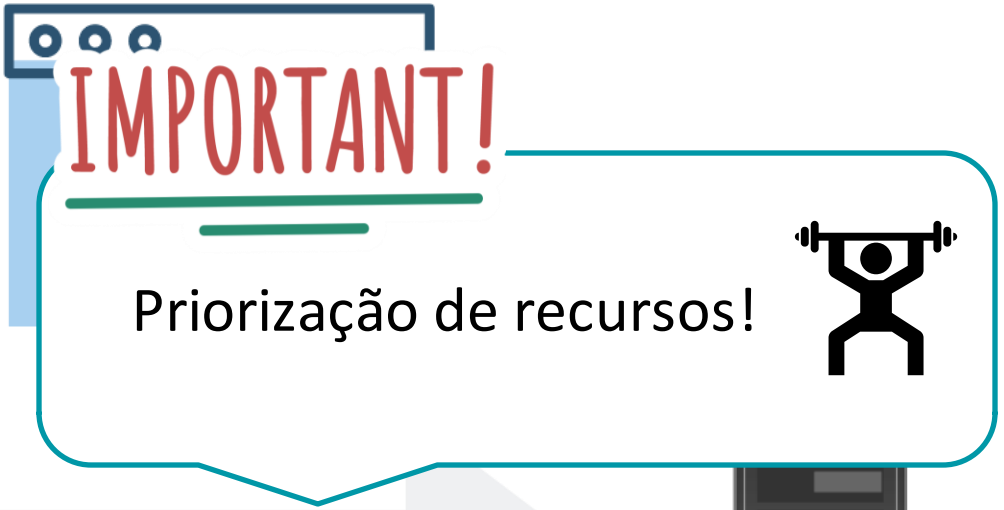
Multiplexação

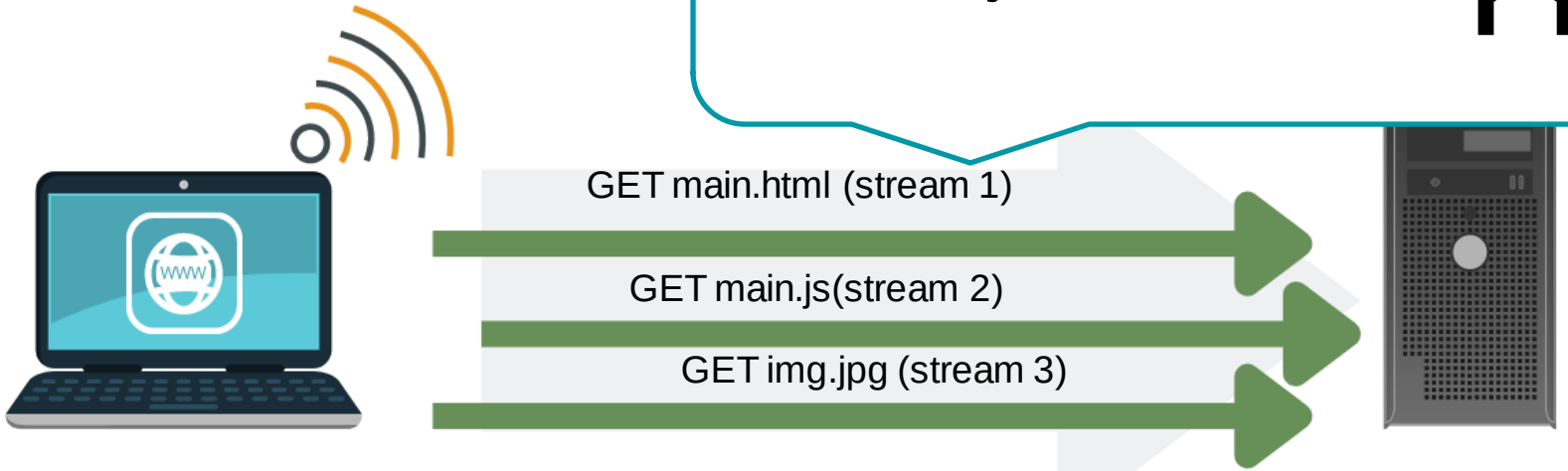




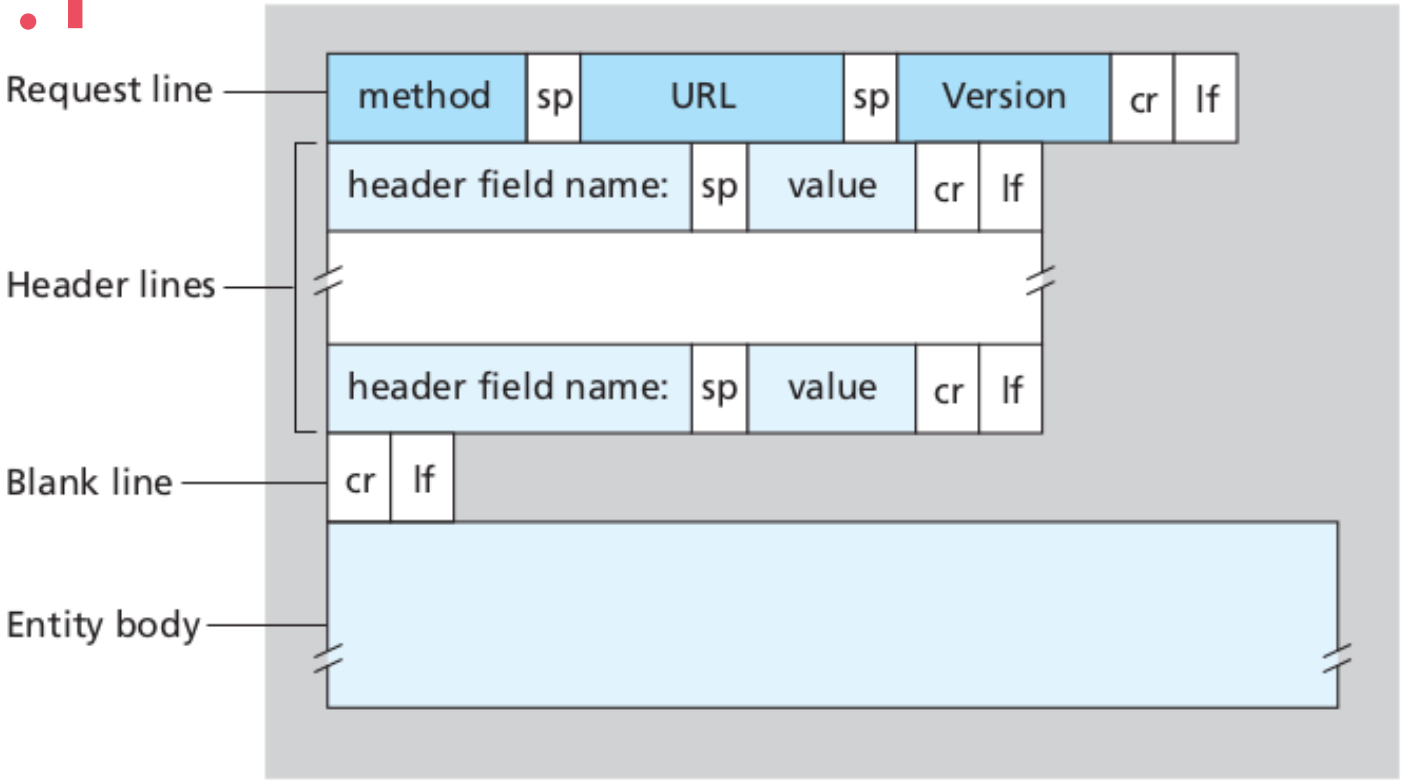
HTTP 2.0

Multiplexação



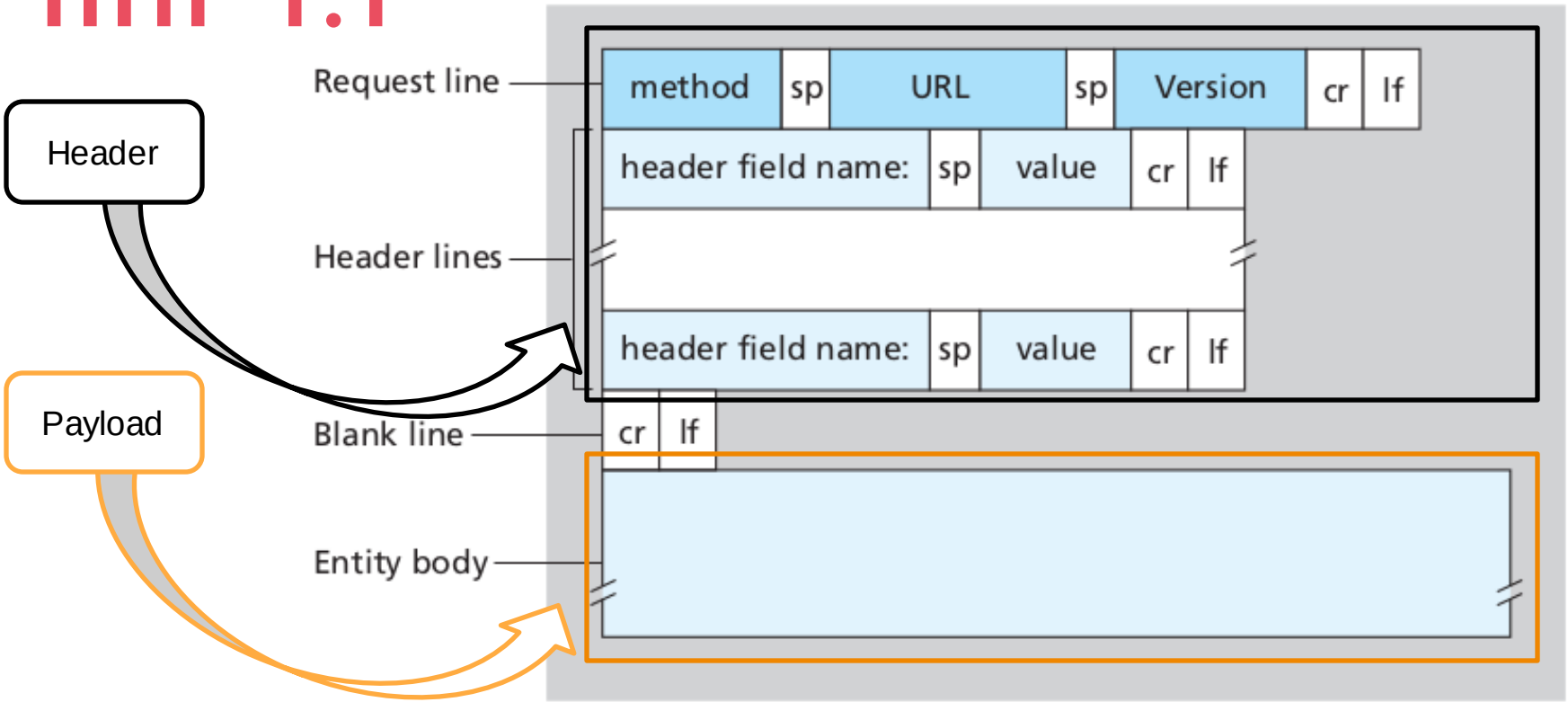


HTTP 1.1



Fonte: Kurose 6° ed.

HTTP 1.1

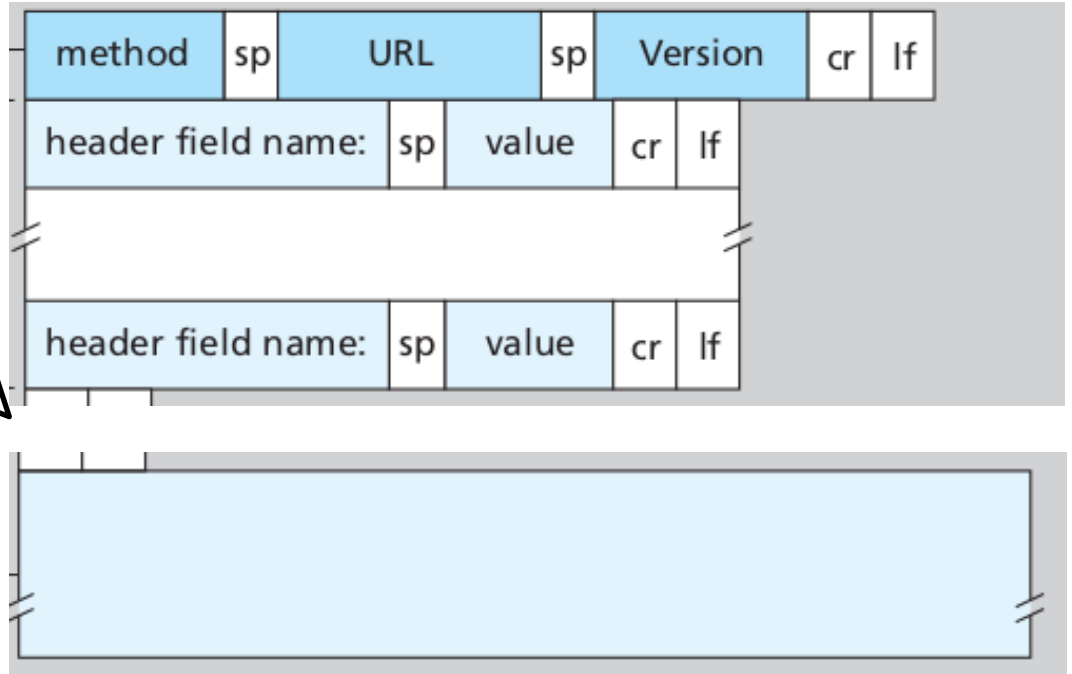


Fonte: Kurose 6° ed.

HTTP 2.0

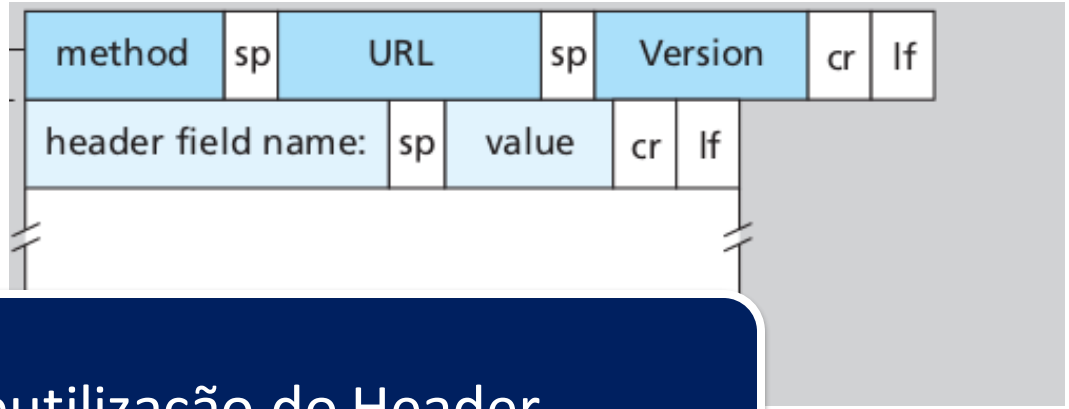
Streaming ID header

Streaming ID data



Segmentação - Streaming

HTTP 2.0



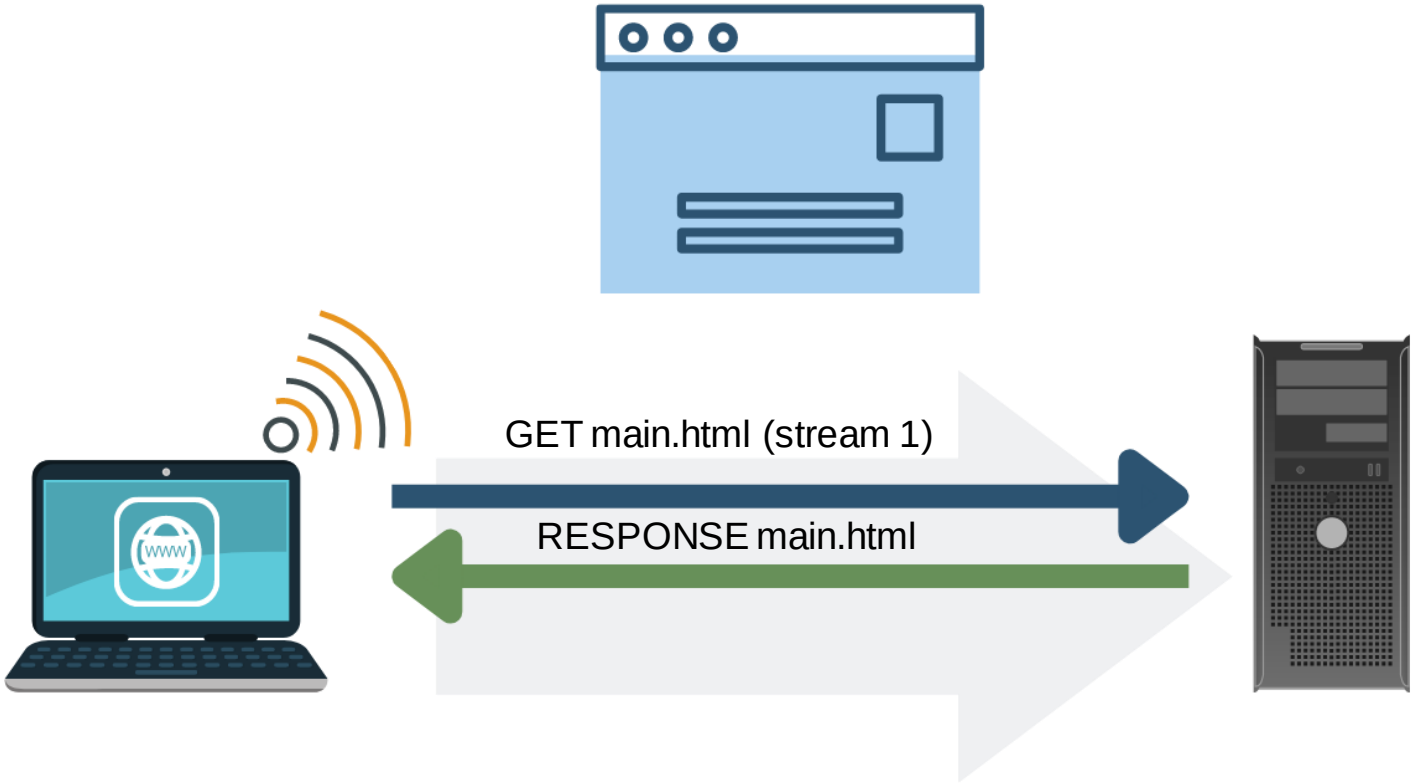
Streaming ID header

Reutilização do Header

Streaming ID data

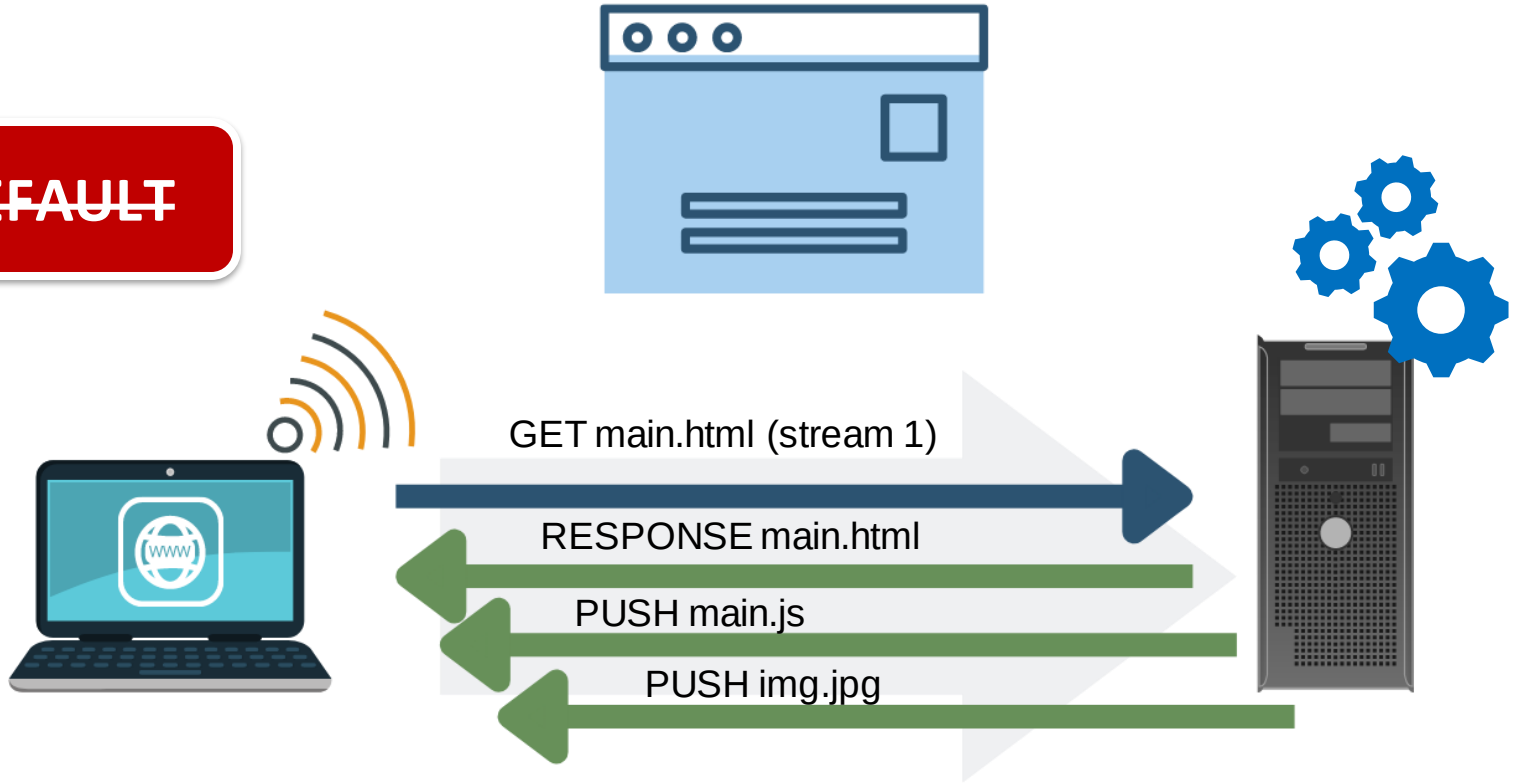
Segmentação - Streaming

HTTP 2.0 - Push

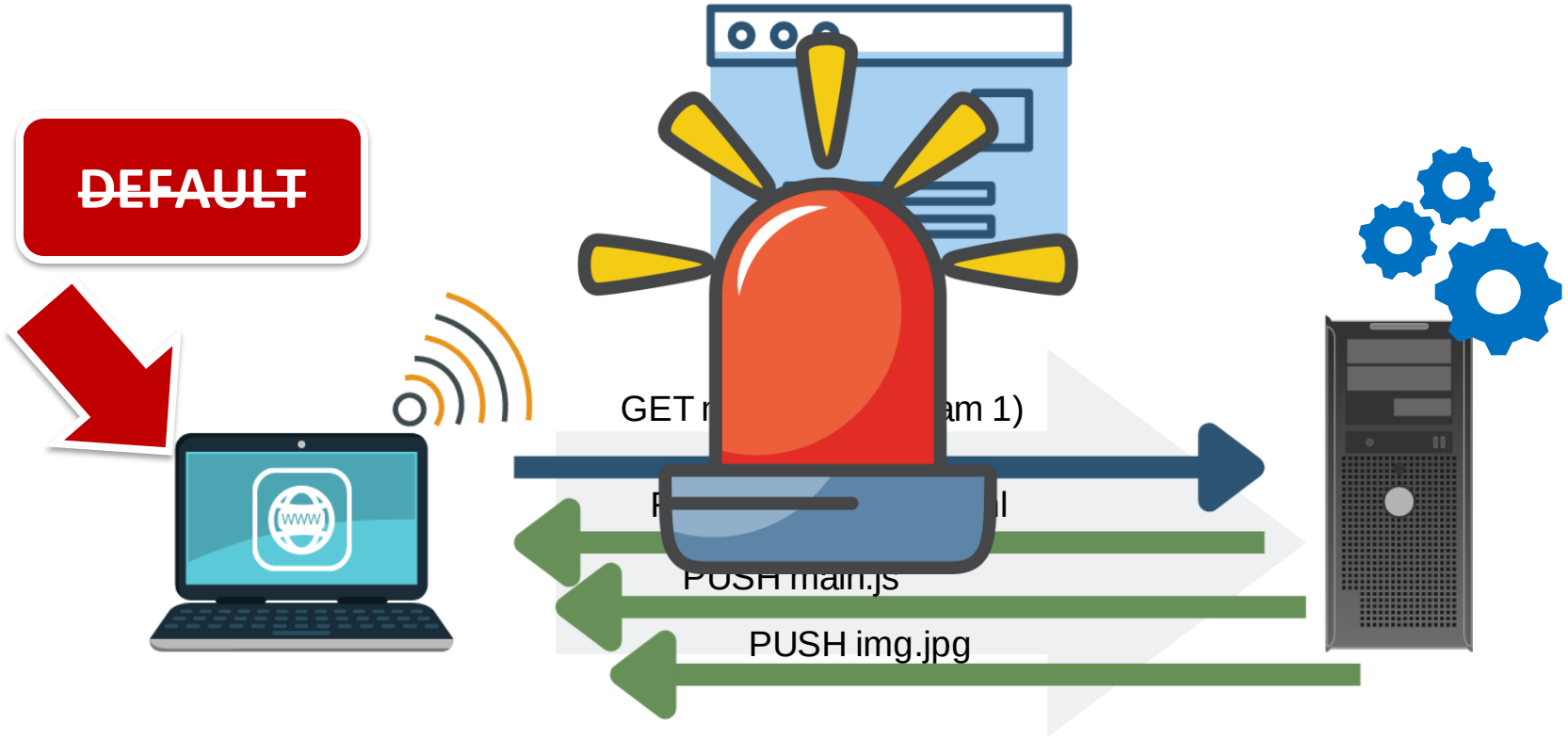


HTTP 2.0 - Push

~~DEFAULT~~



HTTP 2.0 - Push



O cliente suporta?

HTTP 2.0

- Única conexão persistente
- Compressão de header
- Server push
- HTTPS por padrão - TLS
- Negociação no handshake



HTTP 2.0

CONTRAS

PUSH

- Configuração incorreta

Mixe 1.1 e 2.0

- Lentidão
- Load balancer – HTTP 1.1



Etapa 5

Servidores / Sistemas de aplicação

// Conhecendo os principais protocolos de comunicação da internet /Protocolo HTTP

Percurso

~~Etapa 1~~ ~~Como funciona o protocolo HTTP?~~

~~Etapa 2~~ ~~Mensagens HTTP – Request & Response~~

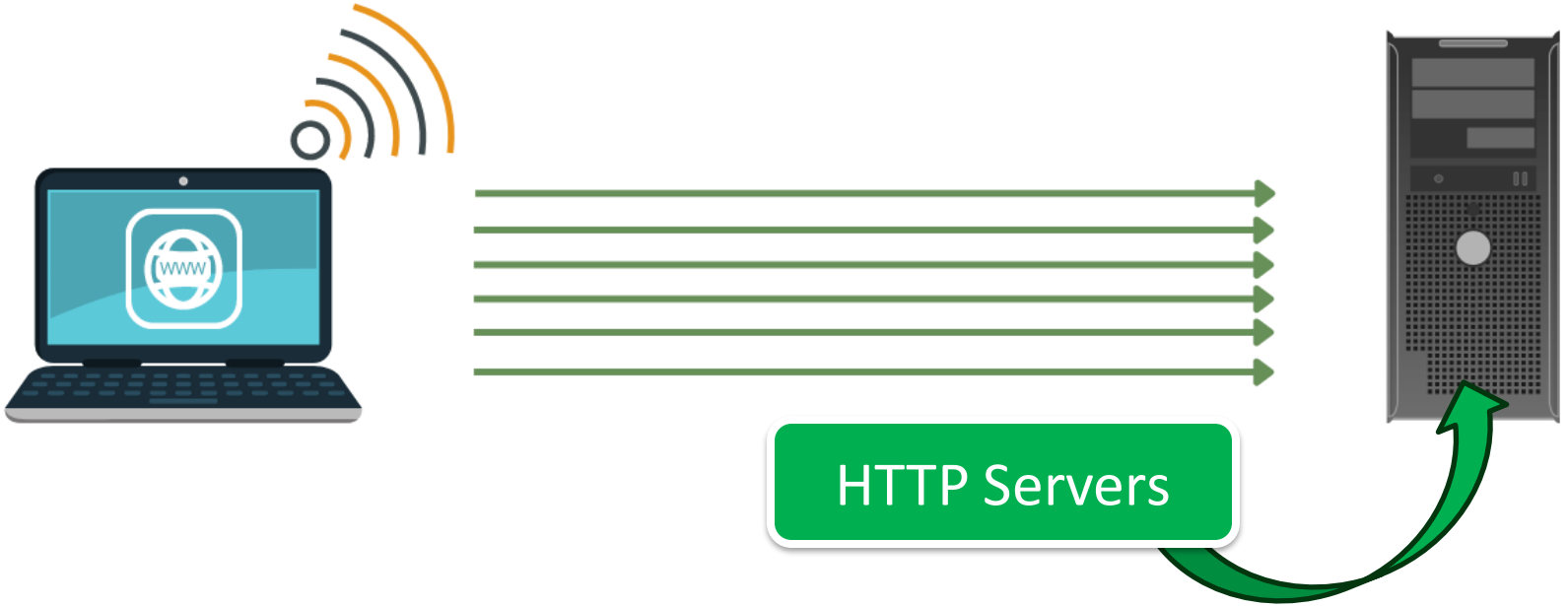
~~Etapa 3~~ ~~Para que servem Cookies e Cache?~~

Percurso

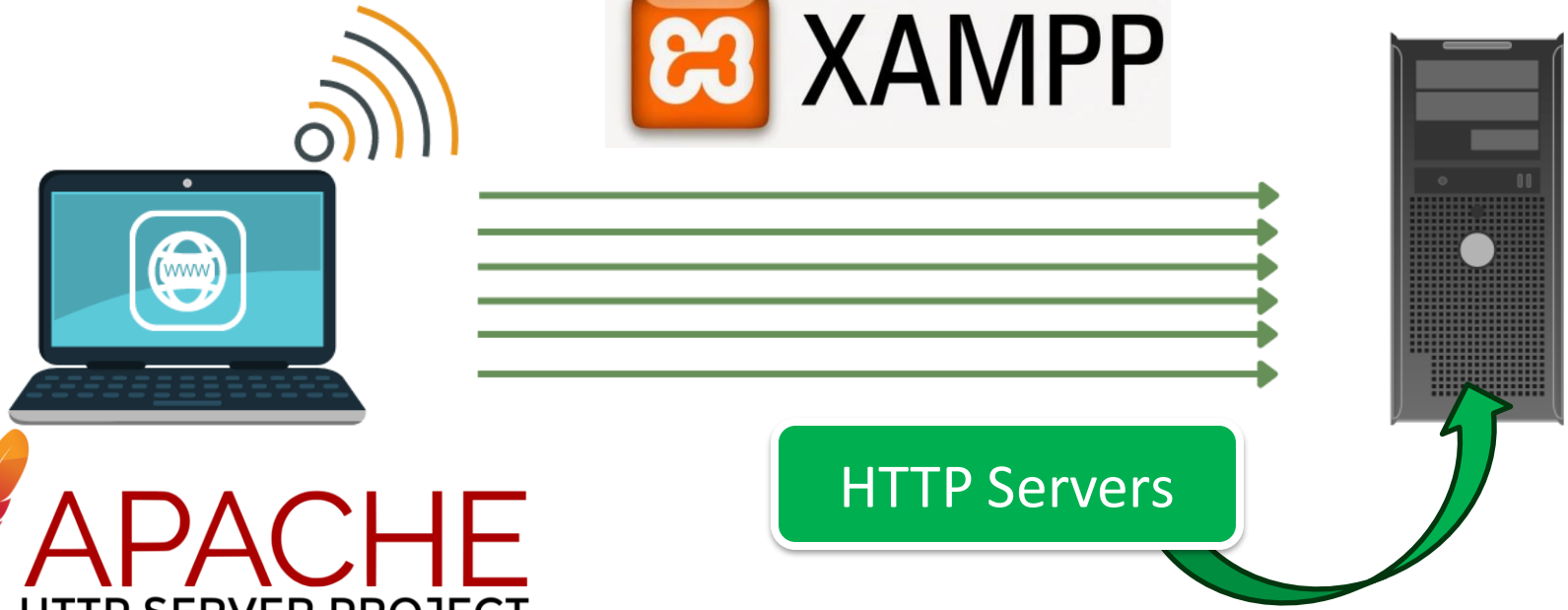
~~Etapa 4~~ ~~HTTP 2.0 - Atualizações do protocolo~~

Etapa 5 **Servidores / Sistemas de aplicação**

Servidores/Sistemas



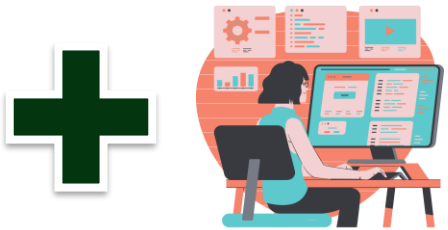
Servidores/Sistemas



Servidores/Sistemas

- Datado de 1995
- Contribuição c/ WWW
- Opensource & free
- Comunidade:

    Apache Software Foundation





Servidores/Sistemas

- Modular, dinâmico e carrefável
- Multiple Request Processing modes (MPMs)
- Altamente escalável (+ 10,000 conexões simultâneas - fácil)
- Lida com arquivos estáticos, index, auto-indexing e negociação de conteúdo



Servidores/Sistemas



- TLS/SSL via OpenSSL ou wolfSSL.
- Servidores virtuais com endereços baseados em IP ou nome
- Compatibilidade com IPV6
- Suporte à versão 2.0 do HTTP
- Proxy reverso, websocket ...





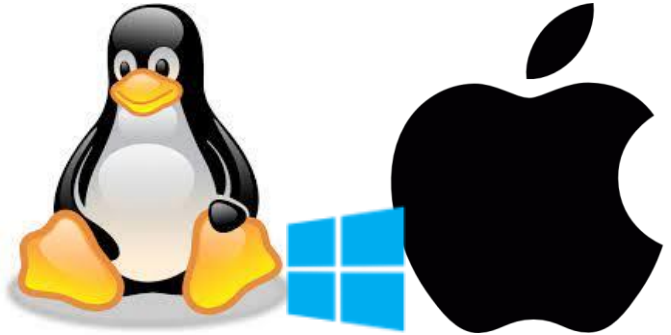
Servidores/Sistemas

Pacote/ambiente

- Apache server
- MariaDB
- Interpretadores script – PHP/Perl

OpenSSL, phpMyAdmin, MediaWki, Joomla, Wordpress ...





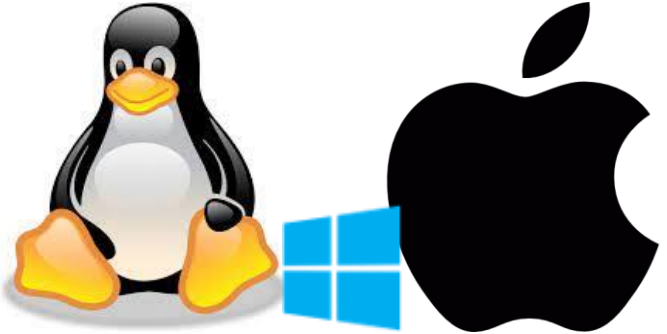
Módulos

Servidores/Sistemas

Pacote/ambiente

- Teste e desenvolvimento
- S/ acesso internet
- Interpreta

OpenSSL, phpMyAdmin, MediaWki, Joomla, Wordpress ...



Módulos

Servidores/Sistemas



Servidor / 2004

- Web server
- Proxy reverso
- **Load balancer**
- Mail proxy
- HTTP cache

Servidores/Sistemas



Servidor / 2004

- Web server
- Proxy reverso
- **Load balancer**
- Mail proxy
- HTTP cache

W3Techs



35%



33%

Servidores/Sistemas



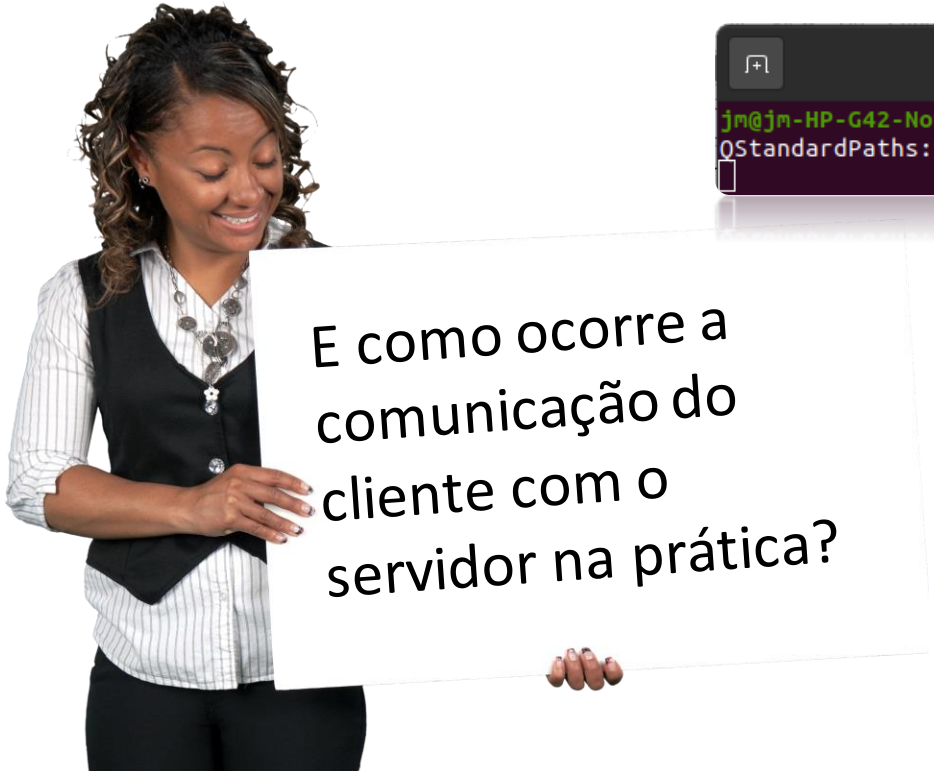
- + 10mil conexões simultâneas
- Lida com arquivos estáticos, index, auto-indexing
- Proxy reverso
- Load balance
- TLS/SSL com SNI via OpenSSL
- Suporte à HTTP 2.0

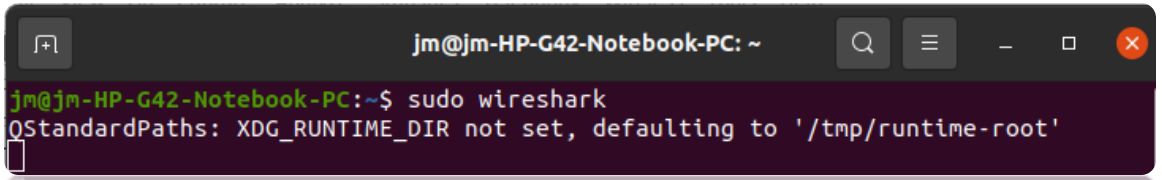
Servidores/Sistemas



- Compatível com IPV6
- FastCGI SCGI, uWSGI com cache
- Suporte a gRPC (v. 1.13.10)
- Servidores virtuais baseados em IP e nome
- WebSocket desde 1.3.13
- URL rewriting and redirection[35][36]

Servidores/Sistemas





WIRESHARK

Aula 2

HTTPS – O que muda no protocolo?

//Conhecendo os principais protocolos de comunicação da internet

Percurso

~~Aula 1~~	~~Protocolo de comunicação Web HTT~~
Aula 2	**HTTPS, o que muda no protocolo?**
Aula 3	Protocolo de comunicação Web Socket

Percurso

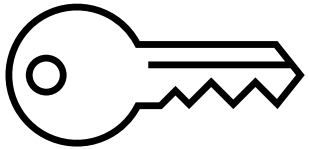
Etapa 1	Conceitos básicos de segurança da informação
Etapa 2	Protocolo SSL – Secure Socket Layer
Etapa 3	Operações do SSL e considerações finais

Etapa 1

Conceitos básicos de segurança da informação

//Conhecendo os principais protocolos de comunicação da internet / HTTPS – O que muda?

Segurança da informação



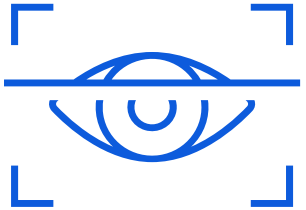
Criptografia por chave



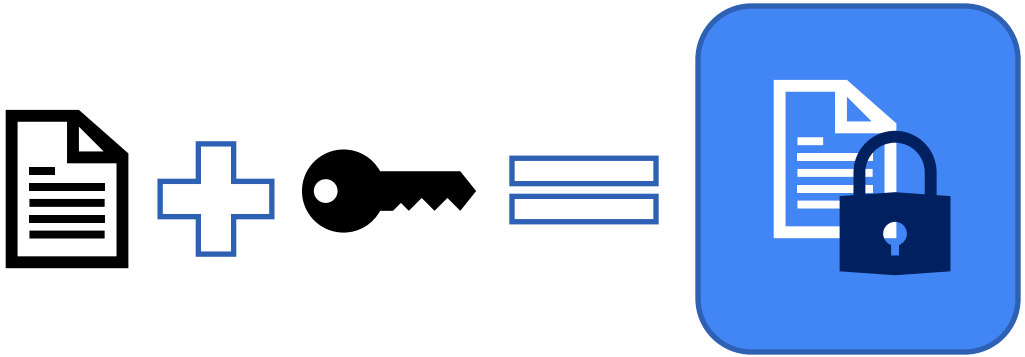
&



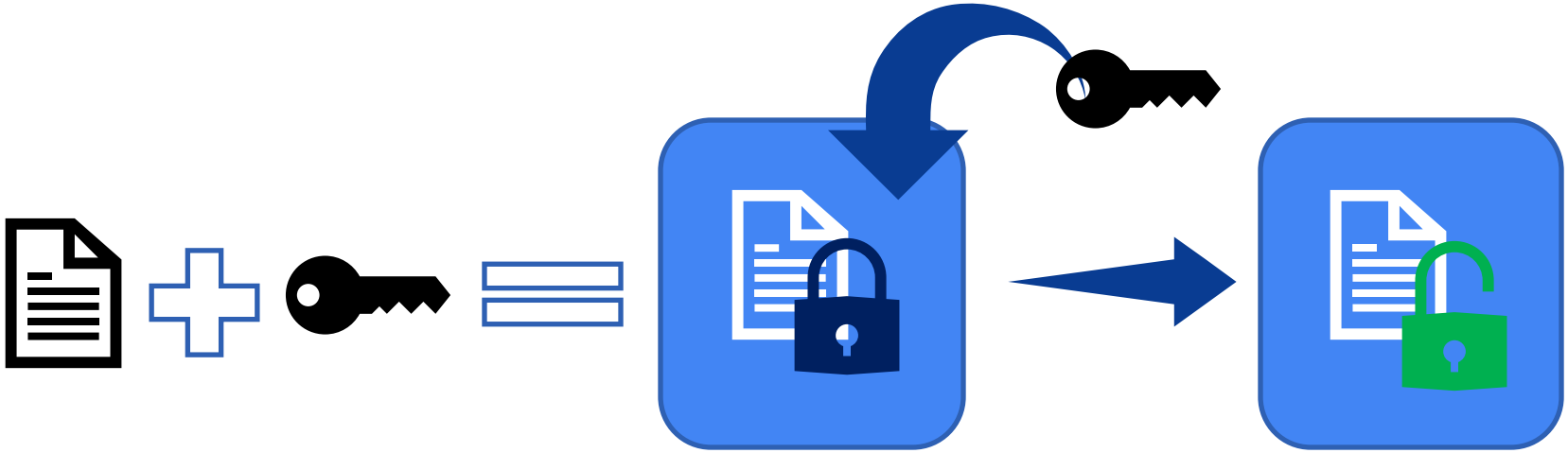
Certificado digital



Criptografia por chave



Criptografia por chave



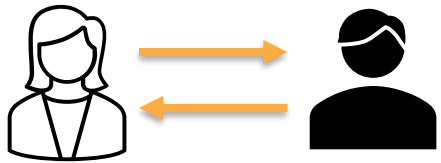
Criptografia por chave

- Assimétrica
- Simétrica

Chave assimétrica

- Assimétrica
- Simétrica

- Chave privada
  Assinatura - criptografia
- Chave pública
  verificação de autenticidade

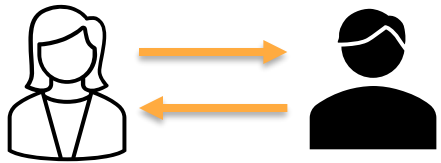


Chave simétrica

- Assimétrica
- Simétrica

Chave única privada

- Conhecimento prévio da chave
- Como transferir a chave?



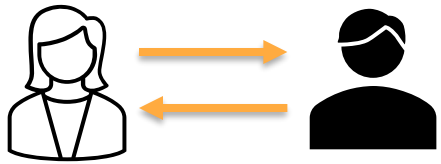
Chave simétrica

- Assimétrica
- Simétrica

Cifra de César

Chave única privada

- Conhecimento prévio da chave
- Como transferir a chave?



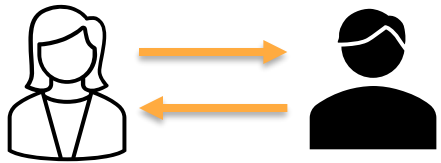
Chave simétrica

- Assimétrica
- Simétrica

Cifra de César

Chave única privada

- Conhecimento prévio da chave
- Como transferir a chave?



Chave simétrica

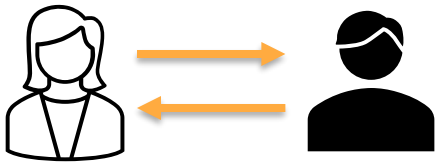
Cifra de César

Criptografia por Chave

Funcionamento

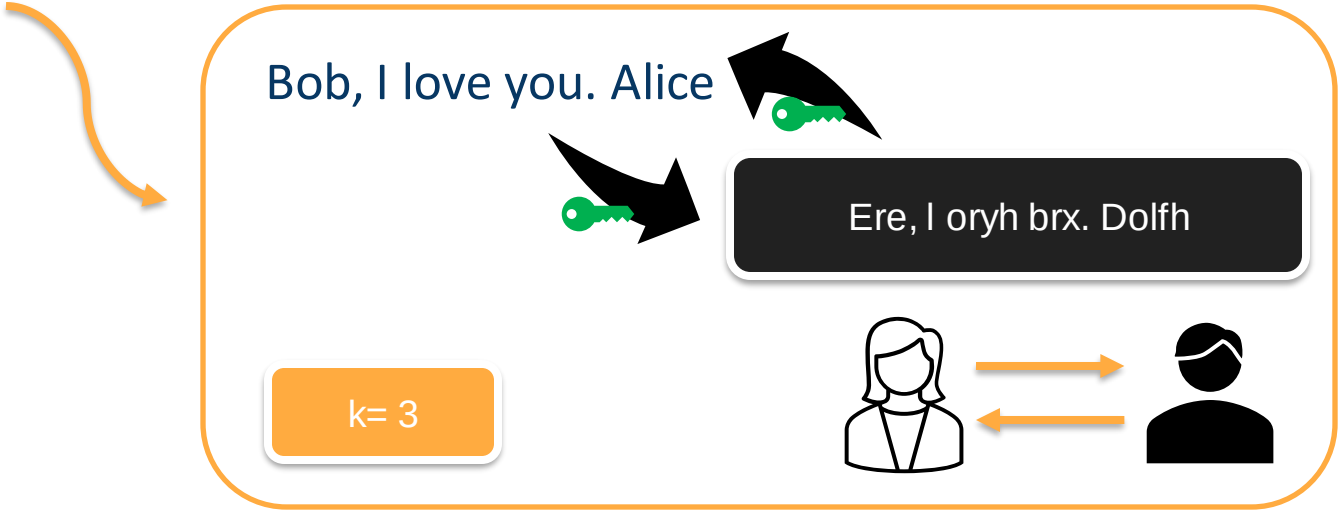
Substituição da letra pela k-ésima do alfabeto
Rotatividade do alfabeto

k= [1,26]



Chave Simétrica

Cifra de César



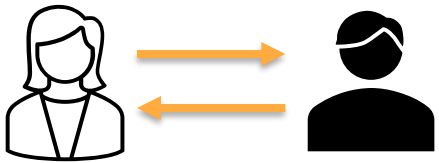
Chave simétrica

Criptografia por Chave

- Assimétrica
- Simétrica

Tipos:

- Cifra de fluxo
- Cifra de bloco



Chave simétrica

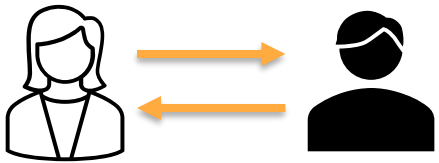
Criptografia por Chave

- Assimétrica
- Simétrica

Tipos:

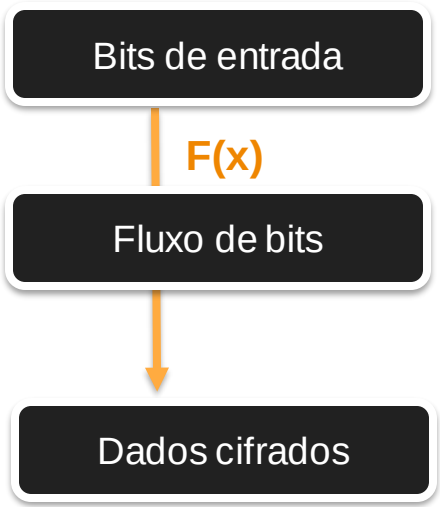
- Cifra de
- Cifra de bloco

SSL, PGP, Ipsec



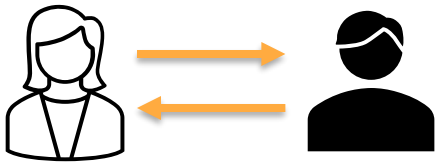
Chave simétrica

101010100
010100101



Cifra de fluxo

- Sequência de bits pseudo-aleatório
- Mapeamento 1 para 1



Chave simétrica

Cifra de bloco

1 2 4 8 16 32 64 128

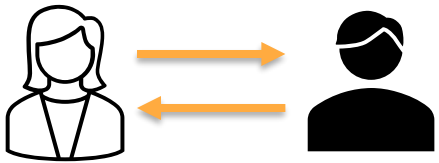
2^0 2^1 2^2 2^3 2^4 2^5 2^6 2^7 2^8

Blocos de bits

- k= número de bits
- Ex:

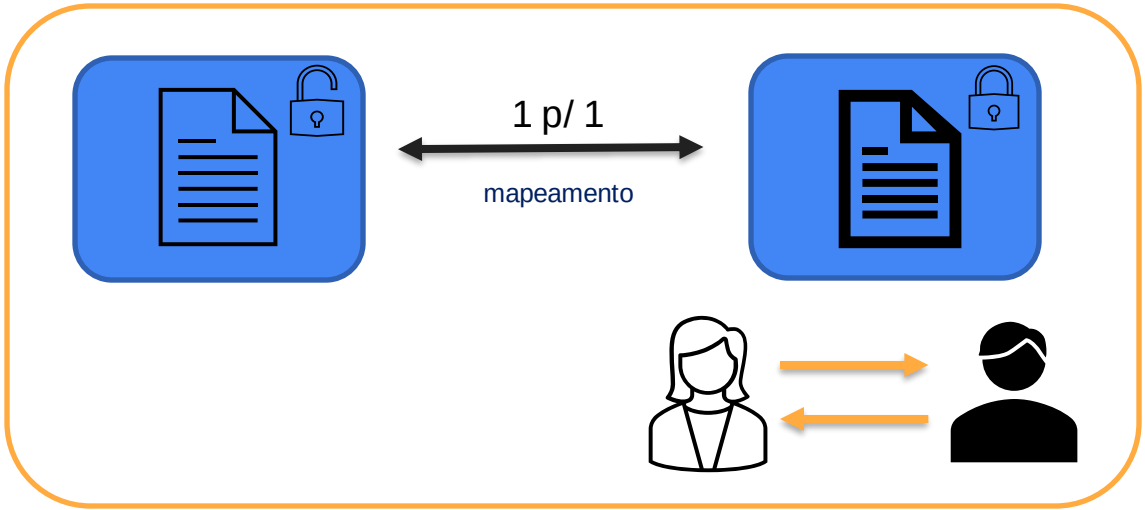
k = 64

Blocos
64 bits



Chave simétrica

Cifra de bloco

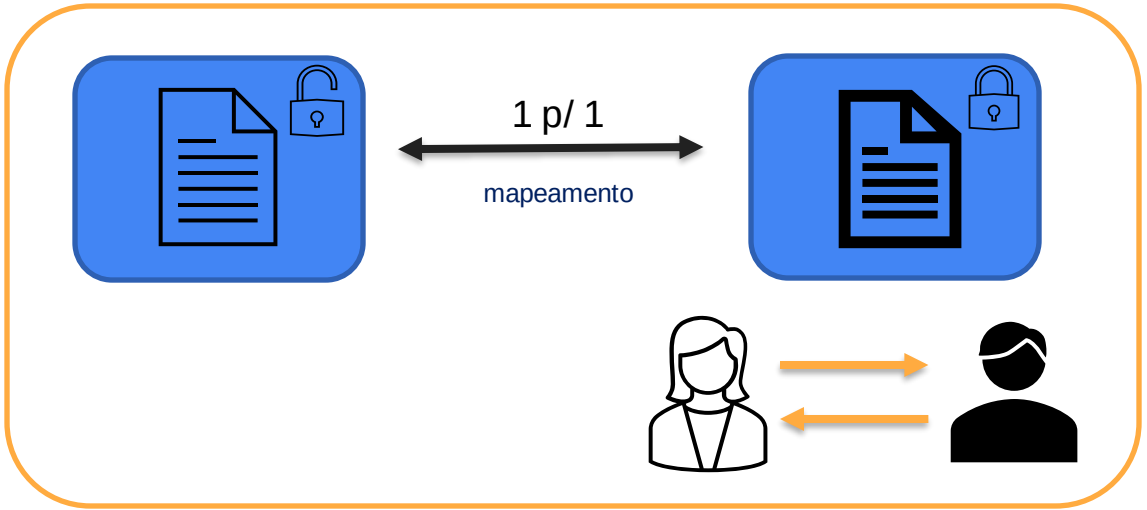


Chave simétrica

k= 3

010110001111

Cifra de bloco



Chave simétrica

k= 3

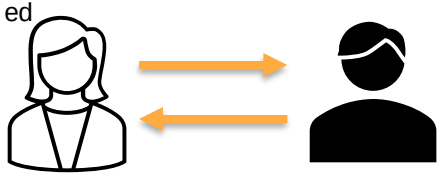
010110001111

101000111001

Cifra de bloco

Entrada	Saída	Entrada	Saída
000	110	100	011
001	111	101	010
010	101	110	000
011	100	111	001

Fonte: kurose - 5° ed



Chave simétrica

k= 3

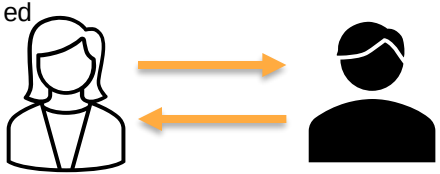
010 110 001 111

101 000 111 001

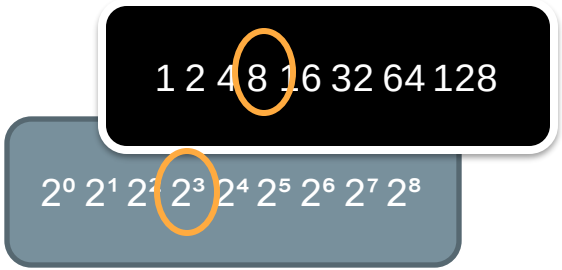
Cifra de bloco

Entrada	Saída	Entrada	Saída
000	110	100	011
001	111	101	010
010	101	110	000
011	100	111	001

Fonte: kurose - 5° ed



Chave simétrica

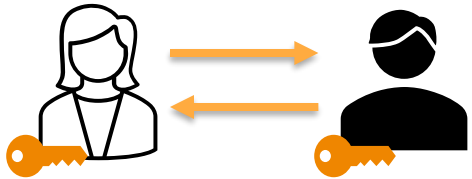


Cifra de bloco

Possibilidades?

- Mapeamento por permutação
- 8!





Chave simétrica

E na prática?



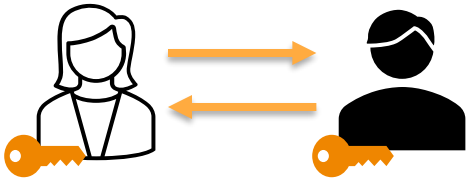
Cifra de bloco

K>64

$1{,}844674407 \times 10^{19}$!

Mapeamento por funções

Permutação



Chave simétrica

DES

AES

3DES

Cifra de bloco

E na prática?

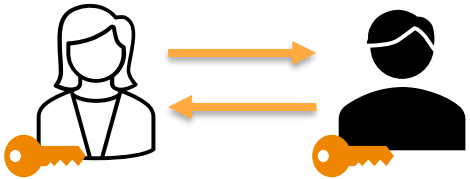


K>64

$1{,}844674407 \times 10^{19}$!

Mapeamento por funções

Permutação



Chave simétrica

DES

AES

3DES

Cifra de bloco

E na prática?

K>64

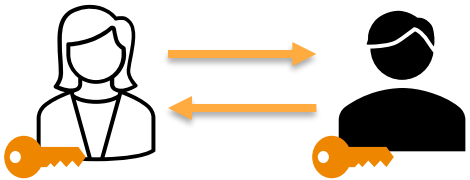
$1{,}844674407 \times 10^{19}$!

Mapeamento por funções

Encadeamento de cifras

Permutação





Certificado digital

IPSec

SSL

TSL

Certificação de chave pública

- O que é certificar uma chave?



Sou bob



Sou bob

? ? ?



Certificado digital

IPSec

SSL

TSL

Certificação de chave pública

- O que é certificar uma chave?



Sou bob



Sou bob

? ? ?



Qual a verdadeira?

Certificado digital

IPSec

SSL

TSL

Certificação de chave pública

- O que é certificar uma chave?
  Comprovar autenticidade

- Entidade certificadora
  Certification Authority - CA

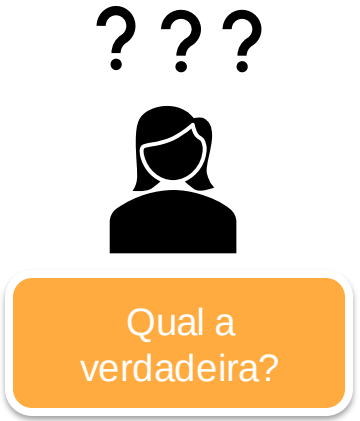


Qual a verdadeira?

Certificado digital

Entidade Certificadora

- Verificação de identidade
- Emissão de certificados



Certificado digital

Alta confiabilidade

Entidade Certificadora



Qual a verdadeira?

- Verificação de identidade
- Emissão de certificados

Vínculo entre chave e entidade

Certificado digital

Entidade Certificadora

Padrões de autoridades certificadoras

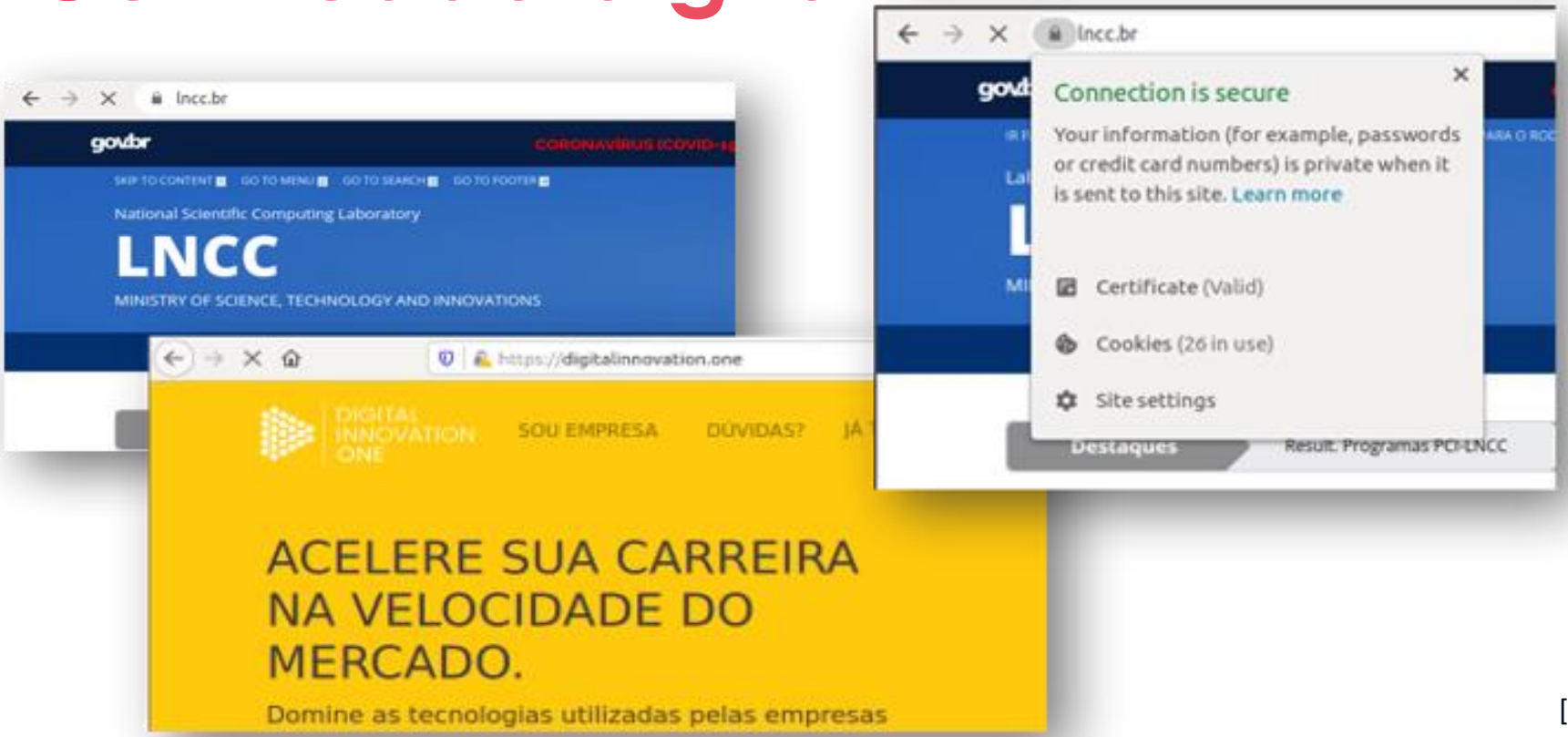
- IETF - recomendação ITUX.509

  Especificação de um serviço de autenticação e sintaxe de certificados

- RFC 1422

  Gerenciamento de chaves baseado em CA em emails seguros

Certificado digital



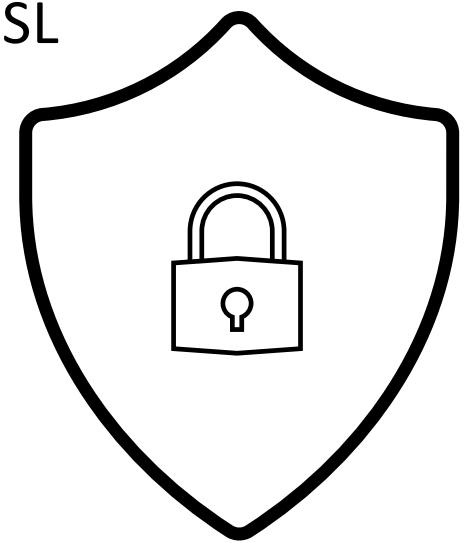
Etapa 2

Protocolo SSL – Secure Socket Layer

//Conhecendo os principais protocolos de comunicação da internet / HTTPS – O que muda?

Protocolo SSL

Secure Socket Layer – SSL

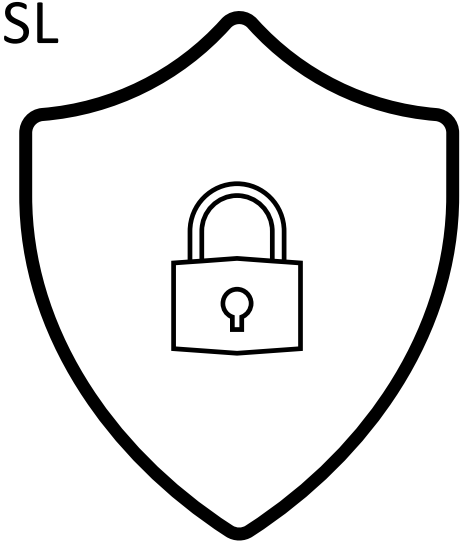


- Segurança para conexões TCP

- Confidencialidade
- Integridade
- Autenticidade end-point

Protocolo SSL

Secure Socket Layer – SSL



- Segurança para conexões TCP

- Confidencialidade
- Integridade
- Autenticidade end-point

Versão 3: Transport Layer Security (TLS) → RFC 4346

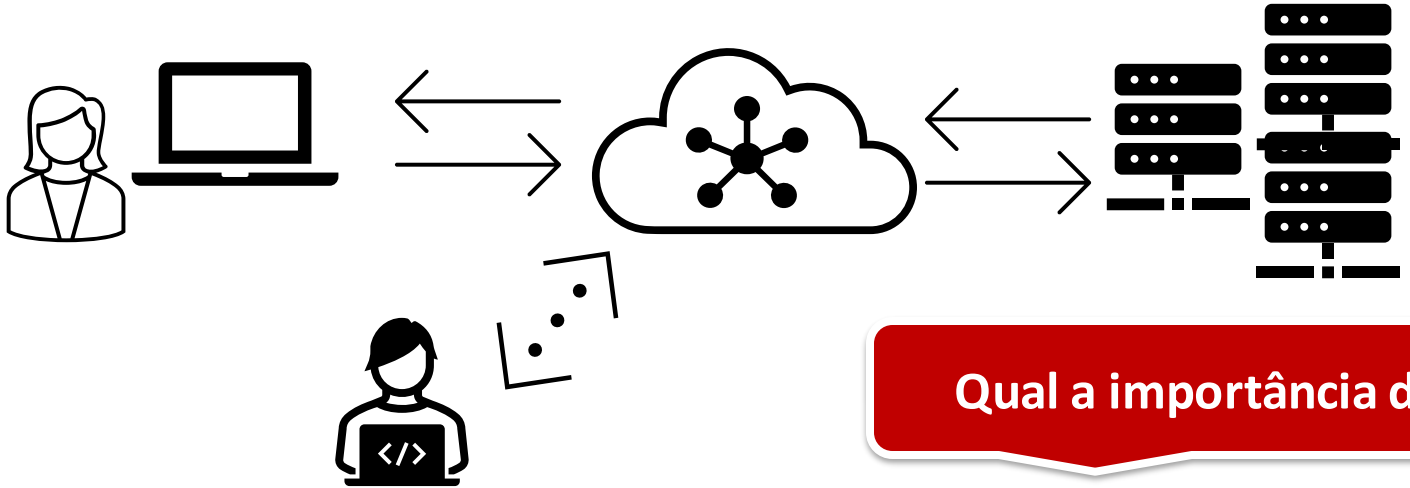
Protocolo SSL

Secure Socket Layer – SSL



Protocolo SSL

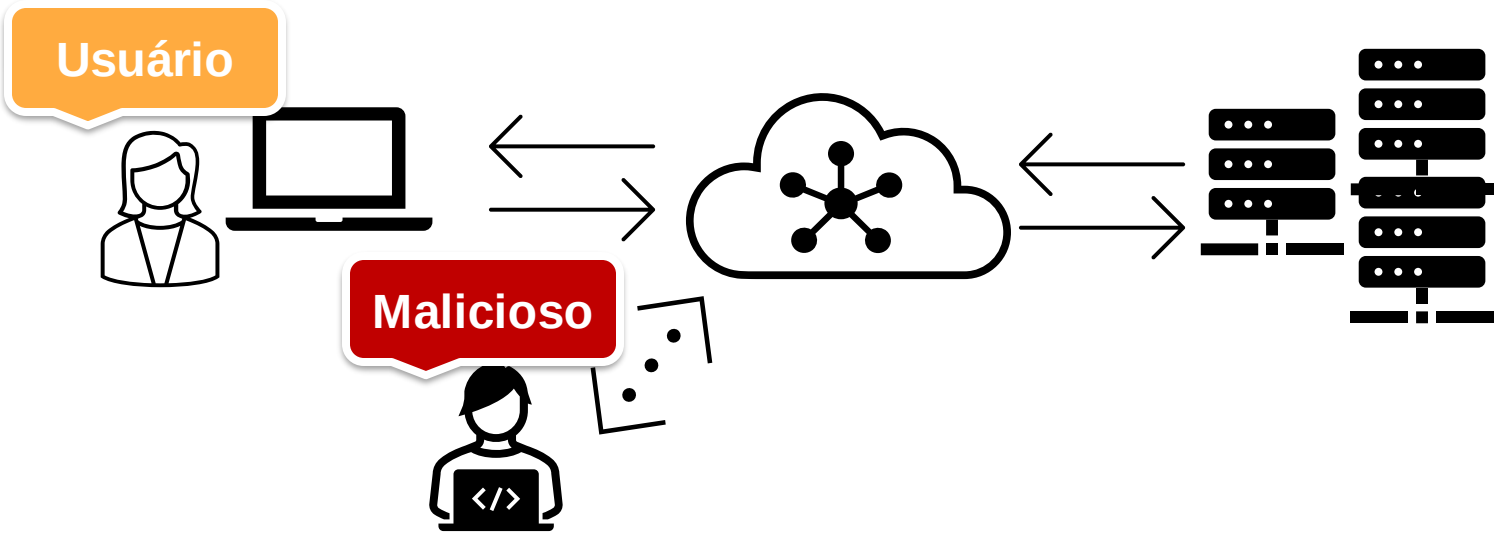
Secure Socket Layer – SSL



Qual a importância do SSL?

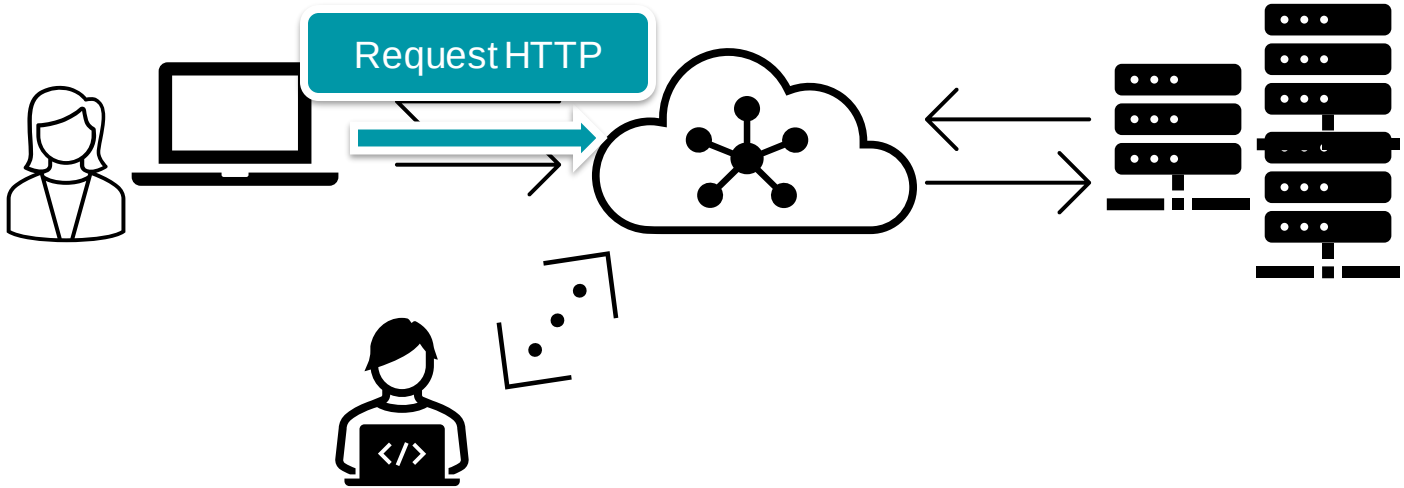
Protocolo SSL

Secure Socket Layer – SSL



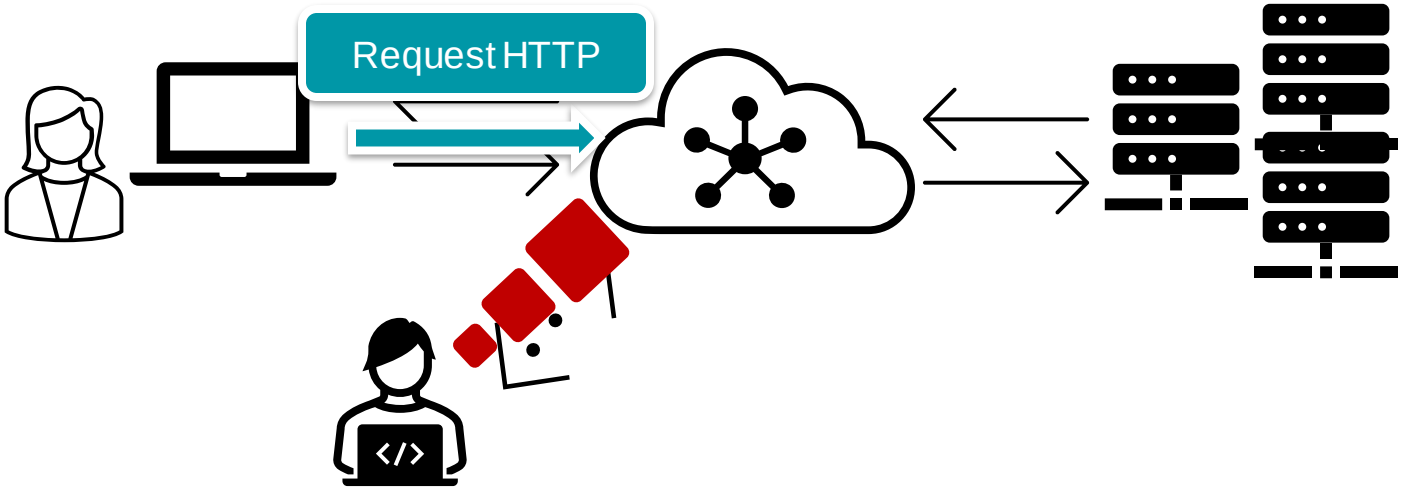
Protocolo SSL

Secure Socket Layer – SSL



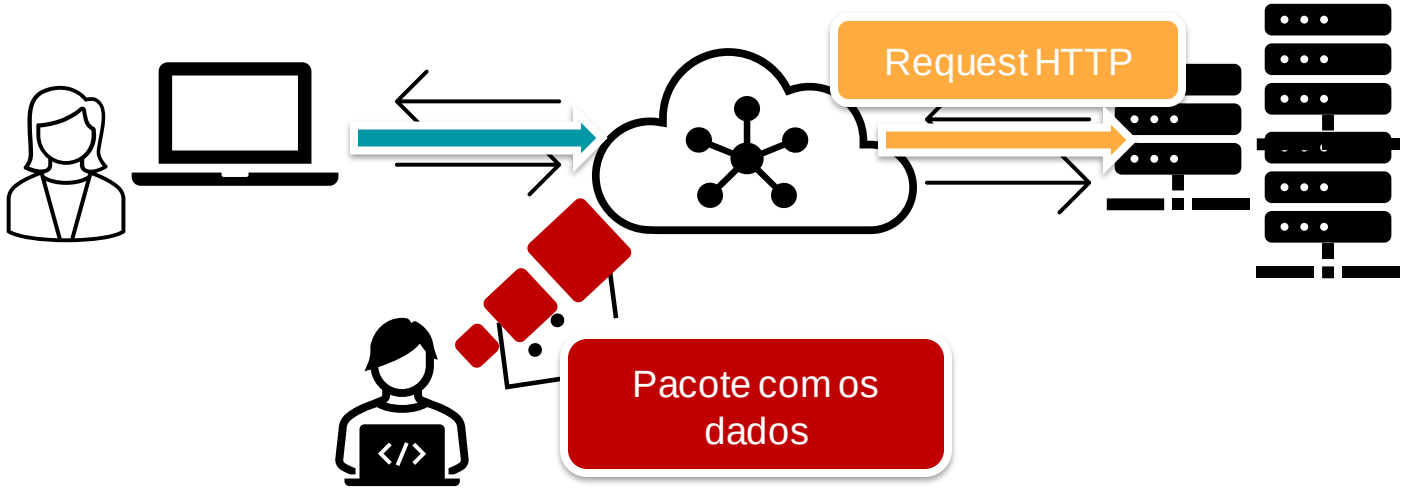
Protocolo SSL

Secure Socket Layer – SSL



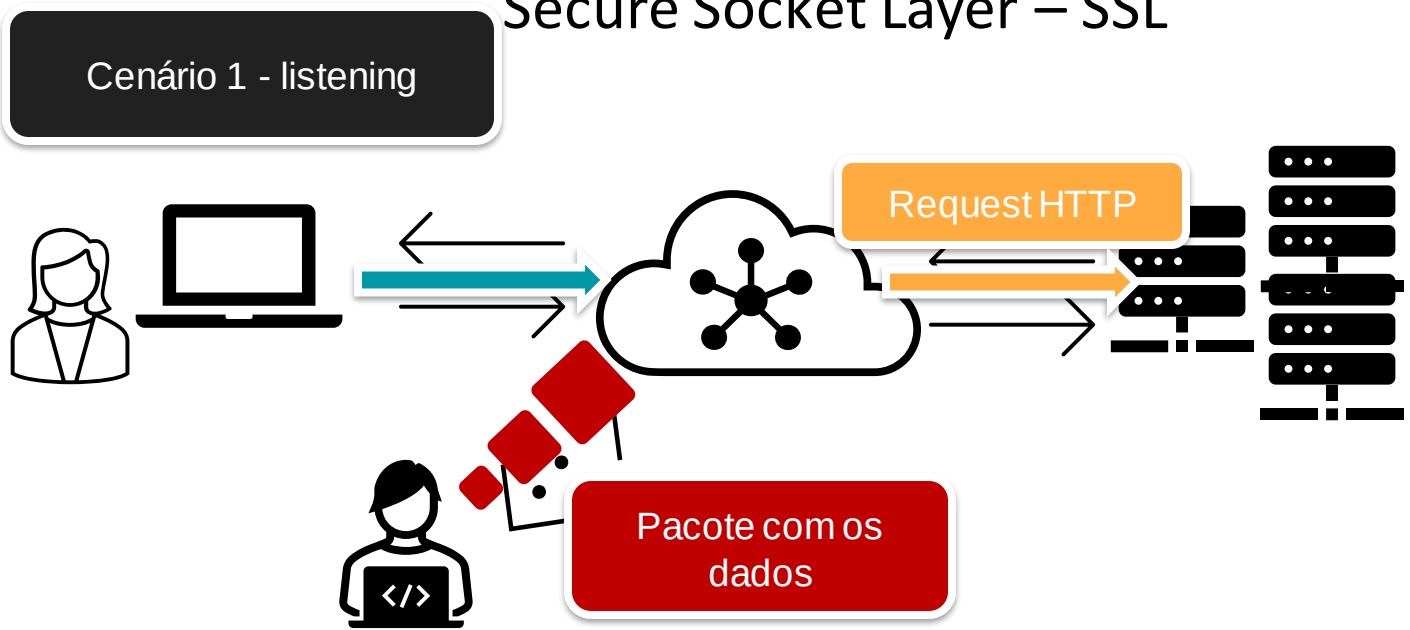
Protocolo SSL

Secure Socket Layer – SSL

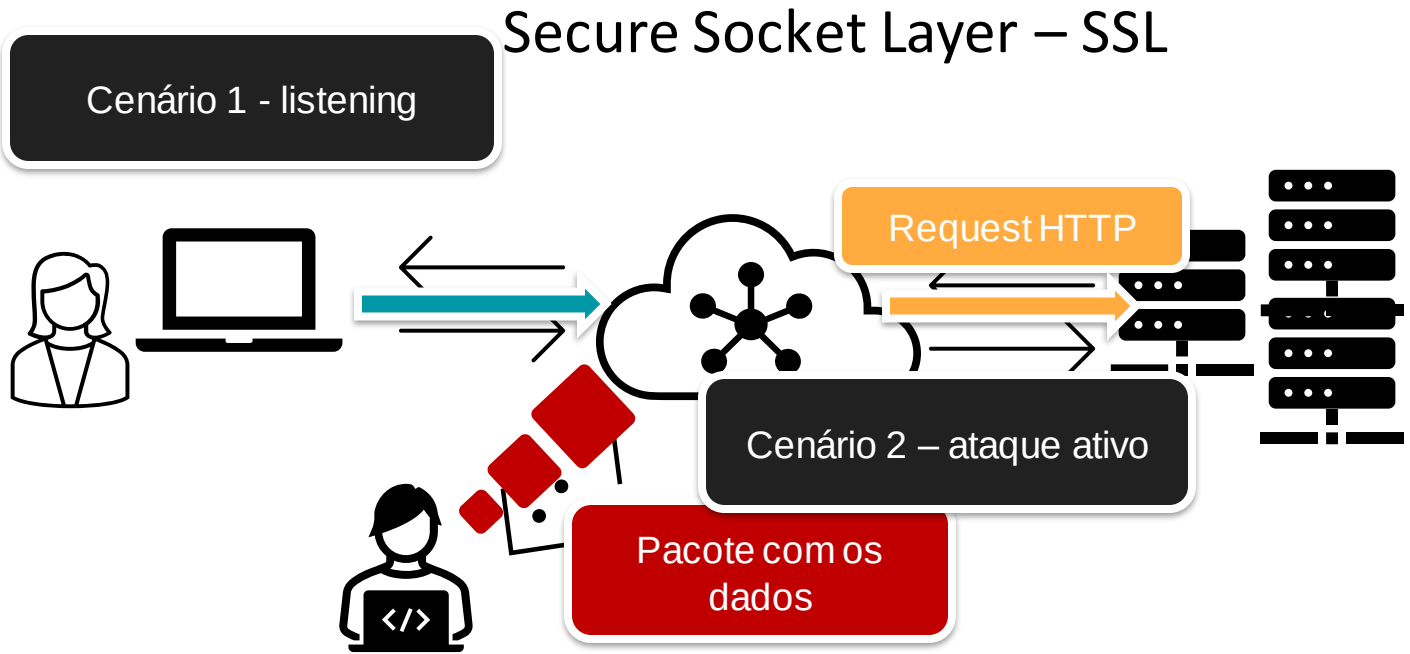


Protocolo SSL

Secure Socket Layer – SSL

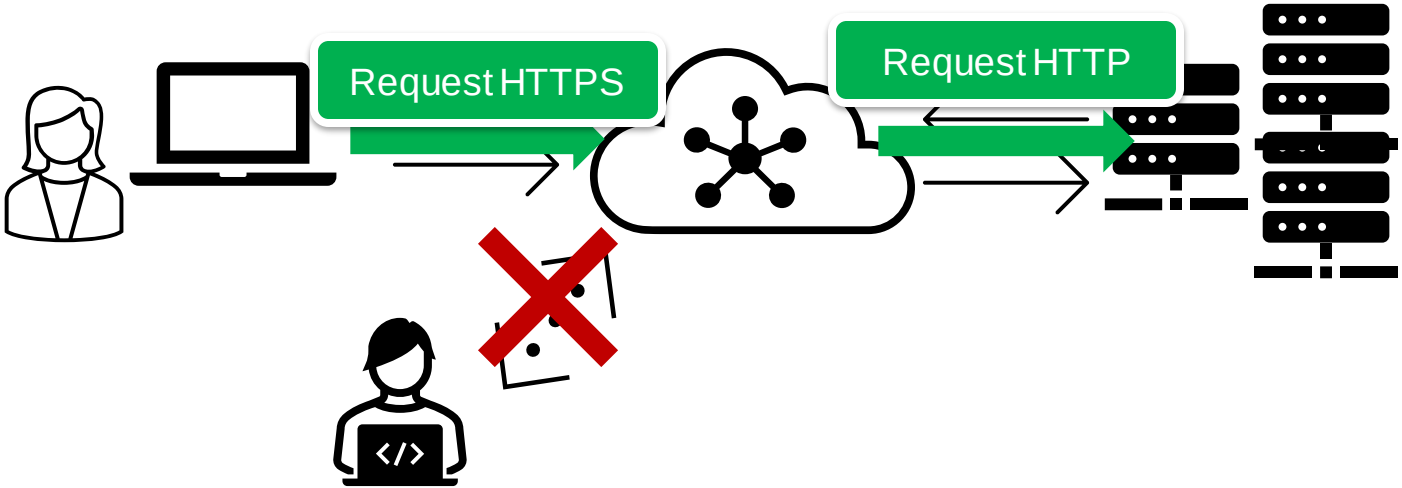


Protocolo SSL



Protocolo SSL

Secure Socket Layer – SSL



Protocolo SSL

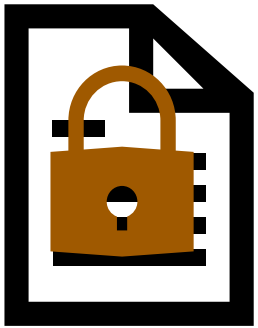
Secure Socket Layer – SSL

Read: unable





Edit: unable



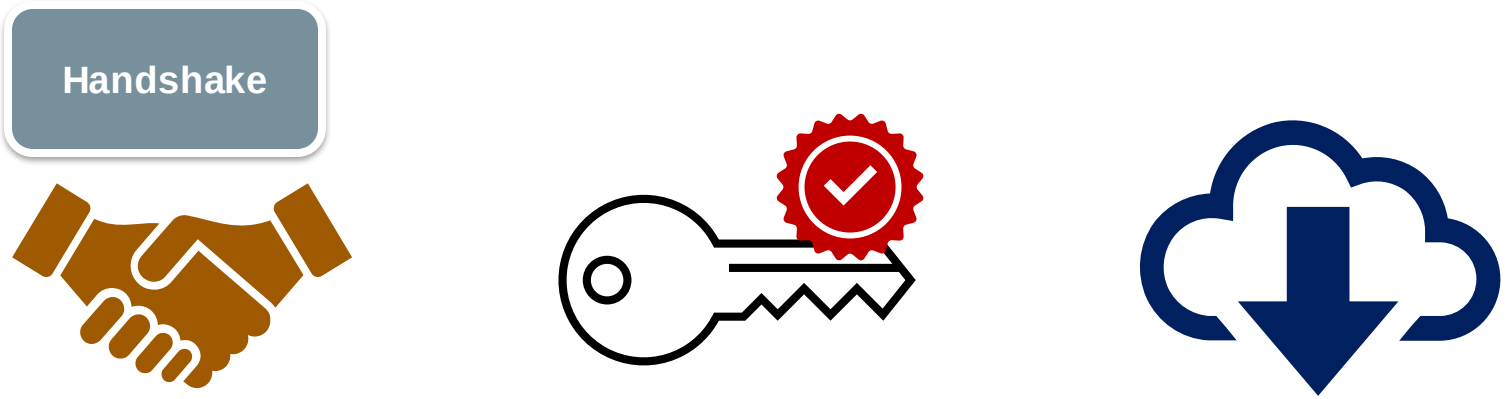
Etapa 2

Operações do SSL & HTTPS

//Conhecendo os principais protocolos de comunicação da internet / HTTPS – O que muda?

Operações do SSL

Secure Socket Layer – SSL



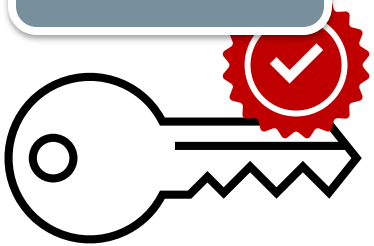
Operações do SSL

Secure Socket Layer – SSL

Handshake



Key Derivation





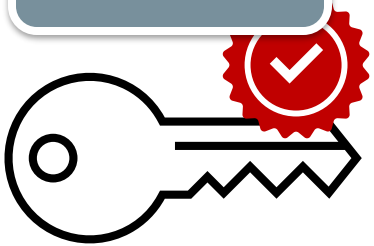
Operações do SSL

Secure Socket Layer – SSL

Handshake



Key Derivation



Data Transfer

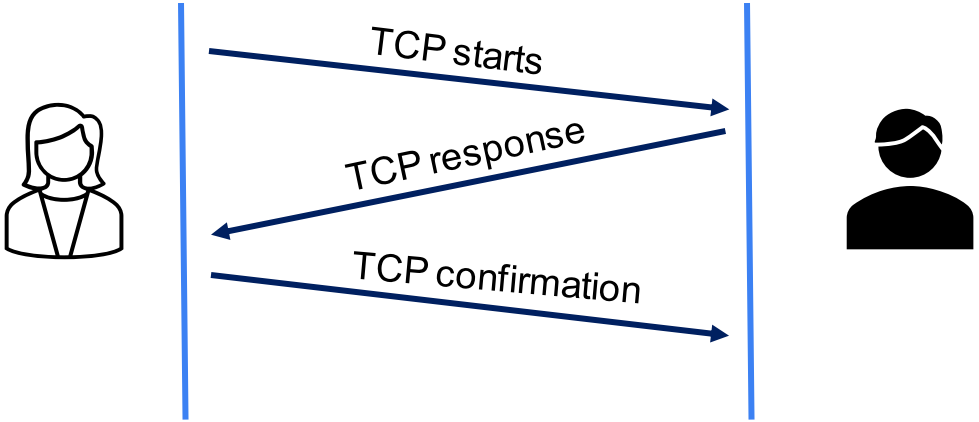


Operações do SSL

Handshake



Estabelecendo conexão TCP

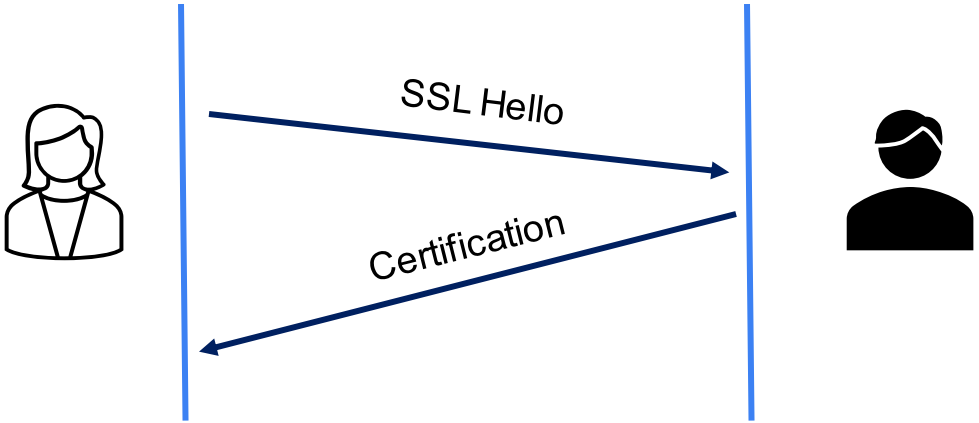


Operações do SSL

Handshake



Verificação de Autenticidade

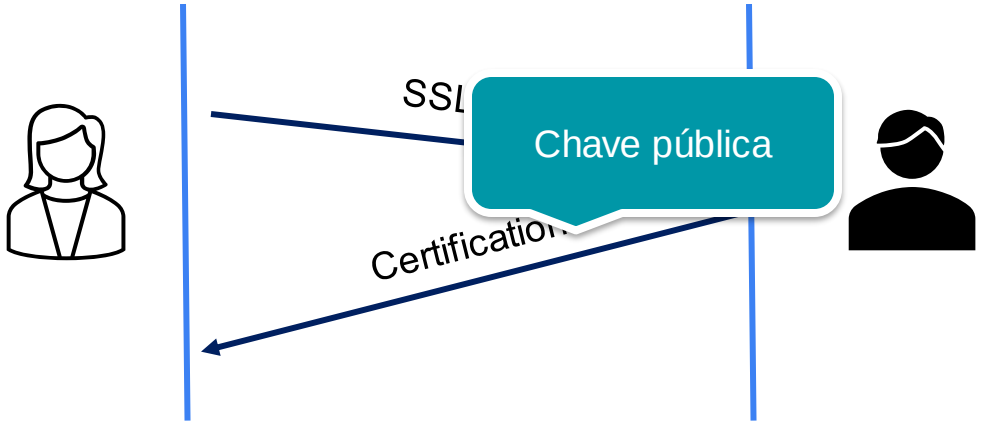


Operações do SSL

Handshake



Verificação de Autenticidade

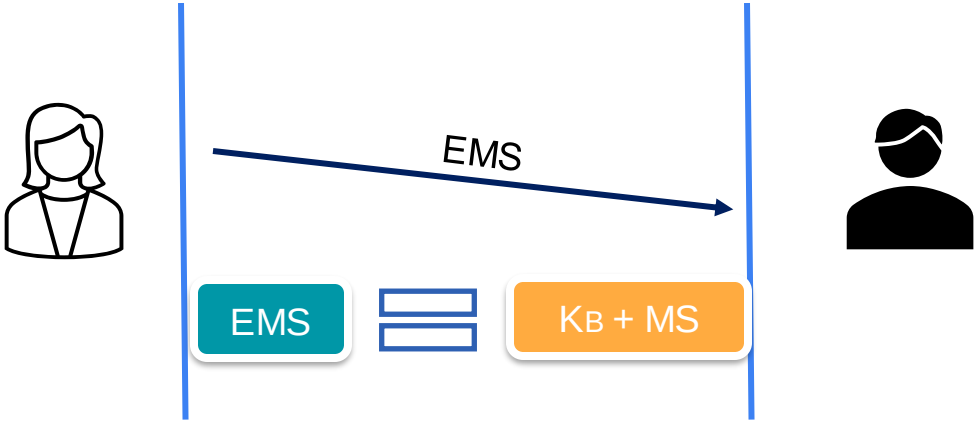


Operações do SSL

Handshake



Envio da Master Secret Key

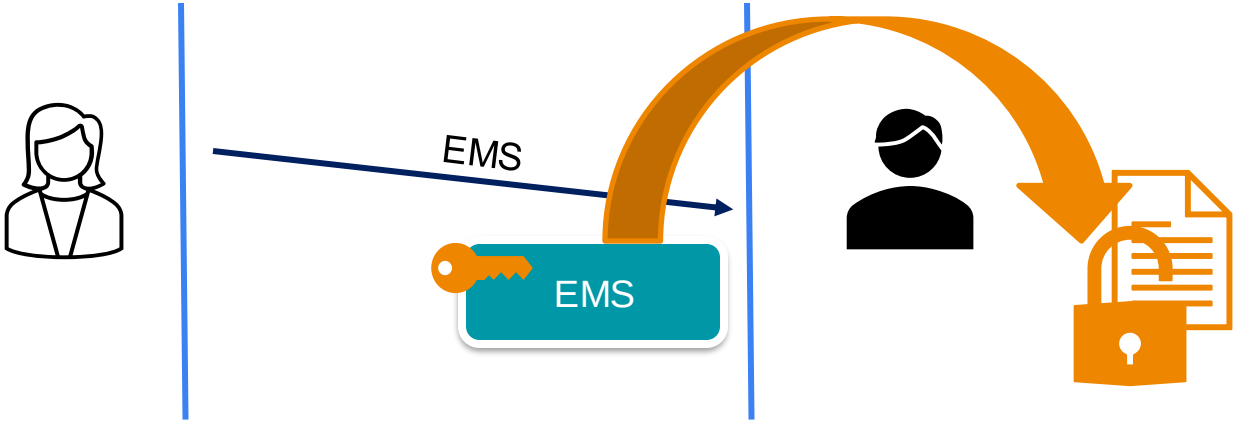


Operações do SSL

Handshake



Envio da Master Secret Key



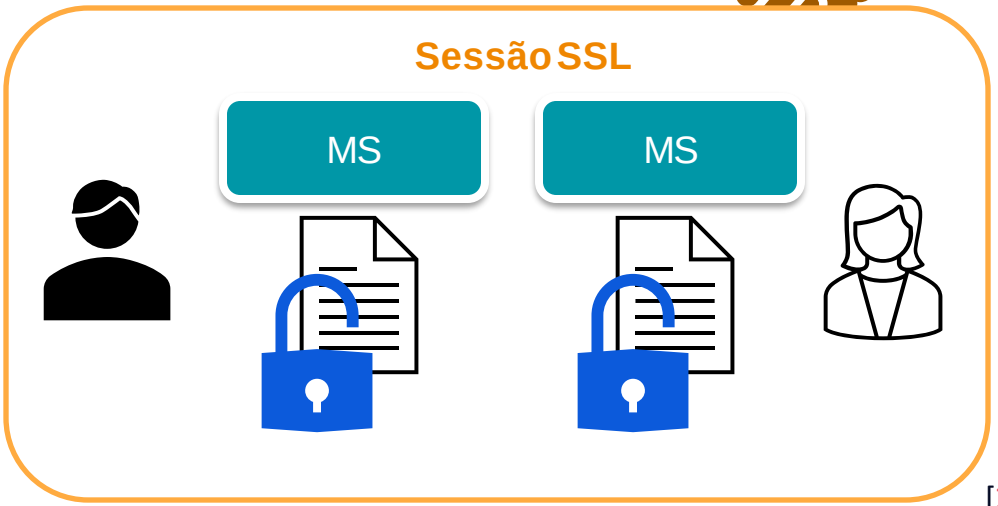
Operações do SSL

Handshake



Master Secret Key





Operações do SSL



Master Secret Key

- MS - Chave simétrica
- Derivação:

Operações do SSL



Master Secret Key

- MS - Chave simétrica
- Derivação:

MS

E_A = SEK de Alice para Bob
M_A = MAC de Alice para Bob
E_B = SEK de Bob para Alice
M_B = MAC de Bob para Alice

Operações do SSL



Master Secret Key

- MS - Chave simétrica
- Derivação:

MS

Session Encryption Key

E_A = SEK de Alice para Bob
M_A = MAC de Alice para Bob
E_B = SEK de Bob para Alice
M_B = MAC de Bob para Alice

Operações do SSL



Master Secret Key

- MS - Chave simétrica
- Derivação:

MS

Verificação da chave MS

M_A = MAC de Alice para Bob
E_B = SEK de Bob para Alice
M_B = MAC de Bob para Alice

Operações do SSL



Operação - fases

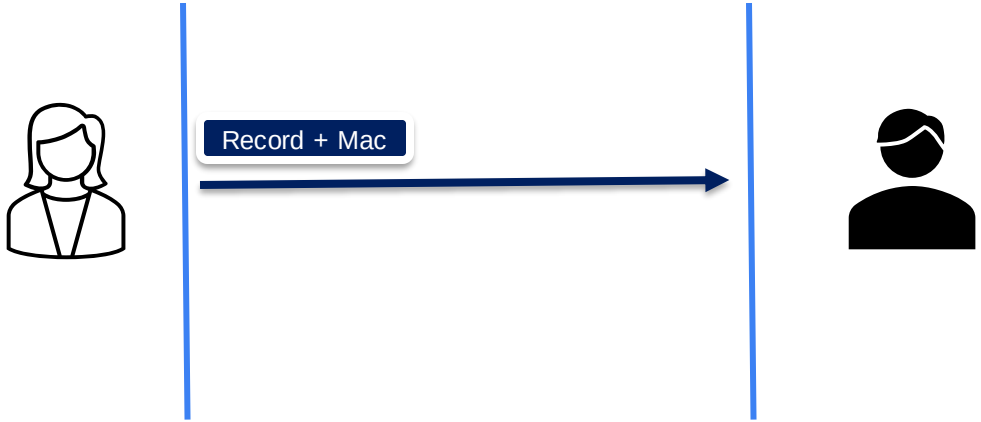
- Transferencia efetiva de dados
- Record+Mac
  verificação de integridade da mensagem



Operações do SSL







HTTPS e breve descrição da LGPD

HTTP + SSL

Segurança na comunicação - HTTP
Over TCP

Verificação da autenticidade por certificados digitais
Porta 443



HTTP + SSL

- Há sites que não utilizam?

https://whynohttps.com/

- Maioria dos site utilizam a versão segura

Proteção contra fishing

Privacidade



LGPD

LGPD – Lei Geral de Proteção de Dados

- Promulgada em 2018
- Entrou em vigor em Agosto de 2020

  Lei de segurança da informação



- N° 13.709/2018

LGPD

LGPD – Lei Geral de Proteção de Dados

- GDPR – General Data Protection Regulation
  legislação europeia

- Legislação do estado da Califórnia - EUA



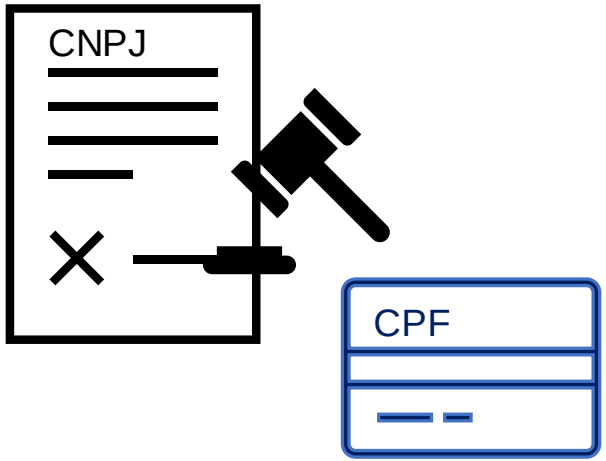
LGPD



LGPD – Lei Geral de Proteção de Dados

O que a lei protege?

- Dados de identificação dos usuários
- Dados sensíveis
  Religião, etnia,

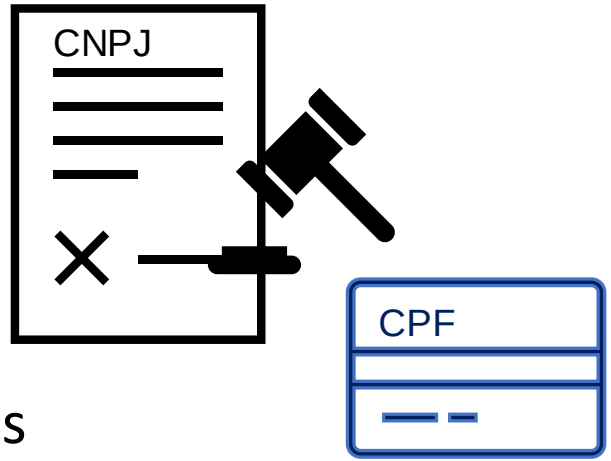


LGPD



LGPD – Lei Geral de Proteção de Dados

- Define o tratamento de dados
  Diversos tipos de operações



- Livre conssentimento
- Direitos do titular das informações
- Sações aos que descumprirem as regras

LGPD



LGPD – Lei Geral de Proteção de Dados

Criação da ANPD (Agencia Nacional de proteção de dados)

- Zelar pela proteção dos dados;
- Elaboração de diretrizes para politica nacional de proteção;
- Promover conhecimento das normas
- Editar regulamentos
- Realizar auditorias

LGPD



LGPD – Lei Geral de Proteção de Dados

Quem deve seguir a lei?

- Empresas que precisam manter um BD
  Funcionários e Usuários

Devem garantir ao titular sigilo das informações

Aula 3

Protocolo de comunicação WebSocket

//Conhecendo os principais protocolos de comunicação da internet

Percurso

~~Aula 1~~ ~~Protocolo de comunicação Web HTT~~

~~Aula 2~~ ~~HTTPS, o que muda no protocolo?~~

Aula 3 **Protocolo de comunicação Web Socket**

Percurso

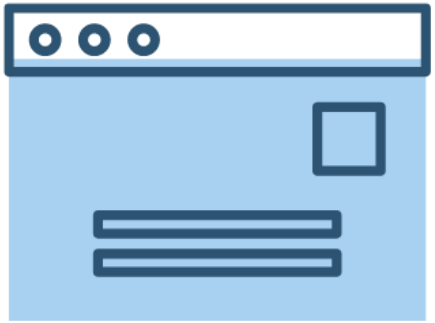
Etapa 1	**Contextualização**
Etapa 2	Como funciona o WebSocket
Etapa 3	Como funciona a comunicação na internet?

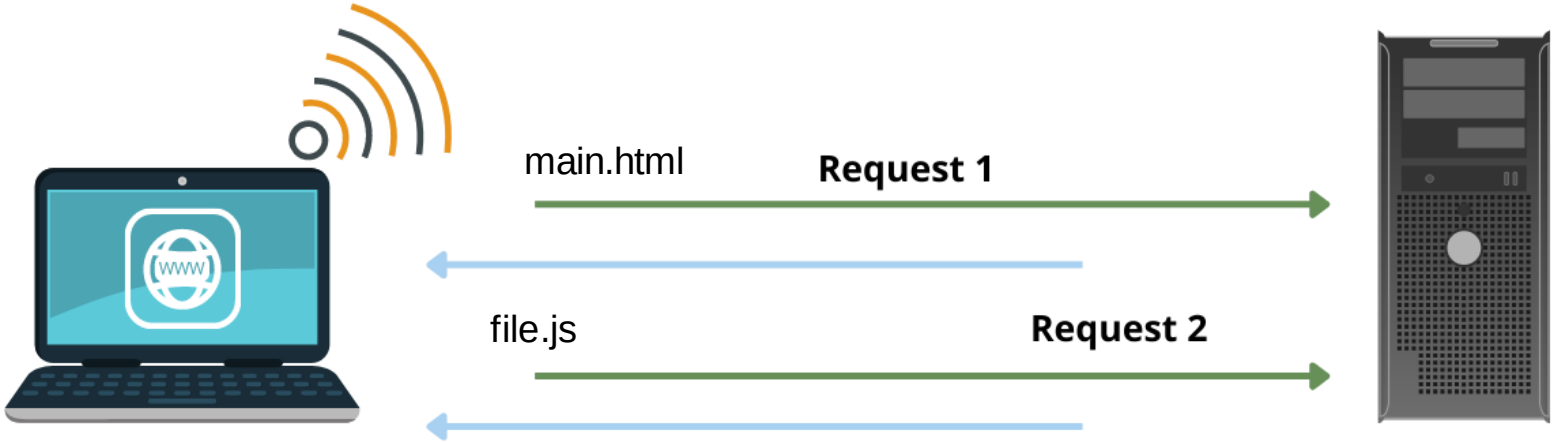
Etapa 1

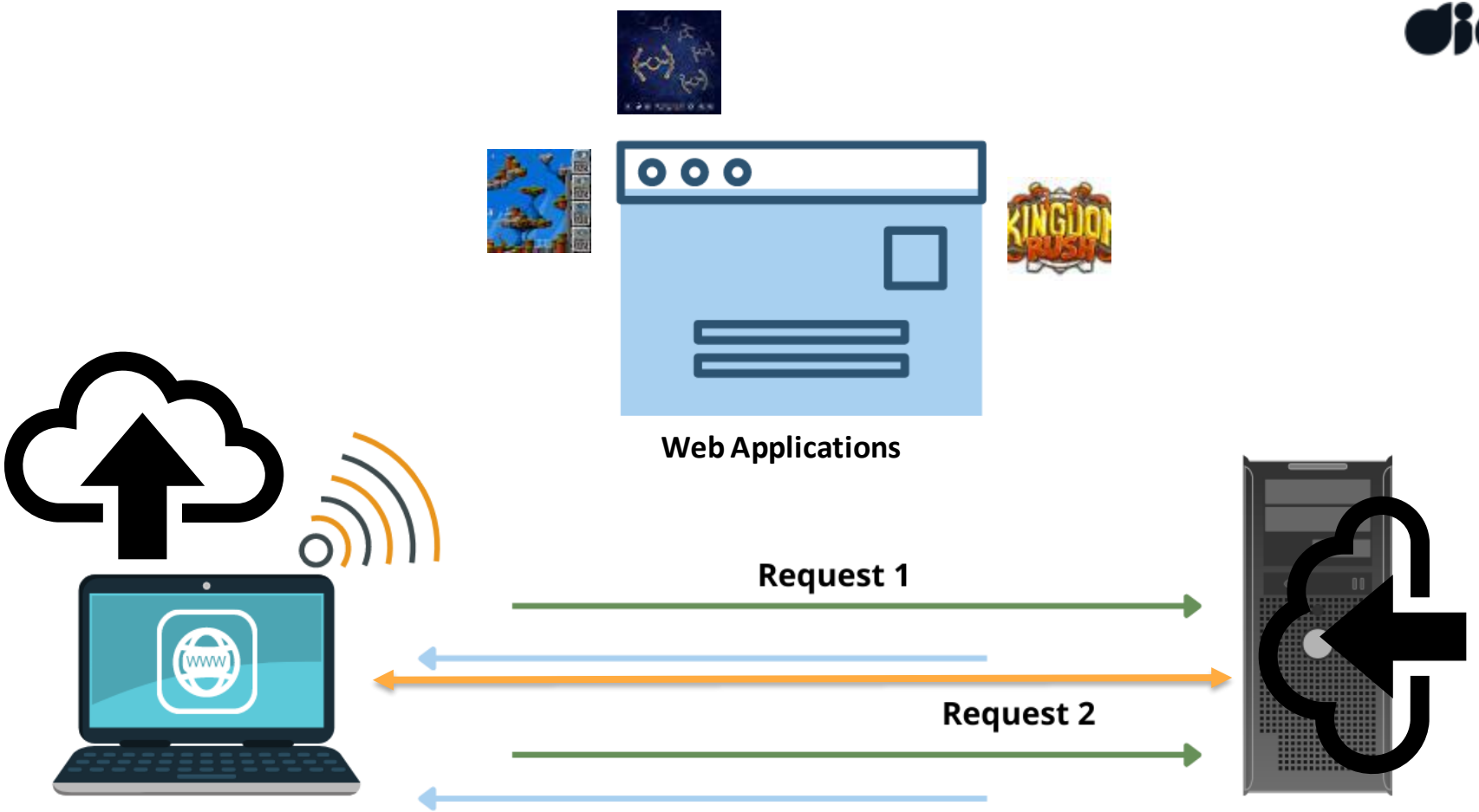
Contextualização

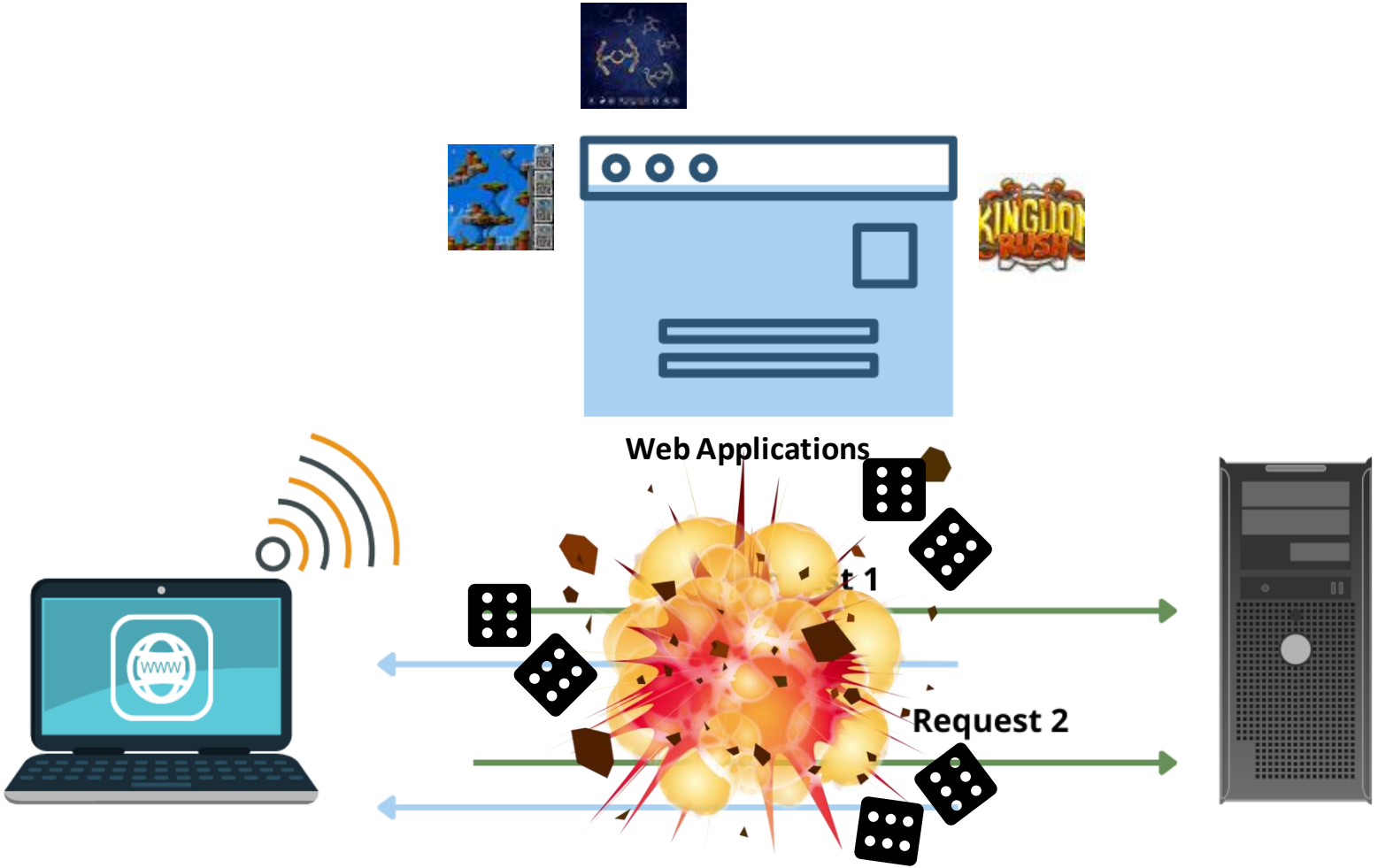
//Conhecendo os principais protocolos de comunicação da internet / Protocolo de comunicação Websocket

HTTP



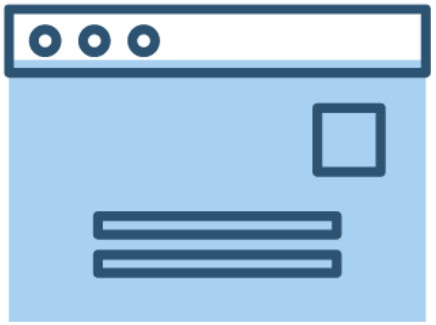



Web Applications
Request 1
Request 2


Web Applications
t 1
Request 2









Web Applications



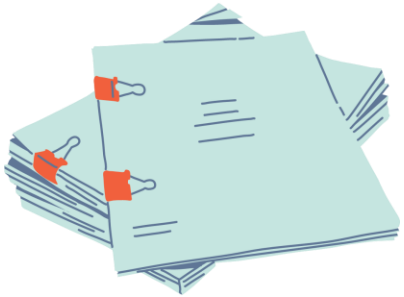
Conexão principal

Chamadas HTTP



Web app

Problemas



1. Conexões subjacentes
2. Alto overhead
3. Mapeamento vis script





HTTP HEADER

Web app



WebSocket

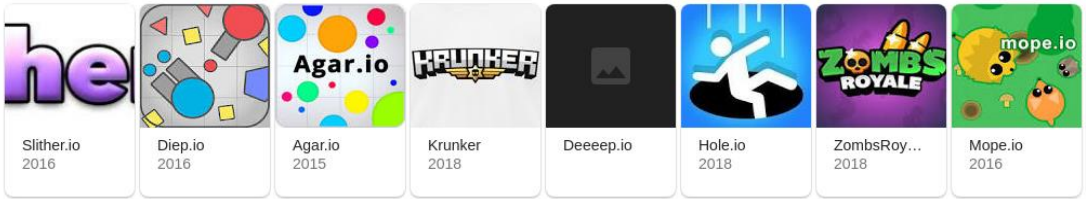


WebSocket

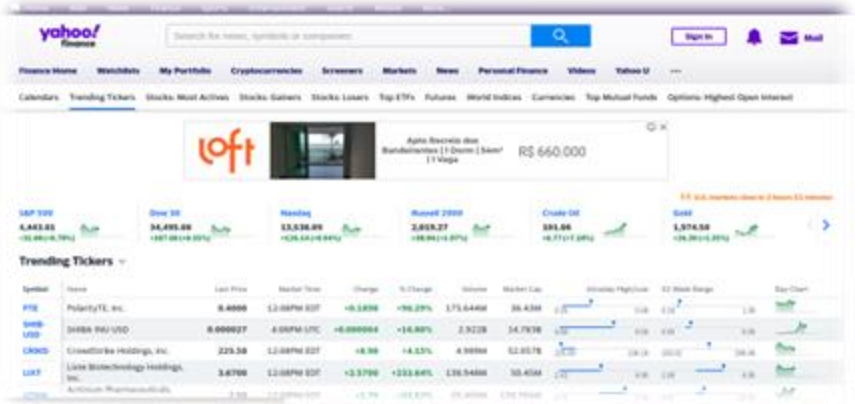


WebSocket API





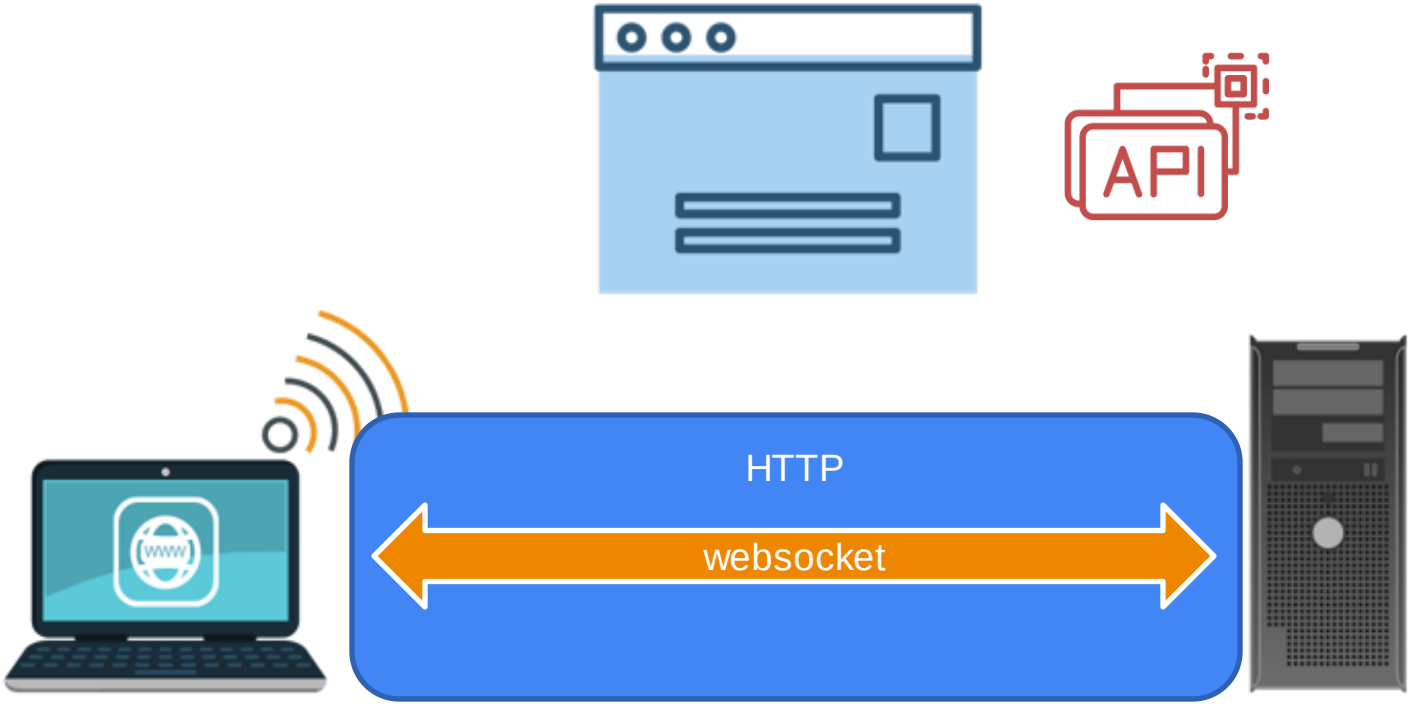
Slither.io 2016 | Diep.io 2016 | Agar.io 2015 | Krunker 2018 | Deeeep.io | Hole.io 2018 | ZombsRoy… 2018 | Mope.io 2016



stock tickers

Trello

WebSocket



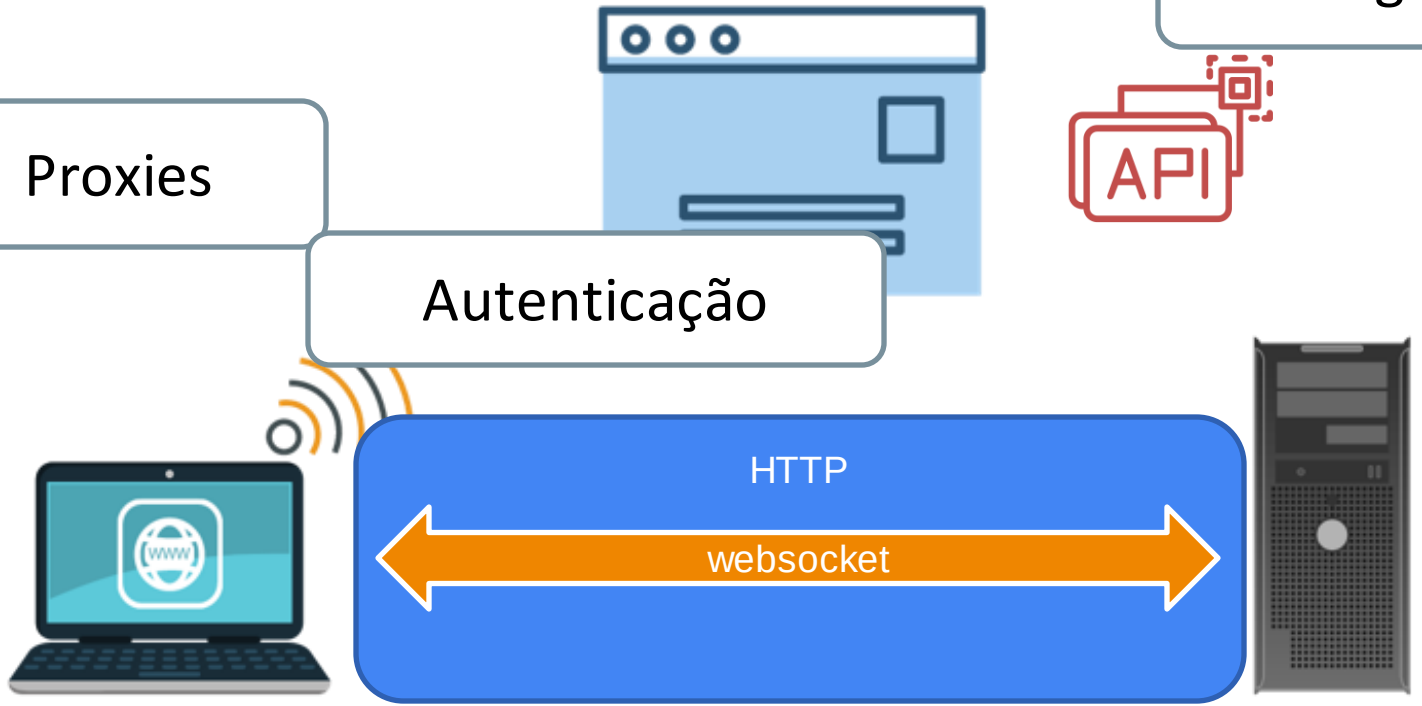
PORTS: 80 E 443

WebSocket

Filtragem

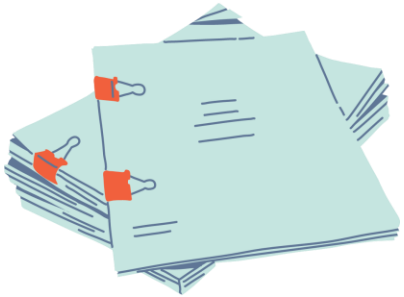
Proxies

Autenticação



Web app

Problemas



1. Conexões subjacentes
2. Alto overhead
3. Mapeamento vis script





HTTP HEADER

Etapa 2

Como funciona o Websocket?

//Conhecendo os principais protocolos de comunicação da internet / Protocolo de comunicação Websocket

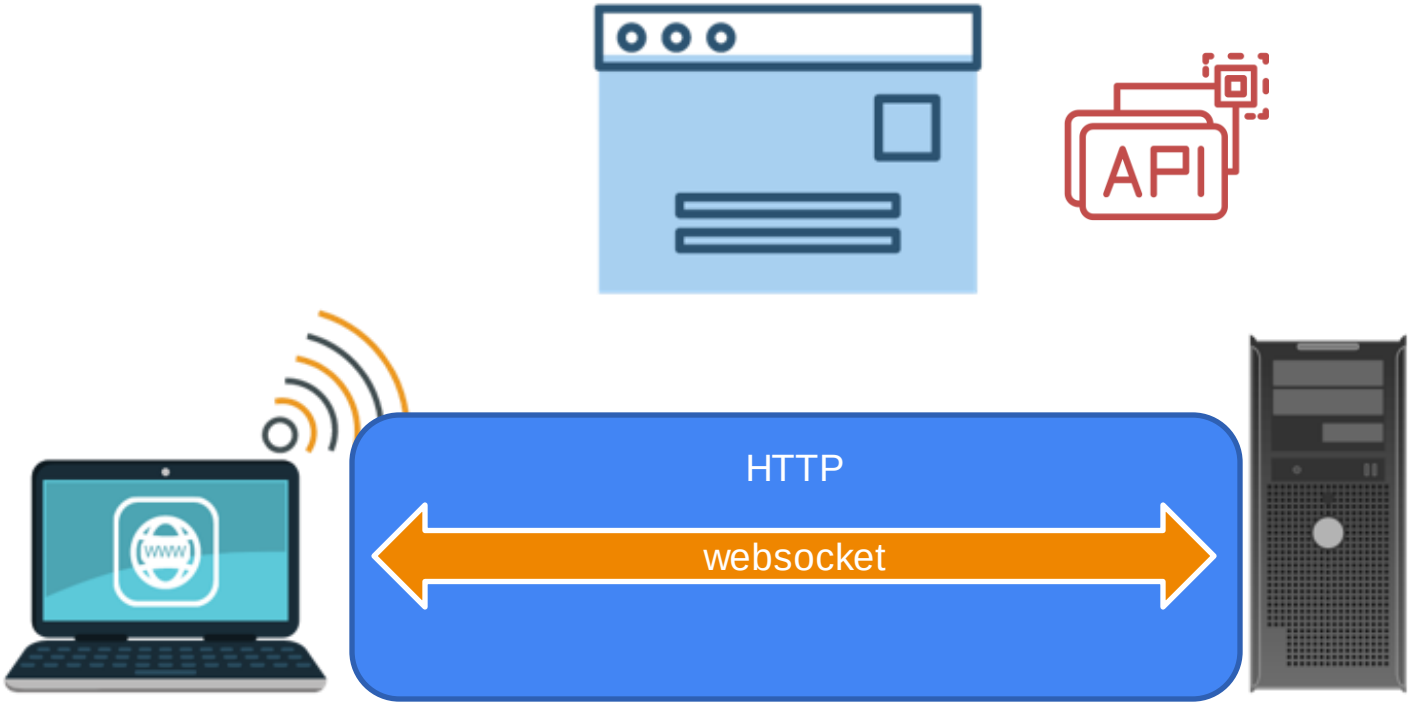
Percurso

~~Etapa 1~~ ~~Contextualização~~

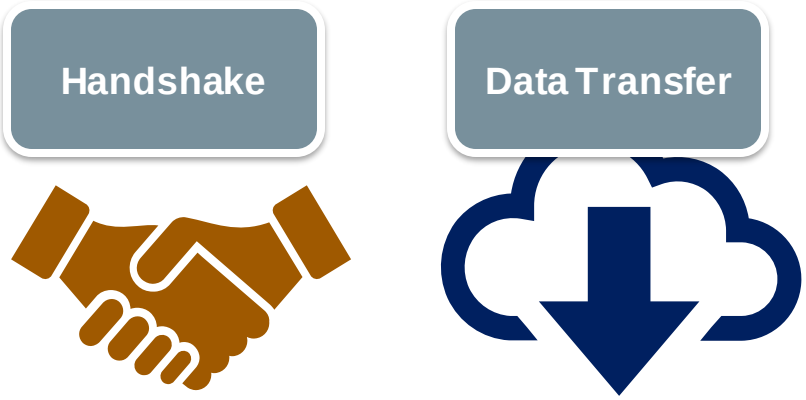
Etapa 2 **Como funciona o WebSocket?**

Etapa 3 Como funciona a comunicação na internet?

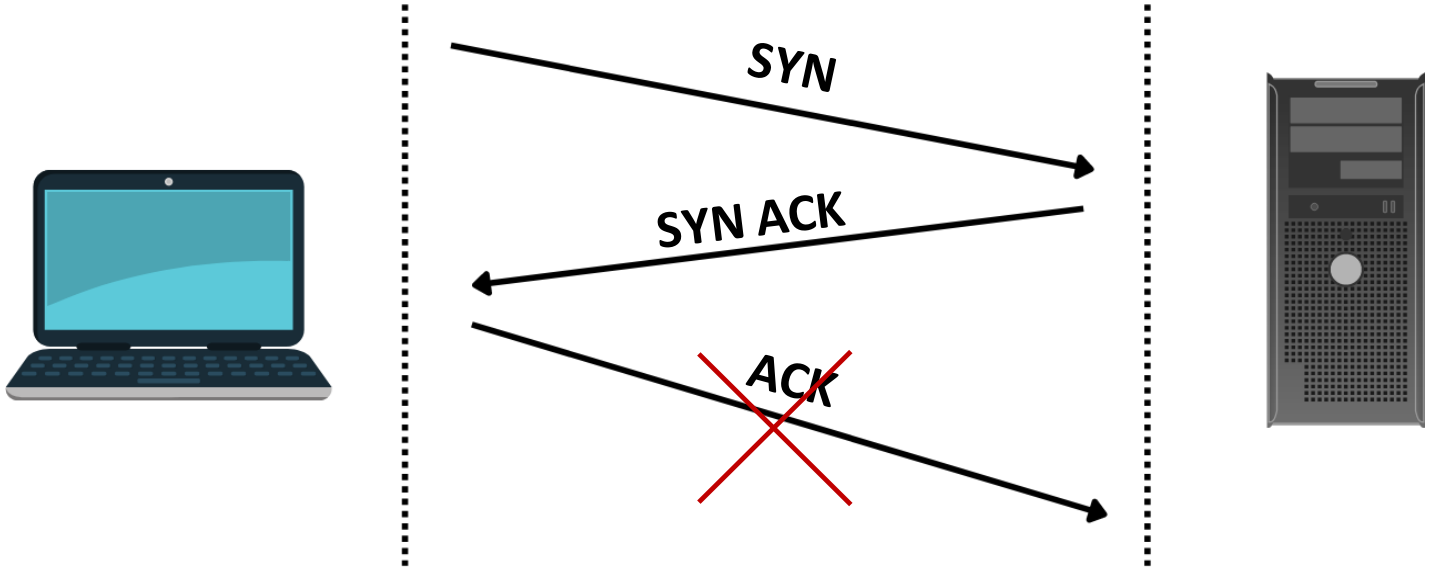
WebSocket



WebSocket



WebSocket



WebSocket







```
GET /chat HTTP/1.1
Host: server.example.com
Upgrade: websocket
Connection: Upgrade
Sec-WebSocket-Key: dGhlIHNhbXBsZSBub25jZQ==
Origin: http://example.com
Sec-WebSocket-Protocol: chat, superchat
Sec-WebSocket-Version: 13
```

```
HTTP/1.1 101 Switching Protocols
Upgrade: websocket
Connection: Upgrade
Sec-WebSocket-Accept: s3pPLMBiTxaQ9kYGzzhZRbK+xOo=
Sec-WebSocket-Protocol: chat
```

WebSocket

Handshake





Websocket server

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WebSocket

Handshake







~~Websocket server~~




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WebSocket

Handshake





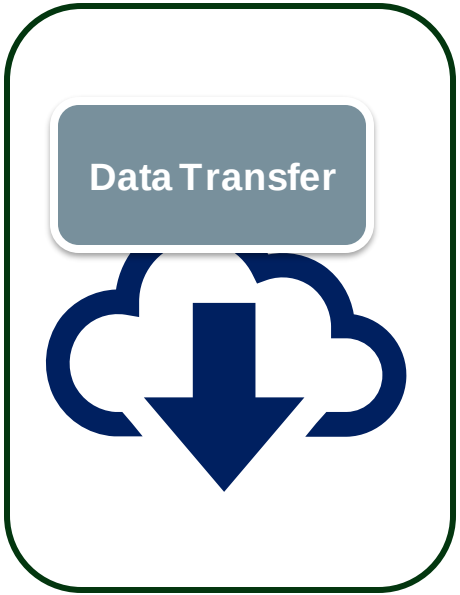


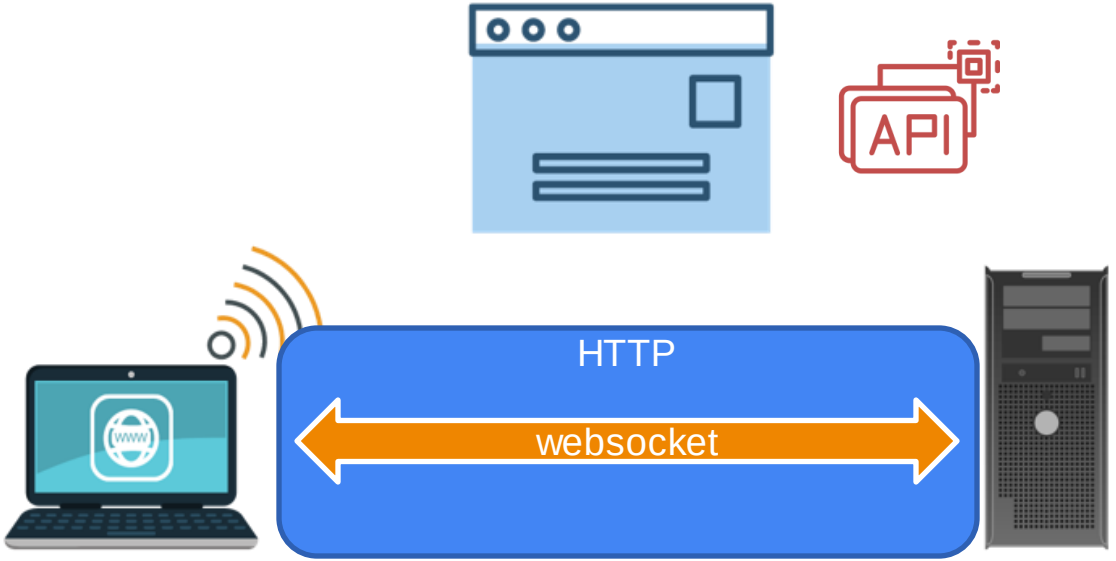
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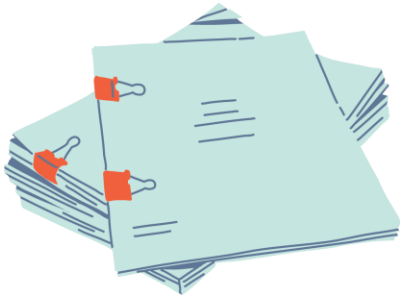
WebSocket





WebSocket

Definição



O protocolo fonece comunicação cliente/servior executando um código não confiável em um ambiente controlado para o host que escolheu essa comunicação.

Modelo origin-base security



WebSocket connection

Subprotocol Selector

Header adicional

routeSelectionExpression

JSON ONLY



```
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WebSocket connection

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Autenticação

WebSocket connection

Subprotocol Selector

Header adicional

routeSelectionExpression

JSON ONLY



```
GET /chat HTTP/1.1
Host: server.example.com
Upgrade: websocket
Connection: Upgrade
Sec-WebSocket-Key: dGhlIHNhbX[illegible]s7SBub25iZQ==
Origin: http://example.com
Sec-WebSocket-Protocol: chat, superchat
Sec-WebSocket-Version: 13
```

CORS não autorizado

WebSocket connection

Subprotocol Selector

Header adicional

routeSelectionExpression

JSON ONLY



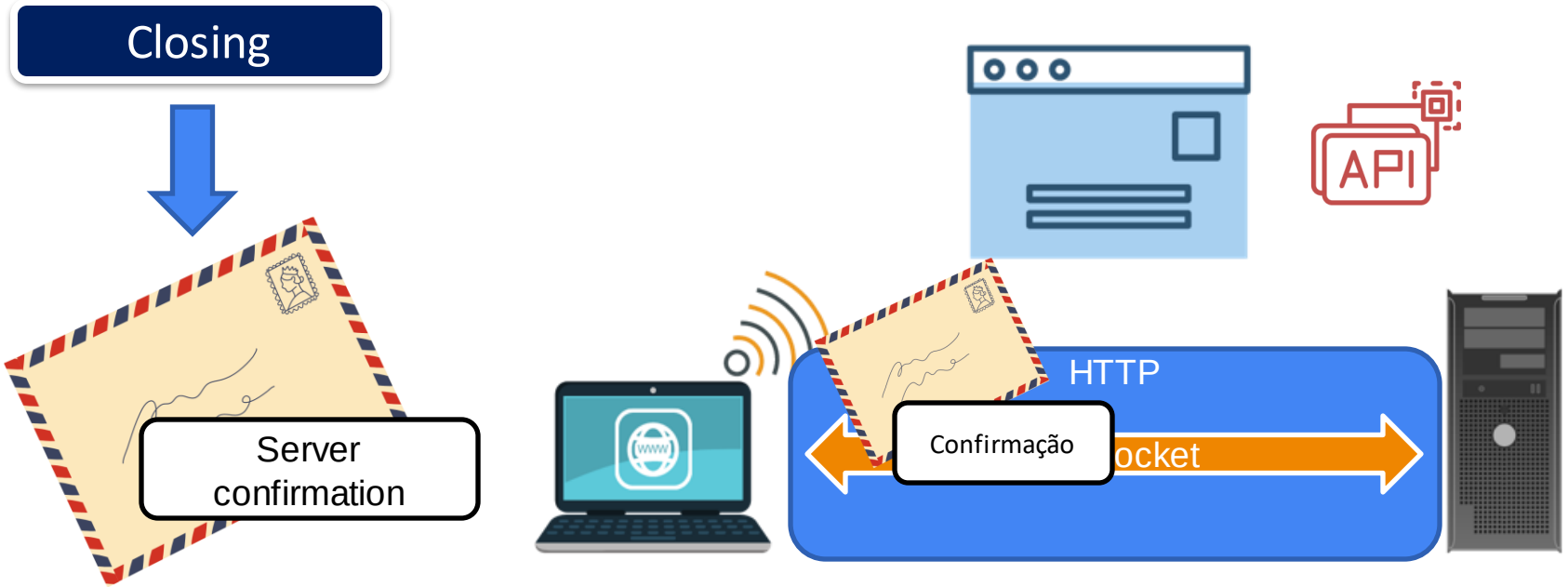
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Sec-WebSocket-Version: 13
```

Layer: aplicação

WebSocket connection



WebSocket connection



Filosofia do WebSocket



ADD

1. Origin-base security model
2. Endereçamento e protocolo
3. Camadas -> enquadramento on top TCP
4. Encerramento do handshake

Filosofia do WebSocket



TCP

ADD

1. Origin-base security model
2. Endereçamento de protocolo
3. [illegible]top TCP
4. Encerramento do handshake

Restringe as páginas web

Outras características



Campos Sec-...: No Edition

Independente do TCP

Interpretação do HTTP server

Porta 443

Etapa 2

Como funciona a comunicação na intenet?

//Conhecendo os principais protocolos de comunicação da internet / Protocolo de comunicação Websocket

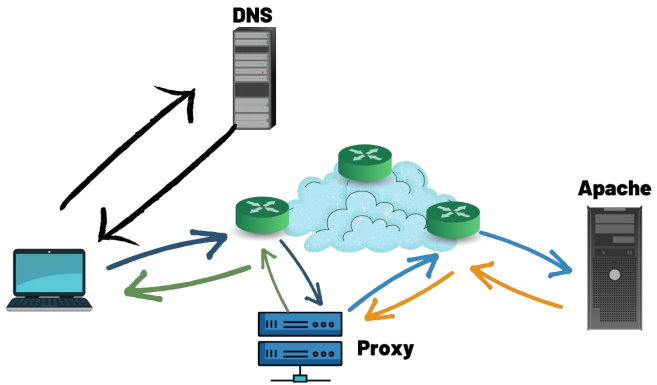
Percurso

~~Etapa 1~~ ~~Contextualização~~

~~Etapa 2~~ ~~Como funciona o WebSocket?~~

Etapa 3 **Como funciona a comunicação na internet?**

Comunicação



- Proxy
- HTTP server

- DNS – Domain Name System
- Socket
- Load balancer

DNS

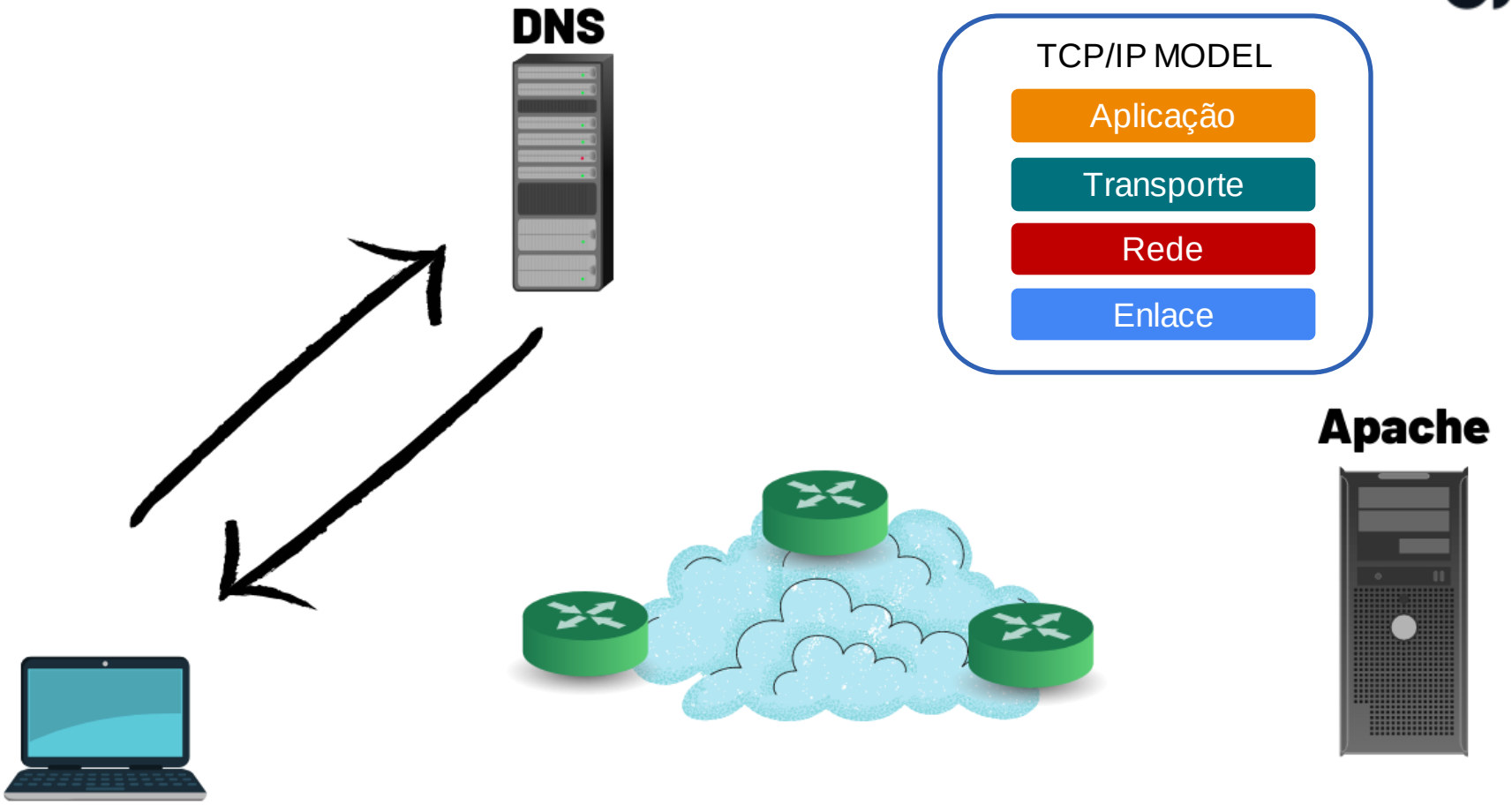


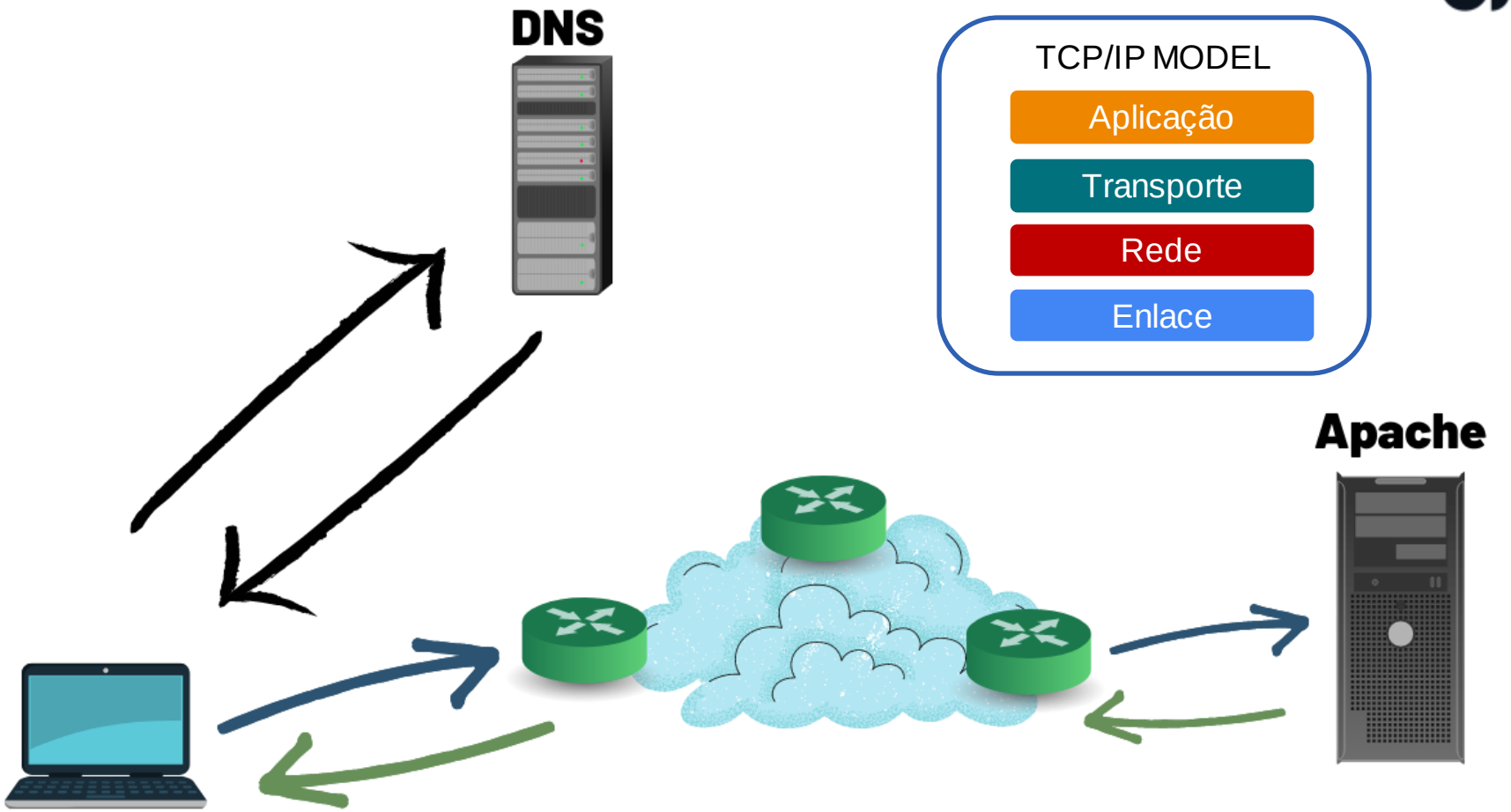


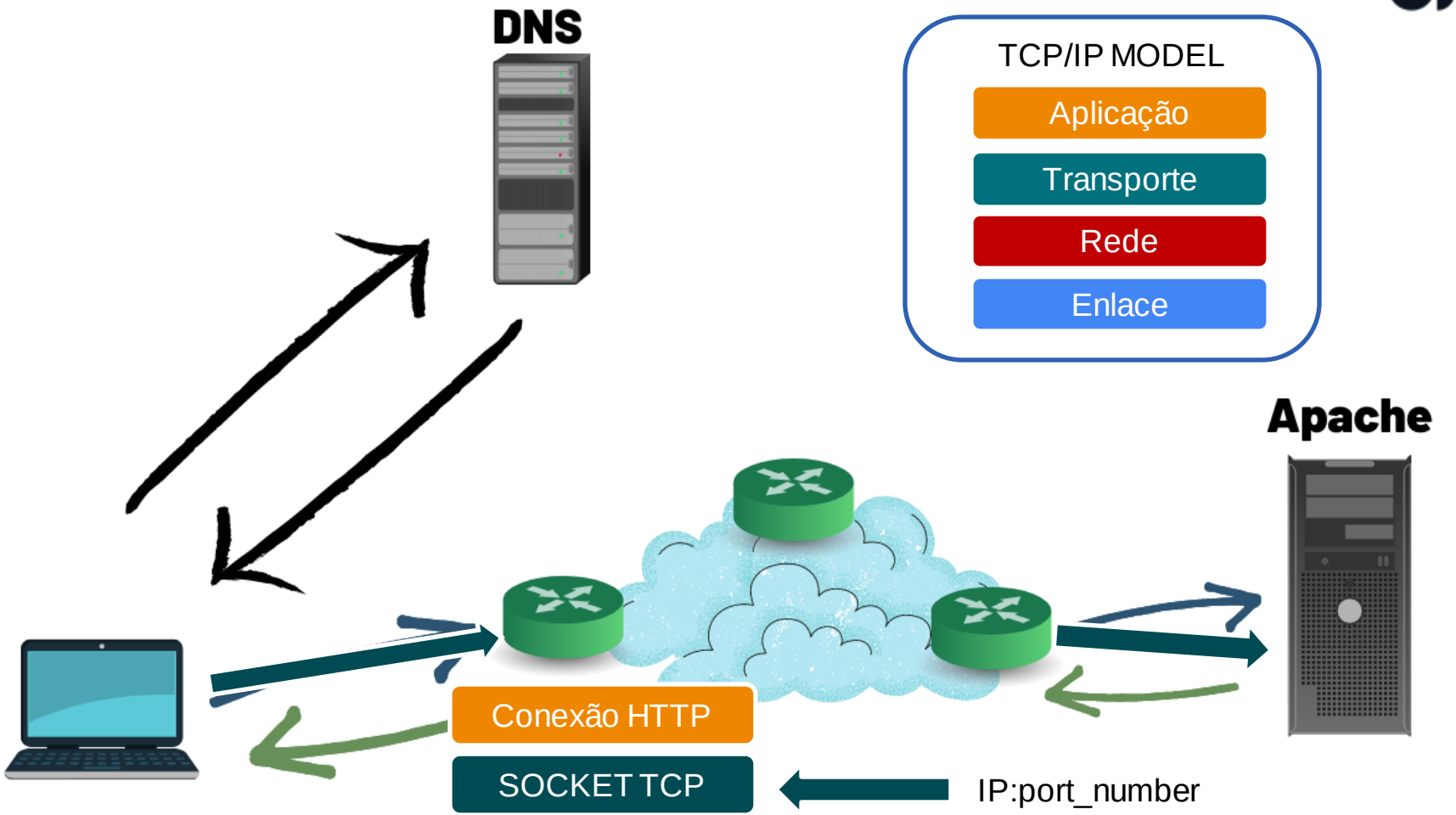
Apache



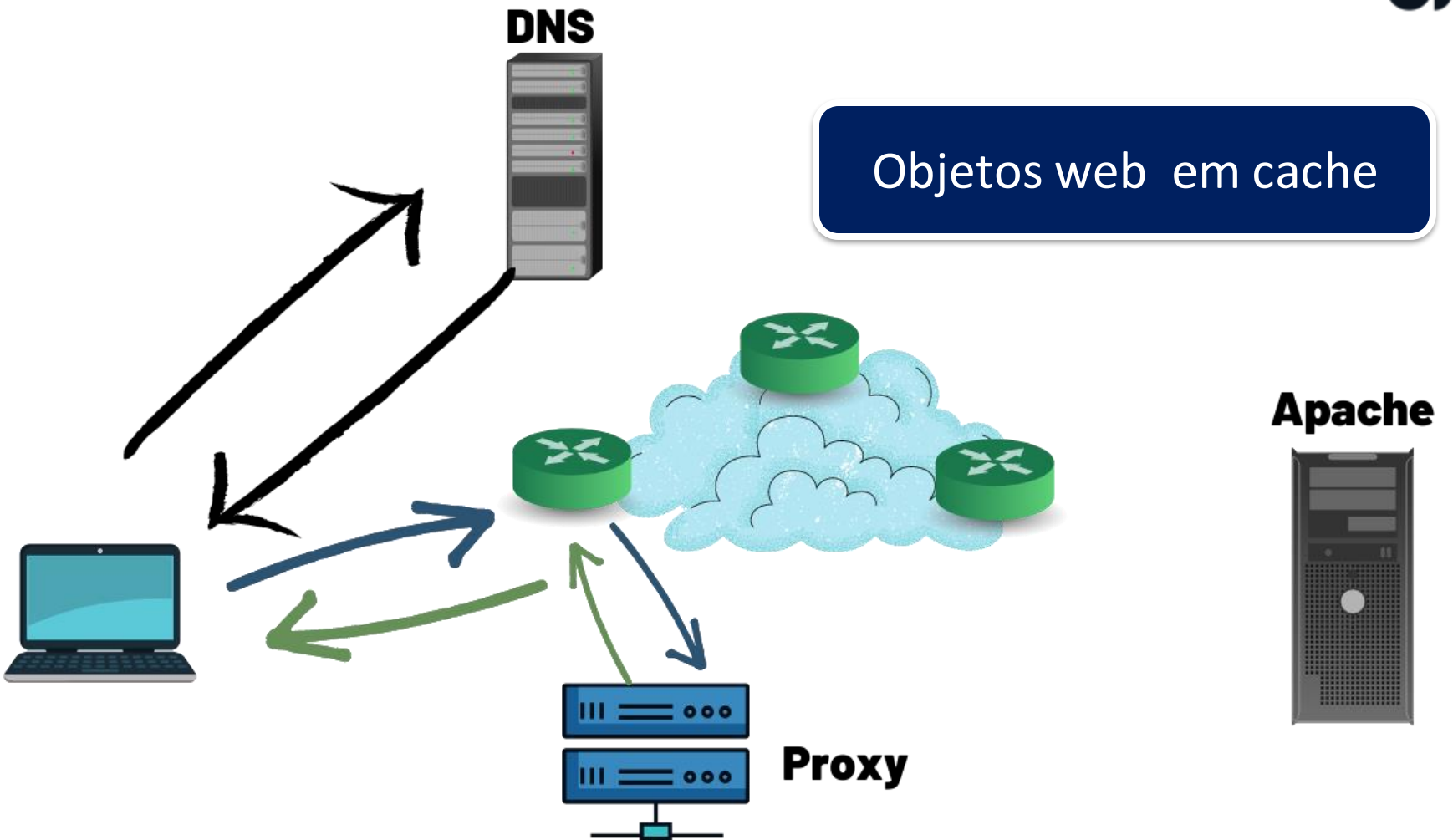


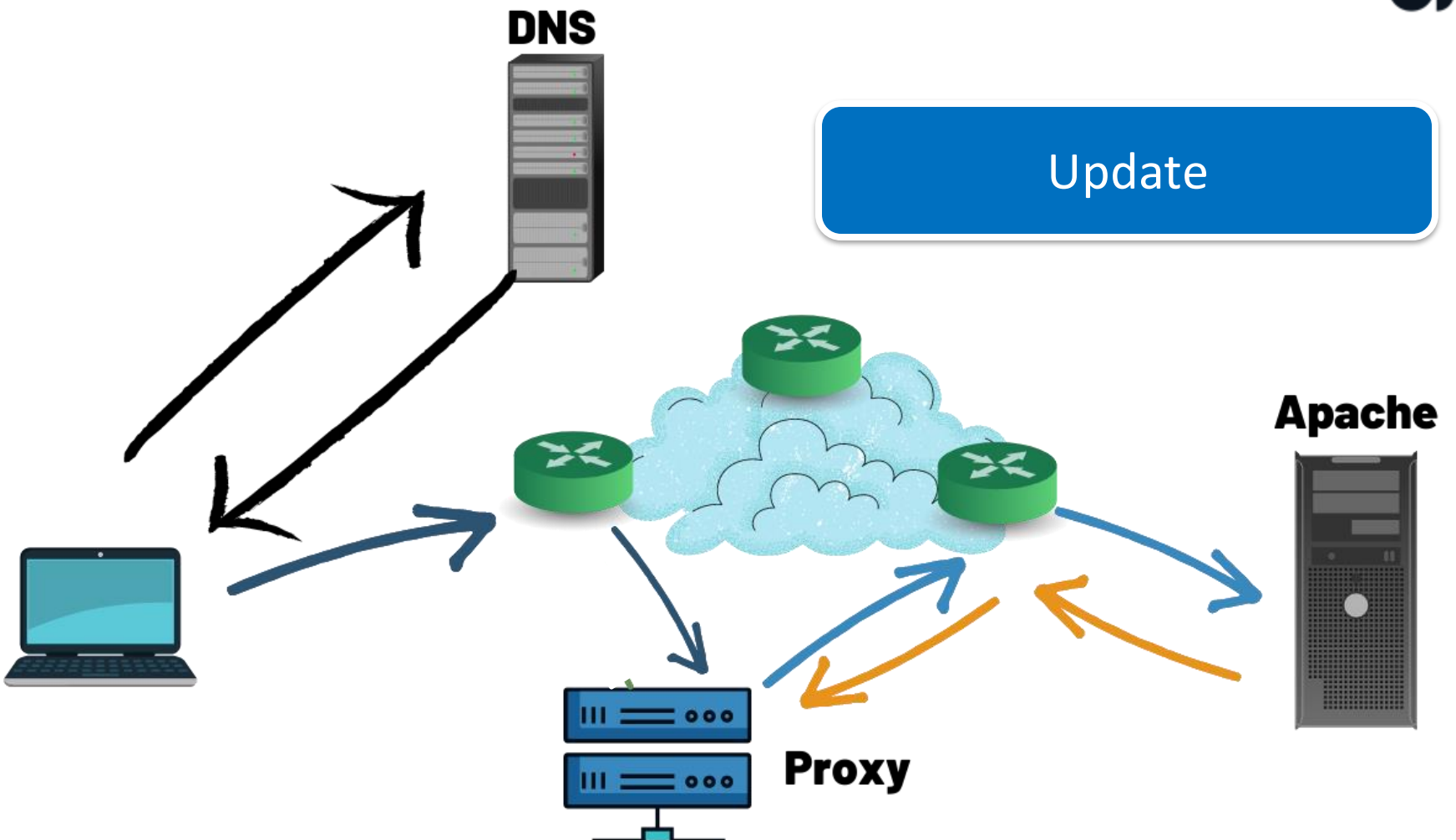

DNS
TCP/IP MODEL
Aplicação
Transporte
Rede
Enlace
Apache

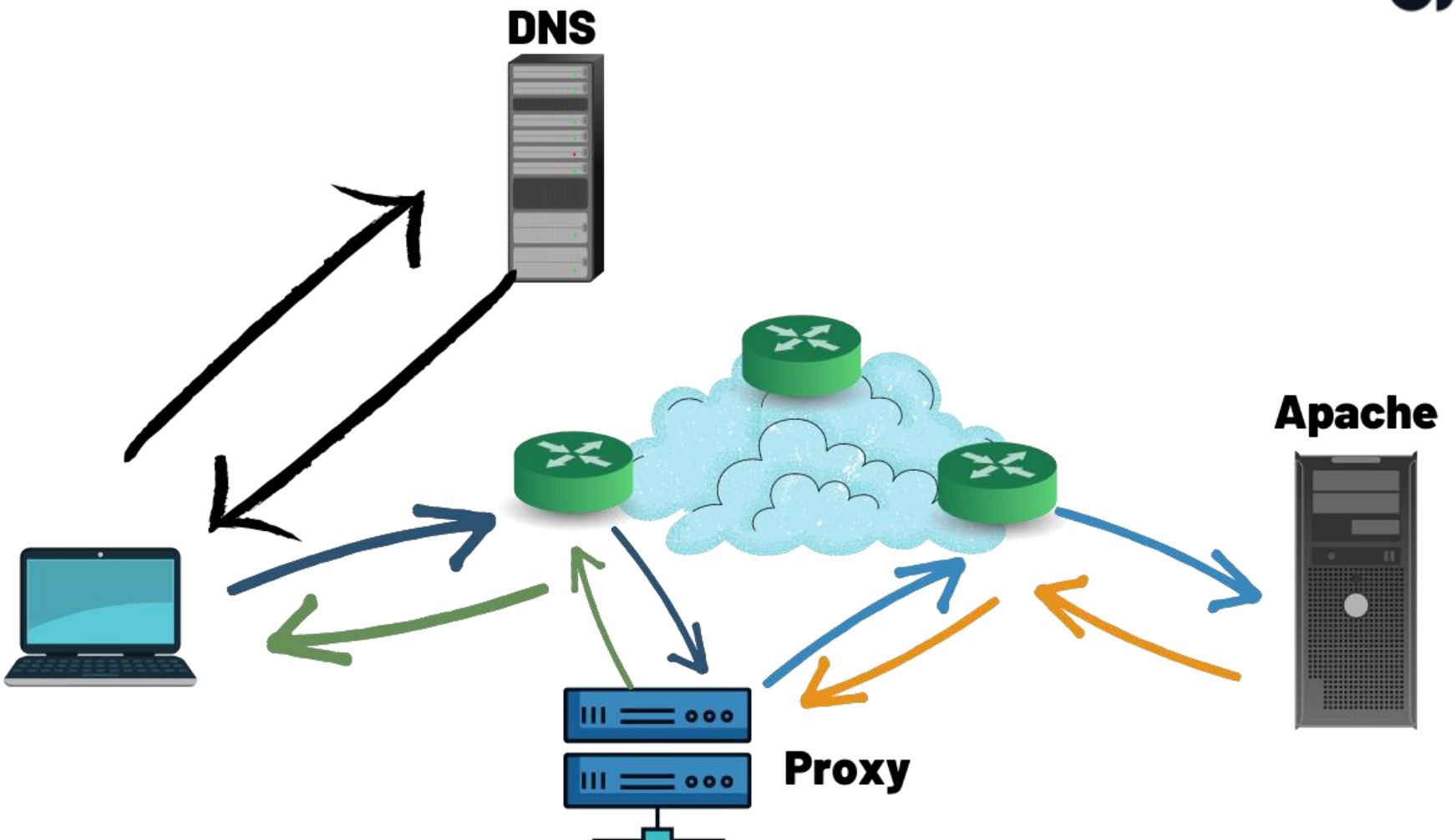

DNS
TCP/IP MODEL
Aplicação
Transporte
Rede
Enlace
Apache


DNS
TCP/IP MODEL
Aplicação
Transporte
Rede
Enlace
Apache
Conexão HTTP
SOCKET TCP
IP:port_number

Objetos web em cache




DNS
Update
Apache
Proxy


DNS
Apache
Proxy







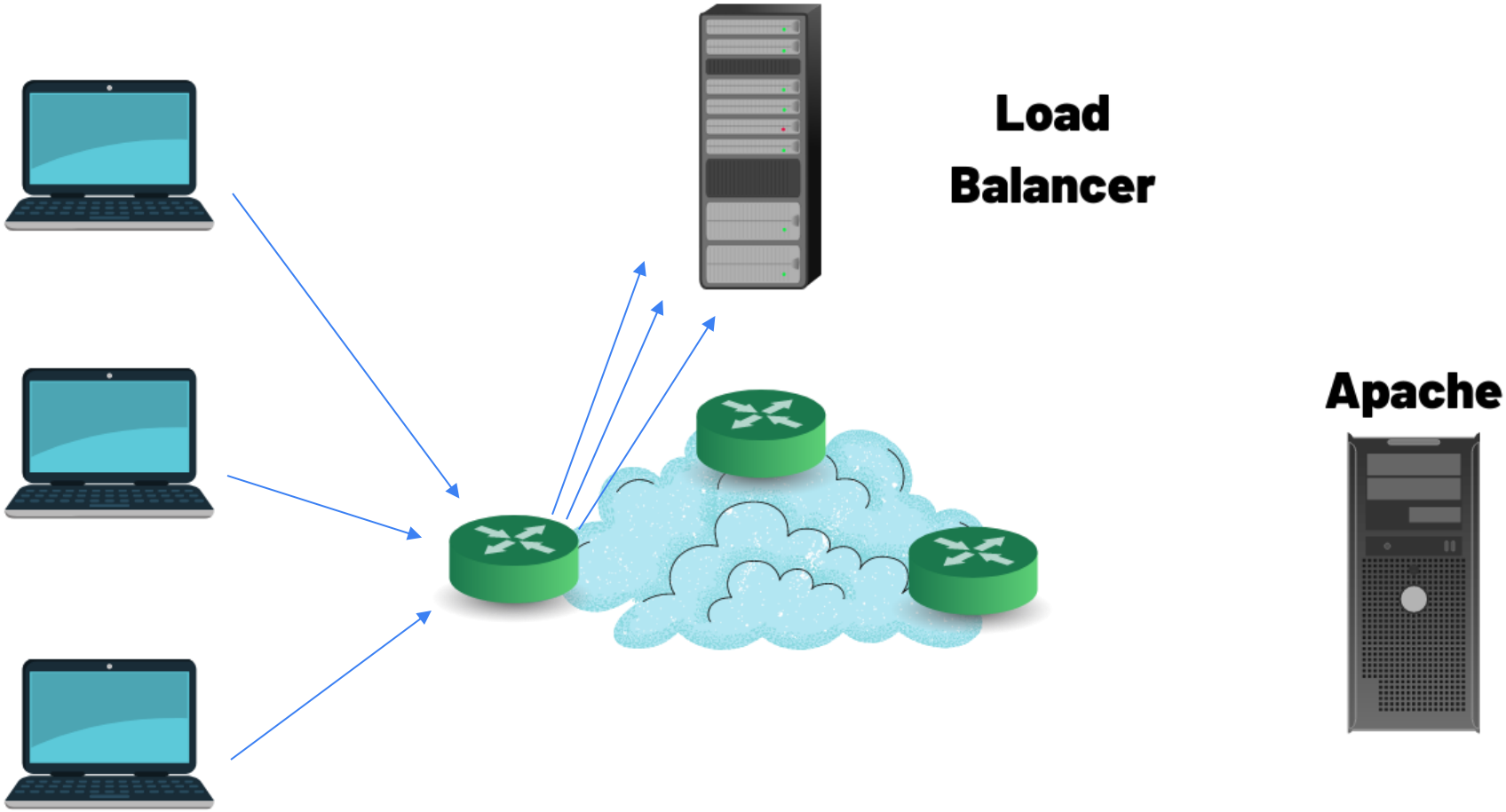


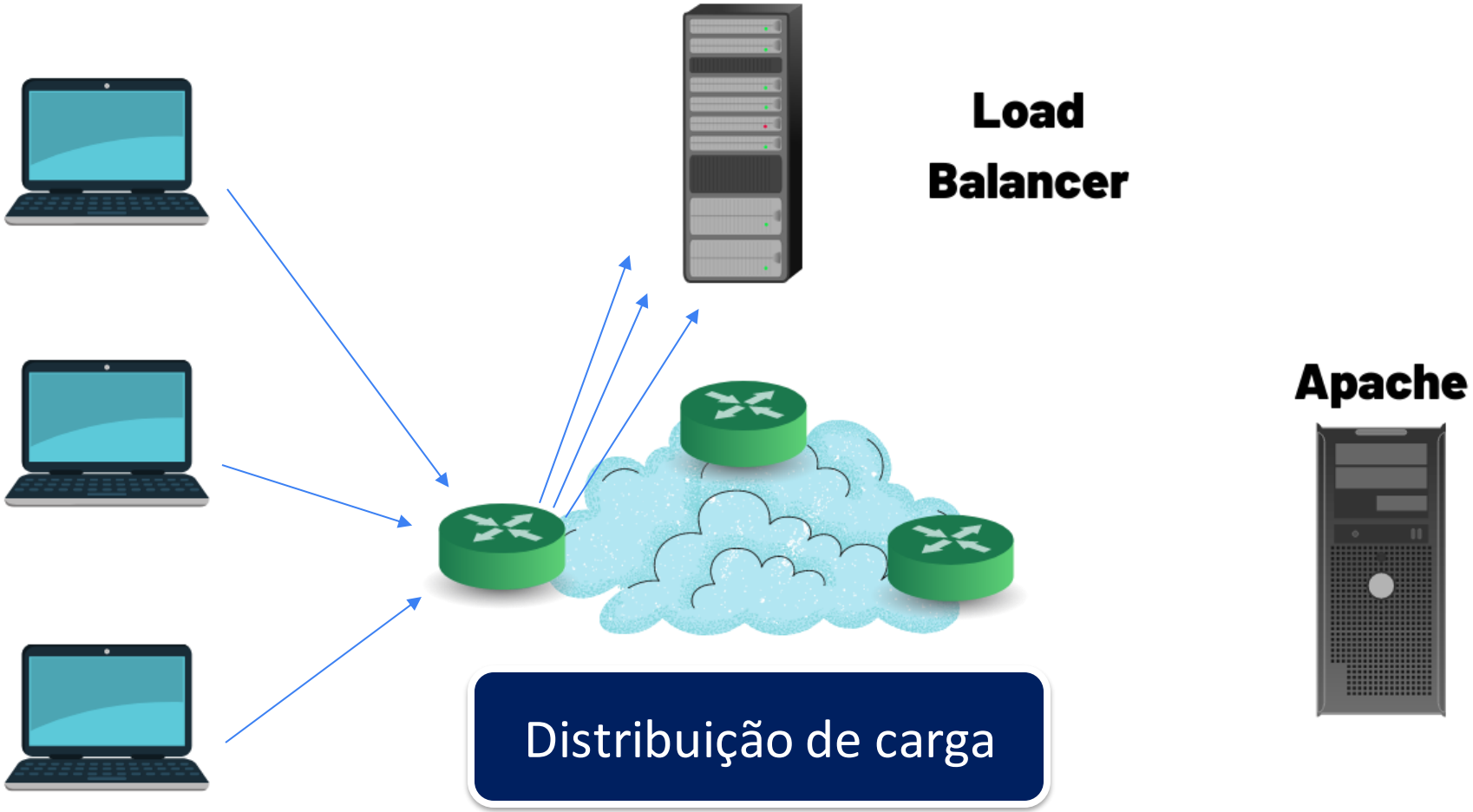
Load Balancer

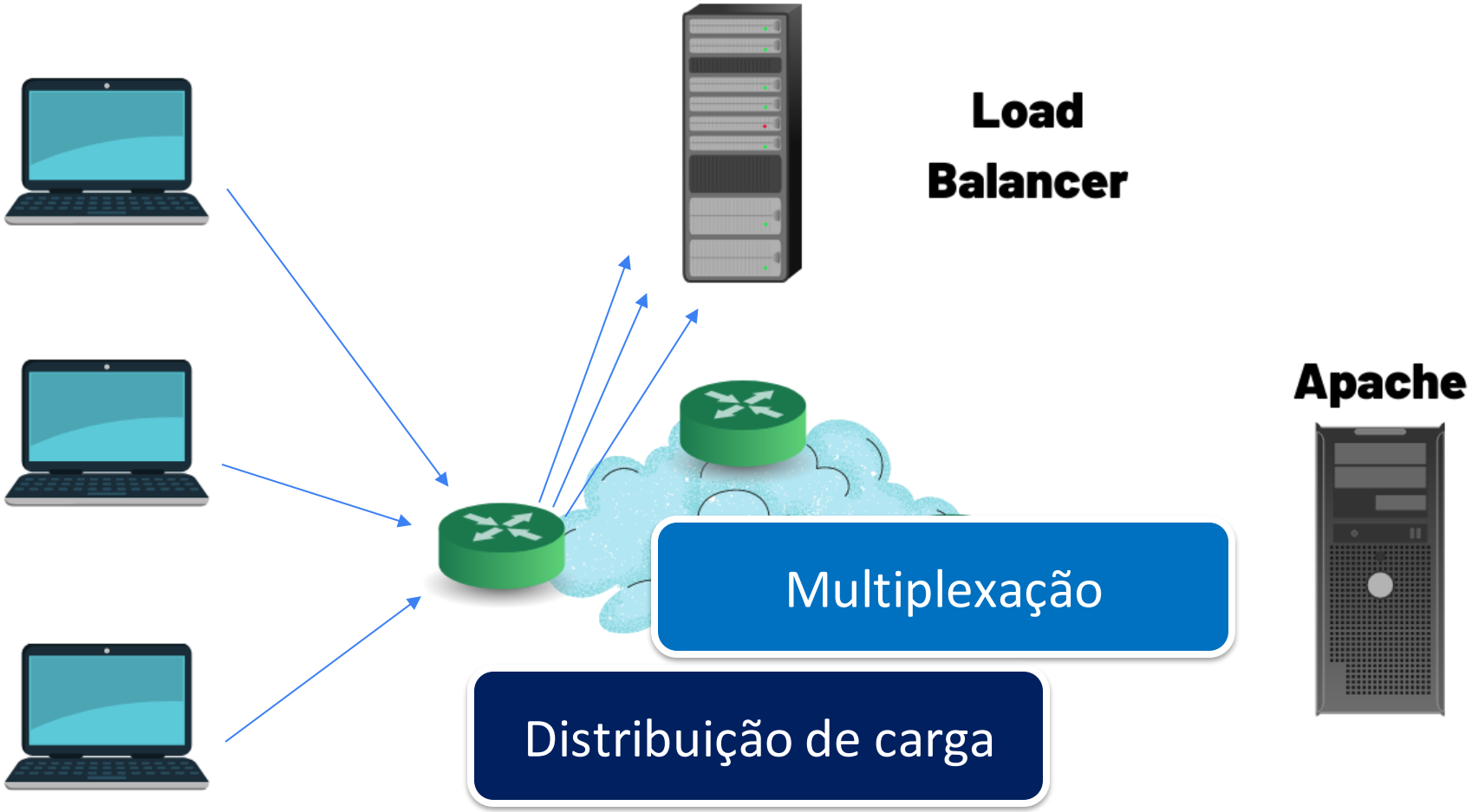


Apache




Load
Balancer
Apache


Load
Balancer
Apache
Distribuição de carga


Load
Balancer
Apache
Multiplexação
Distribuição de carga

Continua ...

Links Úteis

- **Repositório no GitHub** - Java HTTPClient (opcional)
- **Referências oficiais - RFCs**
  - **RFC HTTP 1.1**
  - **RFC HTTP 2.0**
  - **RFC WebSocket**

Para saber mais

Principais livros (HTTP)

- Kurose – Redes de Computadores e a internet
  Capítulo 2 - Camada de aplicação

- Tanenbaum – Redes de Computadores
  Capítulo 7 - Camada de aplicação

Para saber mais

Protocolo HTTP

https://simplificandoredes.com/en/http-methods/
https://simplificandoredes.com/en/cookies-http-protocol/
https://simplificandoredes.com/en/http-protocol-how-does-it-work/
https://simplificandoredes.com/en/apache-https-server-configure/
https://simplificandoredes.com/en/install-apache-on-windows/

Sockets

https://www.youtube.com/watch?v=9ifrhE4UXxA

Para saber mais

Protocolo HTTP

https://datatracker.ietf.org/doc/html/rfc6455
https://developer.mozilla.org/en-US/docs/Web/HTTP/Range_requests#checking_if_a_server_supports_partial_requests
https://developer.mozilla.org/en-US/docs/Web/HTTP/Status
https://developer.mozilla.org/en-US/docs/Glossary/Safe/HTTP
https://testmatick.com/what-are-cookie-files-and-how-to-test-them/#:~:text=Cookie%20files%20can%20be%20considered,html%20page.

Para saber mais

Protocolo HTTP 2.0

https://httpd.apache.org/docs/2.4/howto/http2.html
https://www.youtube.com/watch?v=fVKPrDrEwTI
https://chrome.google.com/webstore/detail/http2-and-spdy-indicator/mpbpobfflnpcgagjijhmgnchggcjblin?hl=pt-BR
https://blog.cloudboost.io/http2-and-spdy-protocols-make-http-faster-and-safer-3b3984987191
https://www.cloudflare.com/learning/performance/http2-vs-http1.1/
https://news.ycombinator.com/item?id=19014117

Para saber mais

Protocolo WebSocket

https://medium.com/@lancers/websocket-api-sec-websocket-protocol-subprotocol-header-support-277e34164537
https://en.wikipedia.org/wiki/WebDAV
https://developer.mozilla.org/en-US/docs/Glossary/WebDAV
https://simplificandoredes.com/como-funciona-o-protocolo-websocket/
https://websockets.spec.whatwg.org//
https://developer.mozilla.org/en-US/docs/Web/API/Web_Workers_API

Para saber mais

XML e JSON

https://en.wikipedia.org/wiki/XML#Applications
https://www.w3schools.com/js/js_json_intro.asp
https://blog.geekhunter.com.br/xml-vs-json-entenda-como-fazer-a-melhor-escolha/

Dúvidas?

> Fórum/Artigos

> Comunidade Online (Discord)

